Pico WiFi DMX User Manual

This manual explains how to use the browser-based DMX controller with the Pico firmware. It is written for daily operation: creating fixtures, controlling lights, saving scenes, building chasers, creating effects, and using GPIO buttons or ADC inputs.

The web interface is normally opened from XAMPP. On the same computer that runs XAMPP this is often:

```
http://localhost/dmx/
```

From an iPad, phone, or another computer, use the XAMPP computer's LAN address or hostname instead, for example `http://192.168.0.50/dmx/`. This XAMPP URL is only the address of the web interface and server-side show storage.

The Pico itself is controlled over the network with its base URL, for example:

```
http://192.168.0.24/
```

Enter the Pico base URL once in any page. The browser stores it and shares it with the other pages. Across the UI, live Pico updates only happen while a Pico base URL is set; clearing the URL puts the page into browser-only editing. The Pico base URL is separate from the XAMPP URL: XAMPP serves the controller pages, while the Pico base URL is the lighting controller hardware API.

This applies to every page that can talk to the Pico:

- **Fixture Controller** edits values locally, and only sends DMX changes when the Pico base URL is set.
- **Chaser** can edit steps locally without a URL; live step preview, Chase Playback, and Pico slot actions need a Pico base URL.
- **Effects** can edit effect recipes locally without a URL; browser preview, scene-to-Pico center updates, and Pico slot actions need a Pico base URL.
- **GPIO Control** can edit and save mappings locally without a URL; upload, readback, and status polling need a Pico base URL.
- **DMX Buffer Monitor** only polls Pico buffers when a Pico base URL is set.
- **Pico Performance Test** only runs when a Pico base URL is set.

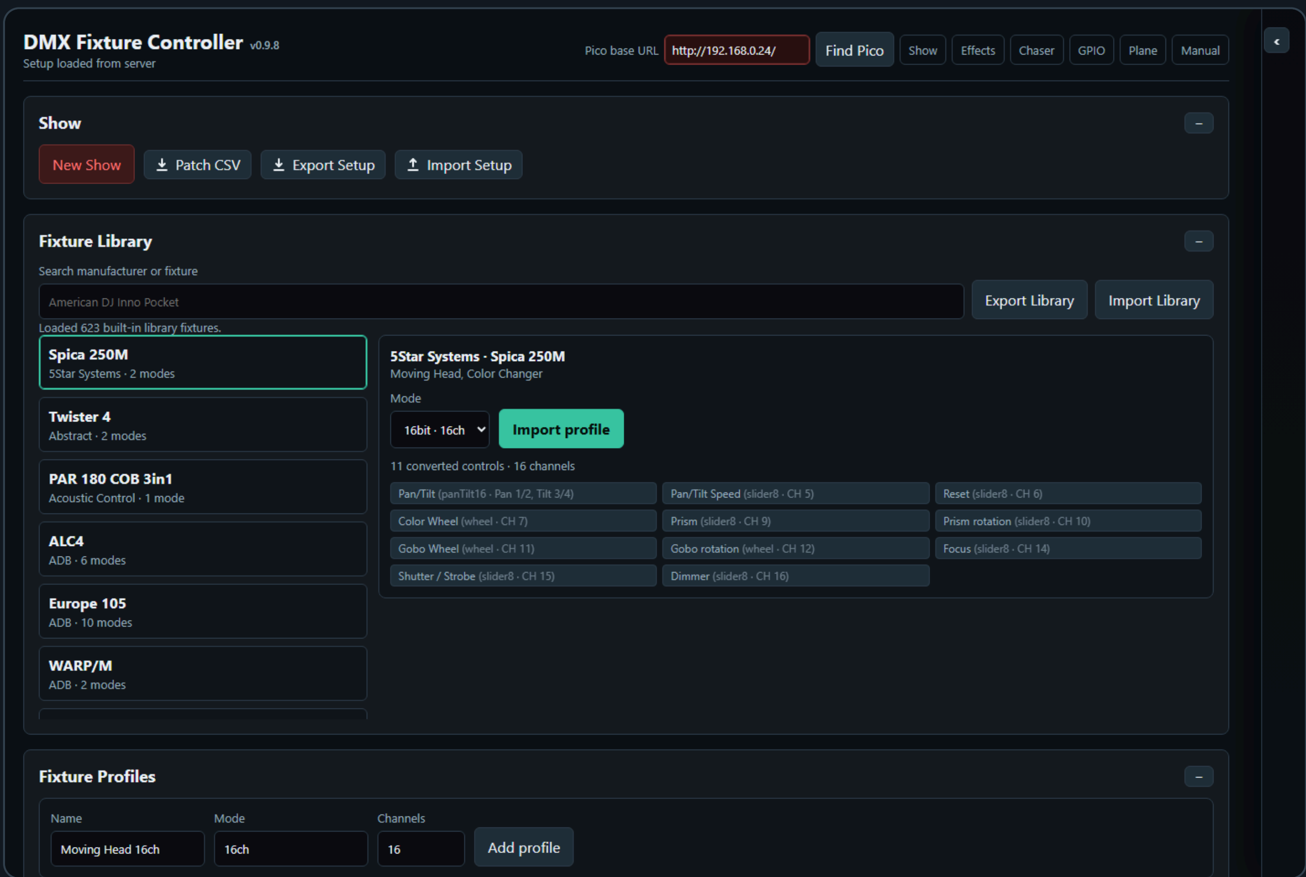
The screenshots in this manual are generated by `scripts/update_user_manual.ps1 -LocalOnly` from the repo-local PHP dev router and the deterministic data in `docs/manual-data/`. Toolbox screenshots are captured as named page elements, so the manual can be regenerated on another development machine without a local XAMPP path or hand-cropping images.

Main Pages

Page	Purpose
Fixture Controller	Fixture profiles, patching, live control, groups, scenes, default and blackout values
Show Run	Operator page for recalling saved groups, scenes, palettes, and Pico chaser/effect slots without editing setup
Chaser	Step-based chases with fade, loop modes, direction, pause/resume, and Pico slot upload
Effects	Continuous effects for pan/tilt pairs or scalar controls such as dimmer, zoom, iris, prism, and gobo
GPIO Control	Map Pico GPIO buttons and ADC inputs to lighting actions
DMX Buffer Monitor	Read all 512 values from either the live output frame or the base/position buffer
Pico Performance Test	Test Pico HTTP/DMX update performance, buffer readback, and core timing logs
Room Plane	Calibrated room-plane coordinate mapping for moving lights

---

1. Fixture Controller

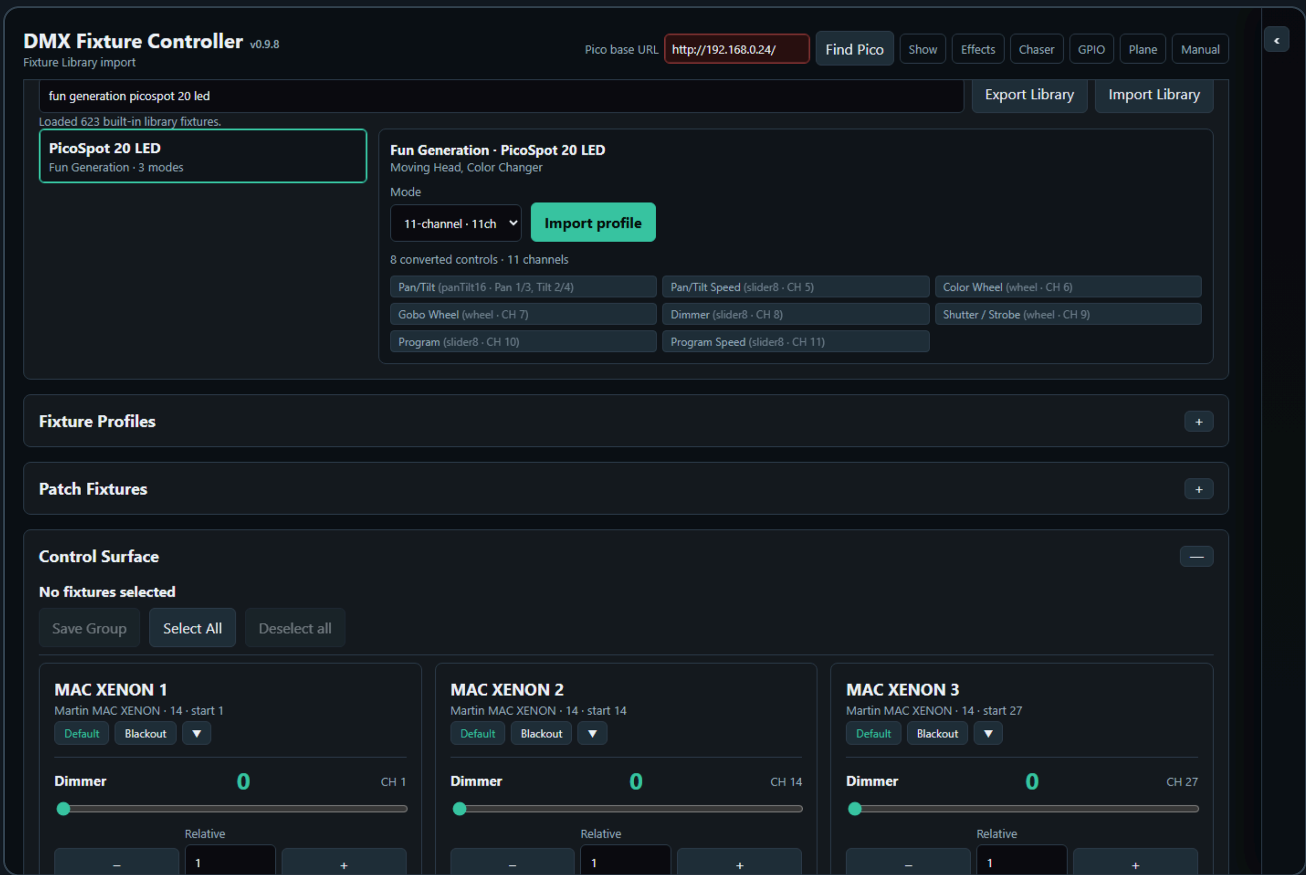


The Fixture Controller is the central page. Use it first when setting up a new show or fixture set.

Set the Pico Base URL

1. Open the XAMPP controller URL, for example `http://localhost/dmx/` on the XAMPP computer or `http://192.168.0.50/dmx/` from another device on the LAN.
2. Enter the Pico base URL, for example `http://192.168.0.24/`.
3. If DHCP changed the Pico address, click **Find Pico**. The XAMPP server listens briefly for the Pico's UDP discovery beacon and fills the URL when a Pico is found on the same LAN. The discovered URL is stored in the browser and saved back to the current page's XAMPP setup file on Controller, Chaser, Effects, and GPIO, so the other pages and other devices can reuse the corrected address after reload.
4. The Pico base URL field checks `/status.json` automatically. It turns dark green when the Pico is reachable, dark red when it cannot connect, and keeps retrying in the background so it can recover after flashing or rebooting the Pico.
5. Control changes are sent to the Pico when a Pico base URL is set. If the field is empty, the page edits locally only and does not send live DMX updates.
6. Fixture setup changes are autosaved to the XAMPP server. Use **Export Setup** before large changes when you want an extra backup of the complete show setup.

Fixture Library



The **Fixture Library** panel imports fixture profiles from the converted Open Fixture Library export. Use it when you want to start from an existing fixture definition instead of building every channel by hand.

The library loads automatically when the controller page opens. While it is loading, the search field is disabled and the panel shows a loading status. If loading fails, the panel shows a **Retry** button.

Command buttons that save, import, export, or update setup data briefly show their result on the button itself. For example, **Update Library** changes to **Updating...** and then **Updated** when the fixture library write is complete.

Use **Export Library** to download the currently loaded fixture catalog as `pico_dmx_fixture_library.json`. Use **Import Library** to upload a converted fixture library catalog to the XAMPP server. After import, the controller loads that custom library first; if no custom library is saved, it falls back to the built-in converted library asset. The development script `scripts/sync_fixture_library_from_xampp.ps1` can refresh that bundled fallback from the current XAMPP catalog after validating the library schema, fixture count, and fixture keys. When fixtures differ, the script asks whether to take each XAMPP change or keep the bundled copy; `-AcceptAllChanges`, `-KeepExistingChanges`, and `-DryRun` are available for intentional automated runs.

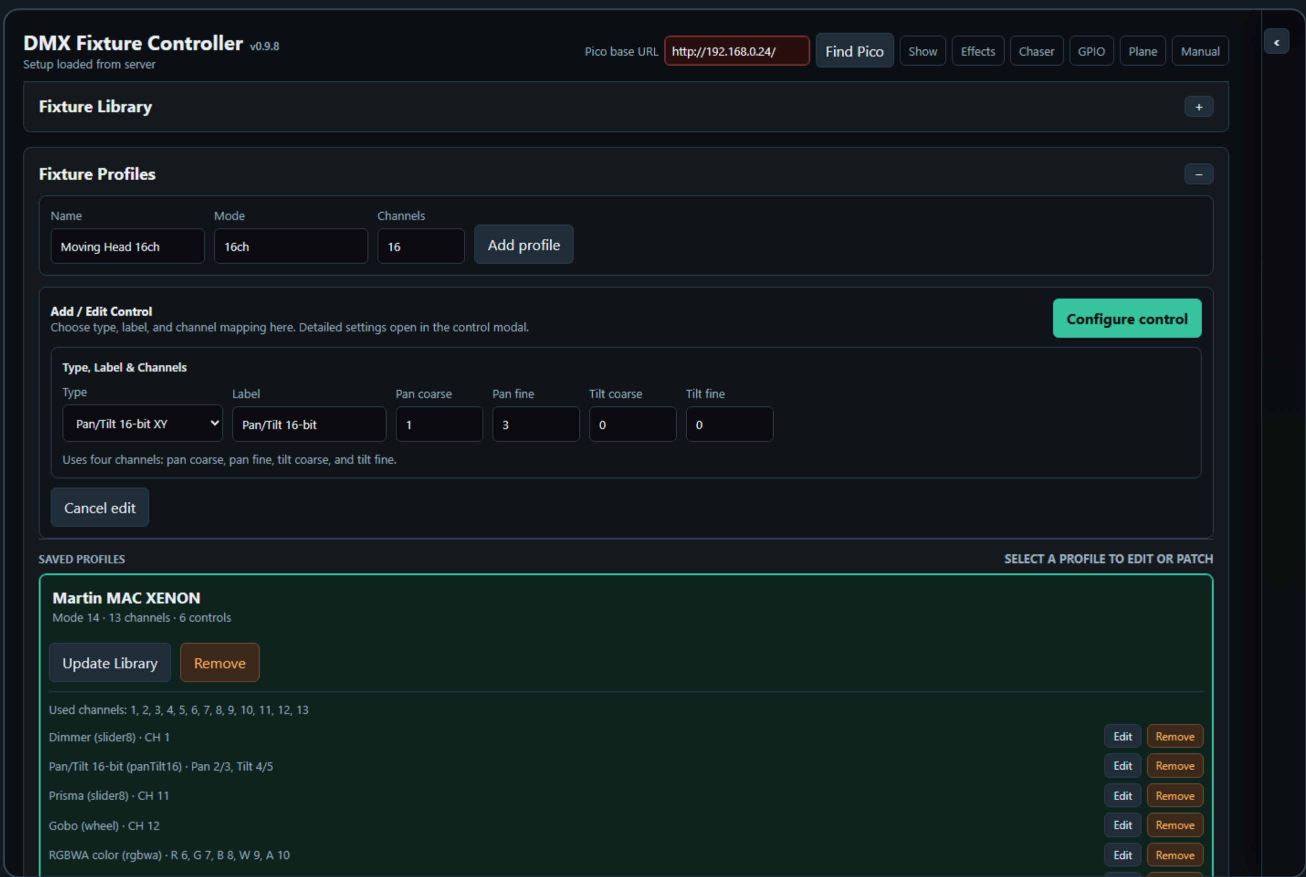
1. Type part of the manufacturer, fixture name, category, or mode in **Search manufacturer or fixture**.
2. Select the matching fixture from the results list.
3. Choose the desired DMX mode from the **Mode** dropdown.
4. Review the converted controls and channel count in the preview.
5. Click **Import profile**.

The imported fixture is added to **Fixture Profiles** and becomes the active profile. You can edit it like any manually created profile, including changing labels, defaults, blackout values, and wheel options.

The converter keeps the richer Open Fixture Library data where the controller can use it:

- Pan/tilt channels are grouped as 8-bit or 16-bit pan/tilt controls when matching channels are present.
- RGB, RGBW, and RGBWA color intensity channels are grouped into color controls.
- Wheel slots keep their OFL names and colors when the library provides them.
- Wheel DMX ranges are preserved. Normal slot buttons send the midpoint of the range.
- Adjustable wheel functions such as `WheelShake`, `WheelRotation`, and `WheelSlotRotation` show a bounded speed/range slider after you select that option.
- Unsupported or ambiguous channels are kept as simple 8-bit sliders so no channel is silently lost.

Create a Fixture Profile



A fixture profile describes the DMX personality of one fixture type. Use this section when the wanted fixture is not available in the **Fixture Library**, or when you want to fine-tune an imported profile.

1. Open **Fixture Profiles**.
2. Enter a profile name, mode name, and channel count. These fields update the selected profile automatically.
3. Click **Add profile**.
4. Select the profile.
5. In **Add / Edit Control**, choose the control type, label, and channel mapping.
6. Click **Configure control** to open the control details modal, then set default/blackout values and type-specific options.
7. Click **Add control** or **Save control** in the modal.

Each control stores its own DMX channel mapping. For example, a moving head can have:

- Dimmer on channel 1
- Pan/Tilt 16-bit on channels 2/3 and 4/5
- RGBWA color on channels 6-10
- Gobo wheel on channel 12

Simple 8-bit and 16-bit sliders only need channel mapping plus optional **Default** and **Blackout** values in the modal. Color controls add a color picker and any extra white, amber, or key channels. Wheel / indexed controls add the wheel editor and guided wheel editor so DMX ranges, colors, icons, and OFL-style function metadata can be entered without writing the raw text by hand.

Pan/Tilt controls also have fixture-profile mapping options in the control details modal:

- **Reverse Pan DMX** sends the logical Pan value as `max - value`.
- **Reverse Tilt DMX** sends the logical Tilt value as `max - value`.
- **Swap Pan/Tilt axes** sends logical Pan through the fixture's Tilt channels and logical Tilt through the fixture's Pan channels.

These options belong to the fixture profile, so every patched fixture using that profile follows the same mapping. The controller, chaser, and browser effect output keep showing logical Pan/Tilt values; only the physical DMX bytes are changed. This means scenes, palettes, chases, and effect centers remain readable even when a fixture is mounted backwards or rotated.

To change an existing control, click **Edit** in the Fixture Profiles list. The compact **Add / Edit Control** fields load the selected type, label, and channels, and the control details modal opens automatically. After editing, click **Save control** in the modal. If a Pico base URL is set, saving an existing control immediately resends the current live value for every patched fixture using that profile/control, so Pan/Tilt reverse or axis-swap changes are applied to the output right away. The **Fixture Profiles** collapse button hides the profile list and the compact Add / Edit Control box together.

Click **Update Library** when the selected profile should become part of the fixture library. The controller saves the current profile name, mode, channel count, and controls into the fixture library catalog on the XAMPP server. If the same fixture and mode already exist in the library, that mode is replaced; otherwise a new custom fixture entry is added. For existing Open Fixture Library modes, Update Library preserves richer wheel metadata such as `WheelShake`, `WheelRotation`, slot numbers, and speed ranges even when the edited controller profile only shows plain wheel values. Pan/Tilt reverse and axis-swap options are not written into the fixture library, because they describe how a fixture is mounted in one show rather than the fixture's DMX personality.

Default and Blackout Values

Each control can store a **Default** value and a **Blackout** value.

Default values are useful for a normal starting look, for example:

- Dimmer open
- Pan/Tilt centered
- RGB white
- Gobo open

Blackout values are useful for quickly shutting down output, for example:

- Dimmer 0
- RGB black
- White and amber 0
- Pan/Tilt at a safe position if needed

For RGB, RGBW, and RGBWA controls, use the color picker for RGB. White and amber channels are edited separately when the fixture type has them.

Patch Fixtures

Patch fixtures after the profile is ready.

1. Open **Patch Fixtures**.
2. Enter a fixture name.
3. Select the fixture profile.
4. Enter the DMX start address.
5. Set **Count** to the number of fixtures you want to patch.
6. Click **Patch fixture**.

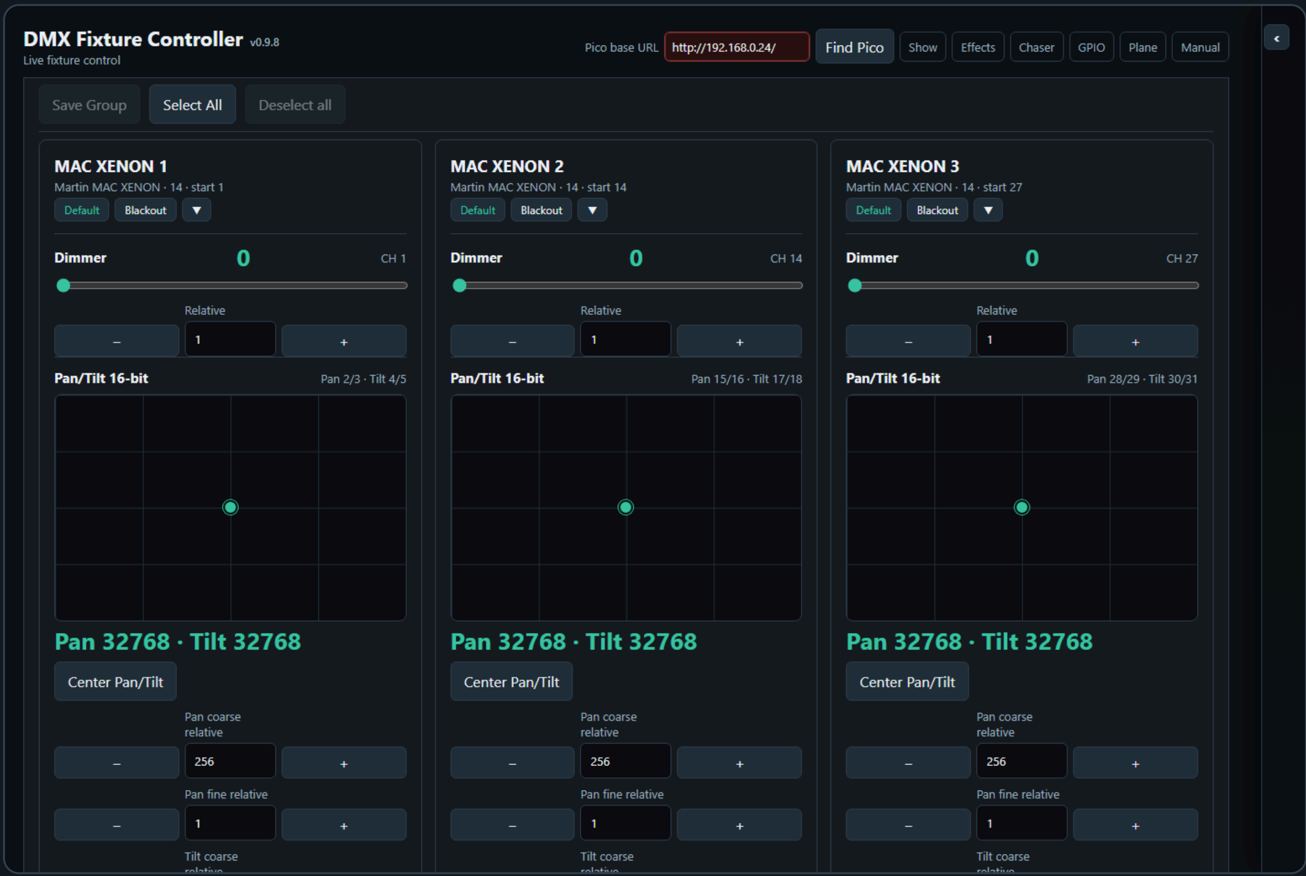
Use **Next free** to find the next available address after already patched fixtures.

When **Count** is greater than 1, the controller creates a numbered run of fixtures. For example, name `RGB Spot` with count `10` creates `RGB Spot 1` through `RGB Spot 10`. The first fixture uses the start address you entered. The following fixtures are spaced by the channel count of the selected fixture profile.

After a multi-fixture patch, the controller asks whether to create a Saved Group for the newly patched fixtures. The suggested group name is the patch name. Press **Cancel** to skip group creation, or enter a name to save the new fixtures as a group.

The patched fixture list is shown as rows of compact cards. Each row represents one consecutive run of the same fixture profile, so two separate MAC runs remain visually separate even when they use the same profile.

Live Fixture Control



The **Control Surface** shows one card per patched fixture.

Each fixture card contains the controls from its profile:

- Sliders for dimmer and simple channels
- XY pads for absolute pan/tilt positioning
- Relative pan/tilt nudge controls for small position corrections
- Color pickers and swatches for color controls
- Wheel buttons, a DMX value slider, and a direct numeric DMX value field for indexed wheel values
- Coarse/fine relative nudges for 16-bit channels

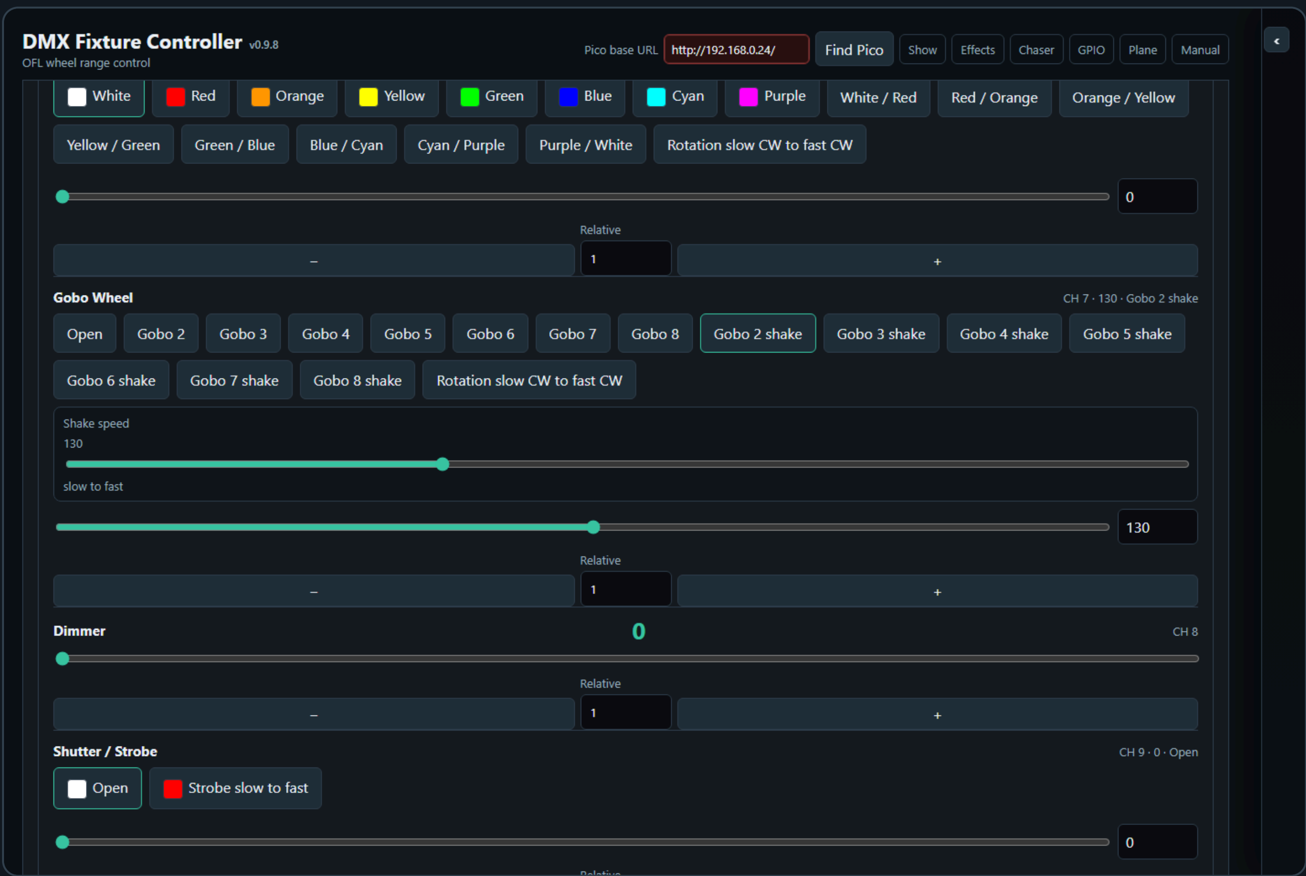
For pan/tilt controls, drag inside the XY pad when you want to move directly to an absolute position. Use **Pan coarse relative**, **Pan fine relative**, **Tilt coarse relative**, and **Tilt fine relative** when you want to adjust from the current position without jumping to a new absolute value. The number field in each relative row sets how many DMX increments the next `-` or `+` click will move. The fine relative buttons move the combined 16-bit value by that many steps, including byte borrow and carry: for example `256 - 1` becomes coarse `0`, fine `255`, and `255 + 1` becomes coarse `1`, fine `0`. The value is clamped at the valid DMX range, so it cannot go below `0` or above `65535`. If the fixture profile reverses or swaps Pan/Tilt, the XY pad and relative controls still show logical Pan/Tilt movement while the DMX output is mapped for the physical fixture.

Wheel / indexed controls require unique DMX option values. If two wheel options use the same DMX value, the control details modal refuses to save the control. This prevents ambiguous wheel buttons where the UI could not know which option should be highlighted after recall or chase editing. On the Controller page and in Controller Group Edit, wheel controls can be changed with the option buttons, the DMX value slider, or the direct numeric DMX value field.

Use **Guided wheel editor** when creating or correcting indexed controls such as color wheels, gobo wheels, and shutter/strobe ranges. The guided table is the normal place to edit wheel rows, including option name, DMX range start/end, function type, slot number, speed labels, background color, uploaded icon, or drawn icon. A wheel option button can show a color swatch, an icon, or an icon on top of the chosen color. This is the recommended way to define rich functions such as `WheelShake`, `WheelRotation`, and `ShutterStrobe` without memorizing the raw text syntax.

The **Wheel options** text box remains editable for advanced use and accepts one option per line. Basic lines use `Name=DMX` or `Name=start-end`, for example `Open=0-15`. Richer OFL-style metadata can be added after pipe characters. Use `kind=WheelSlot` with `slot=2` for a named slot, `kind=WheelShake` with `shake=slow-fast` for a bounded shake speed range, `kind=WheelRotation` with `speed=slow CW-fast CW` for a bounded rotation range, and `kind=ShutterStrobe` with `speed=slow-fast` for a bounded strobe range. For example: `Gobo 2 shake=125-140|kind=WheelShake|slot=2|shake=slow-fast`.

Fixture profiles imported from the **Fixture Library** keep richer Open Fixture Library wheel information when it is available. Wheel slots use their real names and colors, for example `White`, `Red`, or `Gobo 2`. If an OFL wheel option covers a DMX range, the button sends the middle of that range so the value is safely inside the fixture function. Adjustable wheel ranges such as `WheelShake`, `WheelRotation`, `WheelSlotRotation`, and `ShutterStrobe` also show a bounded range slider after you select them. For example, the Fun Generation PicoSpot 20 LED in 11-channel mode imports `Gobo 2 shake` as a wheel button and shows a **Shake speed** slider limited to its real DMX range.



Use **Default** or **Blackout** on a fixture card to recall the stored values for that fixture only.

Use the small button in the **Control Surface** header to collapse all visible fixture cards or expand them again. The button affects only the fixtures currently shown by the active group, scene, or palette filter.

Click a fixture card header or empty card area to include or exclude that fixture for group editing. The selected card uses the same accent outline style as other selectable tiles in the app. Clicking sliders, color controls, wheel buttons, Default, Blackout, or collapse controls does not change fixture selection.

After a hard page reload, no fixture cards are selected automatically. This keeps Group Edit disabled until you deliberately choose the fixtures you want to edit.

Use **Select All** in the group bar above the control surface when you want to select every patched fixture. Use **Deselect all** to clear the current manual selection. When you manually select fixture cards, the control surface stays visible so you can keep building or adjusting the selection. Saved Groups can also be selected from the Groups toolbox to filter the control surface.

Fixture Controller Toolboxes

The Fixture Controller uses four toolboxes in the shared right-side **Toolboxes** sidebar.

Plane Manual

nport Library

CH 5)
7)
CH 9)
CH 11)

+

+

—

CH 14

TOOLBOXES

Groups

Group Edit

Cols 2 Rows 4 Move

MAC XENON Backdrop
6 fixtures

RGB Spot Backdrop
12 fixtures

MAC XENON Front
4 fixtures

RGB Spot Front
8 fixtures

FAL
6 fixtures

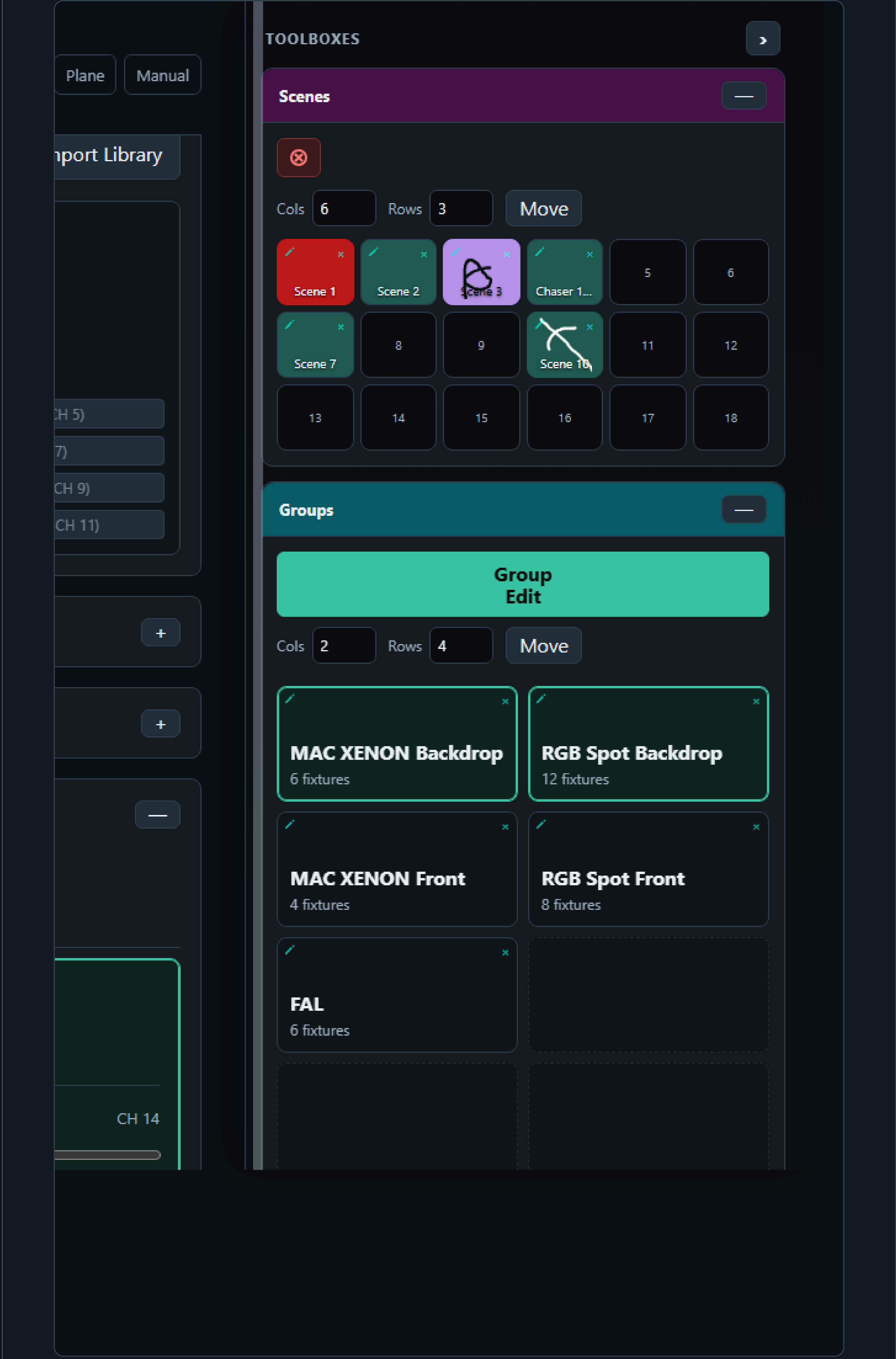
Scenes

Cols 6 Rows 3 Move

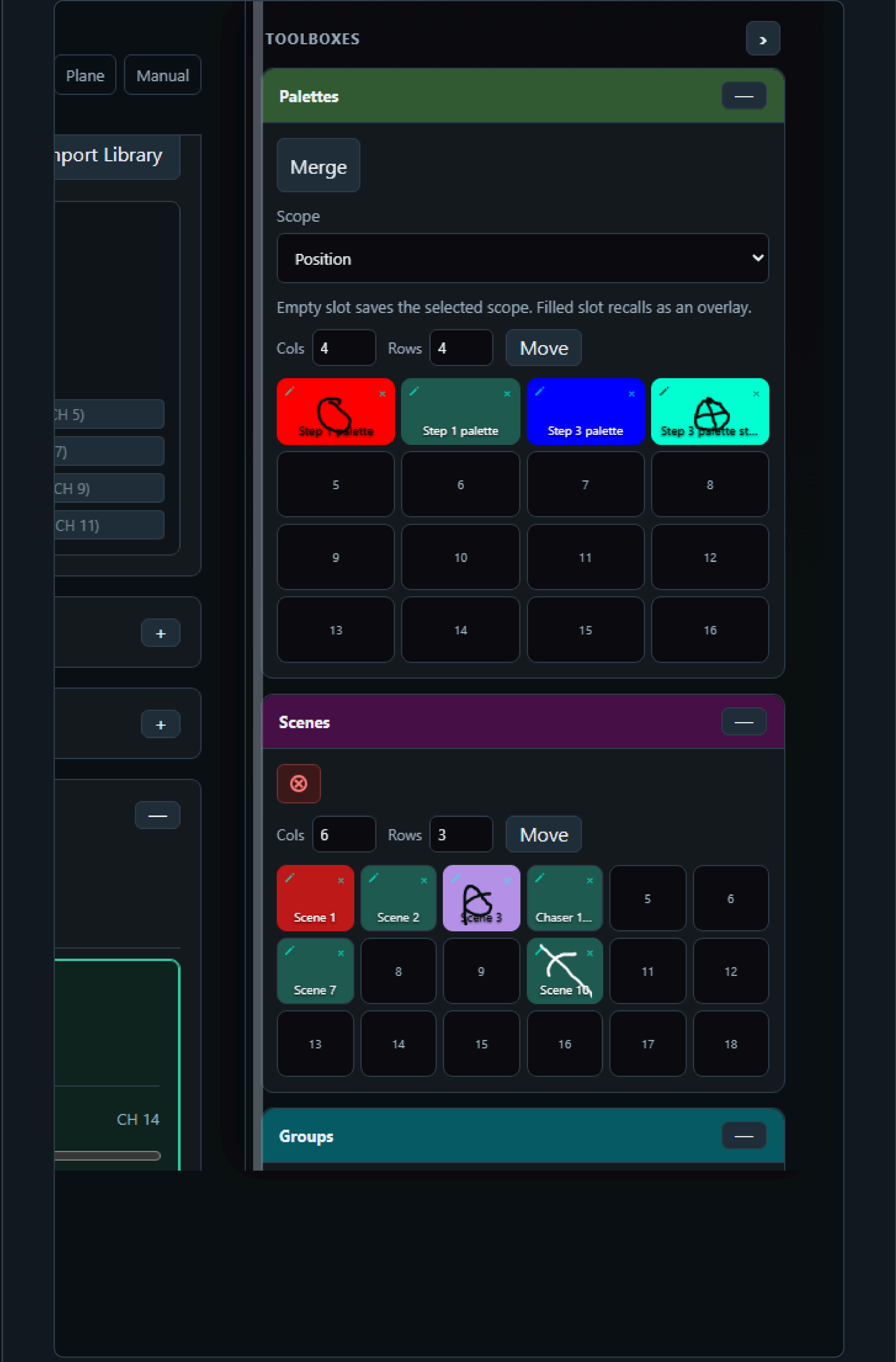
Scene 1	Scene 2	Scene 3	Chaser 1...	5	6
Scene 7	8	9	Scene 10	11	12
13	14	15	16	17	18

Show is the project-level card for setup files. **New Show** starts a fresh show after confirmation, clearing fixtures, live values, groups, scenes, palettes, chases, effects, saved room planes, GPIO mappings, Pico playback slots, Show Run layout, and saved toolbox layout while keeping the reusable fixture library. **Export Setup** downloads the complete show backup, **Import Setup** restores one, and **Patch CSV** exports the patched channel table. The card can be collapsed when you do not need these buttons.

Groups stores fixture groups and shares the selected group filter with other toolbox pages. Use it to select fixtures quickly, edit group tile names and visuals from the small pencil icon, delete groups from the small `x`, import/export group JSON, reorder group tiles with **Move**, and open **Group Edit** when the current scope supports it.

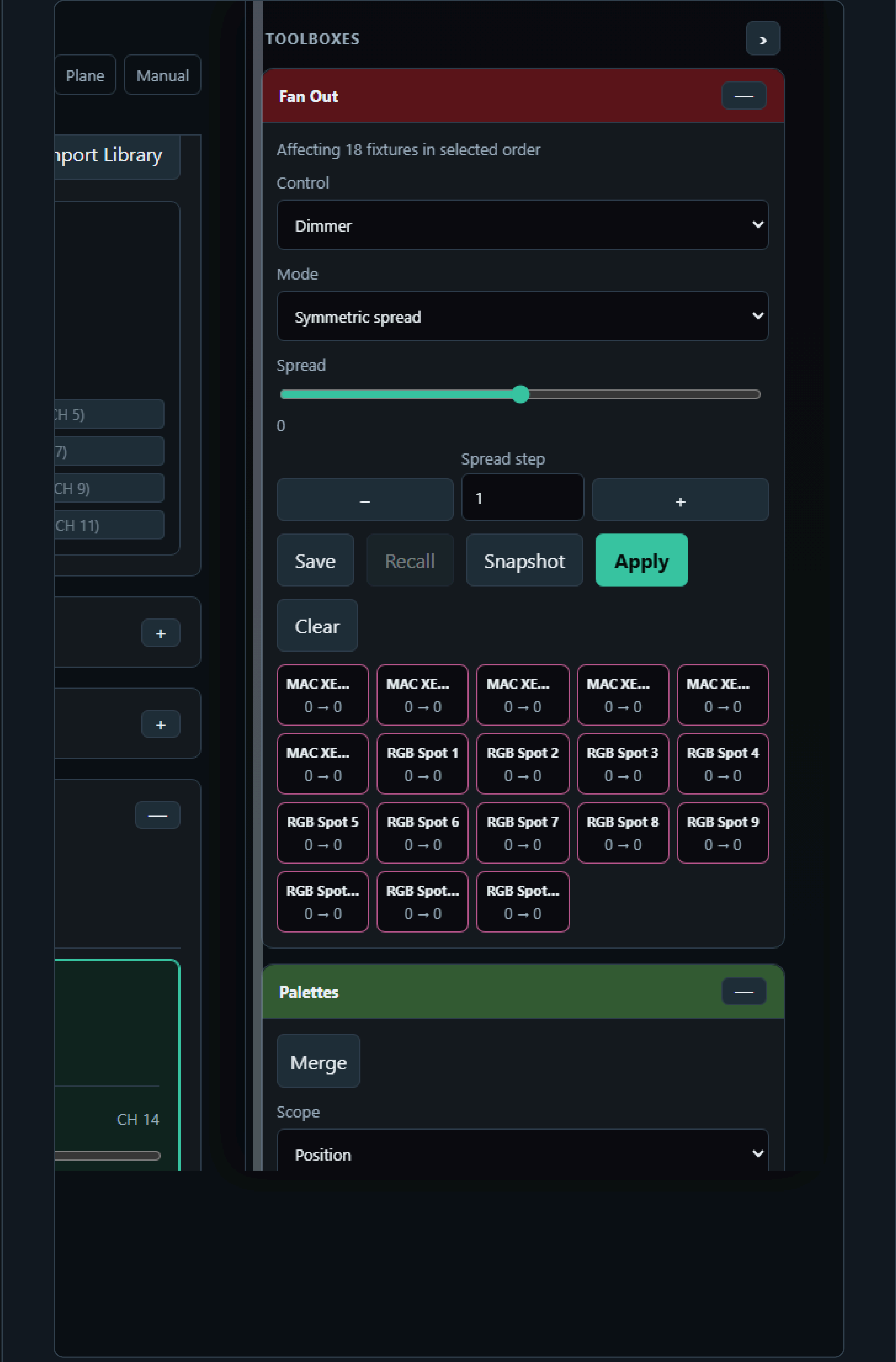

TOOLBOXES
Plane
Manual
port Library
Scenes
Cols 6
Rows 3
Move
Scene 1
Scene 2
Scene 3
Chaser 1...
5
6
Scene 7
8
9
Scene 10
11
12
13
14
15
16
17
18
CH 5)
7)
CH 9)
CH 11)
Groups
Group Edit
Cols 2
Rows 4
Move
MAC XENON Backdrop
6 fixtures
RGB Spot Backdrop
12 fixtures
MAC XENON Front
4 fixtures
RGB Spot Front
8 fixtures
FAL
6 fixtures
CH 14

Scenes stores complete looks for the current working scope. Empty slots save, filled slots recall, the small `x` deletes, **Move** reorders tiles, and the small top-left pencil icon opens **Edit Tile** for that slot's name, background, and optional drawing/upload.


TOOLBOXES
Plane
Manual
nport Library
CH 5)
7)
CH 9)
CH 11)
CH 14
Palettes
Merge
Scope
Position
Empty slot saves the selected scope. Filled slot recalls as an overlay.
Cols 4
Rows 4
Move
Step 1 palette
Step 1 palette
Step 3 palette
Step 3 palette st...
5
6
7
8
9
10
11
12
13
14
15
16
Scenes
Cols 6
Rows 3
Move
Scene 1
Scene 2
Scene 3
Chaser 1...
5
6
Scene 7
8
9
Scene 10
11
12
13
14
15
16
17
18
Groups

Palettes stores reusable value fragments such as positions, colors, gobos, dimmer looks, or Fan Out results. Palette recall applies only the stored controls and leaves unrelated values untouched. **Move** reorders palette tiles without recalling them.

In Groups, Scenes, and Palettes, **Move** sits beside the **Cols** and **Rows** layout controls because it organizes the tile grid. Click **Move** to enable move mode, then drag a filled tile to a new position. While dragging, the source tile gets the same accent outline used when moving toolboxes, and the target position shows an accent insertion line. On touch screens, tap the filled source tile and then tap the destination. Dropping onto an empty slot moves the tile there; dropping onto another filled tile swaps the two slots. Groups use the same tile-slot behavior as Scenes and Palettes; the fixture order stored inside a group is not changed by moving the group tile.


TOOLBOXES
Plane
Manual
nport Library
CH 5)
7)
CH 9)
CH 11)
CH 14
Fan Out
Affecting 18 fixtures in selected order
Control
Dimmer
Mode
Symmetric spread
Spread
0
Spread step
–
1
+
Save
Recall
Snapshot
Apply
Clear
MAC XE... 0 → 0
MAC XE... 0 → 0
MAC XE... 0 → 0
MAC XE... 0 → 0
MAC XE... 0 → 0
MAC XE... 0 → 0
RGB Spot 1 0 → 0
RGB Spot 2 0 → 0
RGB Spot 3 0 → 0
RGB Spot 4 0 → 0
RGB Spot 5 0 → 0
RGB Spot 6 0 → 0
RGB Spot 7 0 → 0
RGB Spot 8 0 → 0
RGB Spot 9 0 → 0
RGB Spot... 0 → 0
RGB Spot... 0 → 0
RGB Spot... 0 → 0
Palettes
Merge
Scope
Position

Fan Out shapes the currently selected fixtures or groups by spreading one compatible control through the selected order. It is documented in more detail below because its output can be saved as scenes or palettes.

Fan Out Toolbox

The Fixture Controller includes a **Fan Out** toolbox in the shared Toolboxes sidebar. It shapes the current controller values for the selected group or selected fixture cards, so the result can immediately be saved as a scene.

1. Select one or more Saved Groups, or select fixture cards manually.
2. In **Fan Out**, choose the control to shape, for example Dimmer, Pan, Tilt, or another compatible single-value control.
3. Click **Snapshot** to use the current controller values as the fan base.
4. Adjust **Spread**, or use **Start to end** mode with From/To offsets.
5. In **Symmetric spread** mode, `0` is the center of the slider. Positive and negative Spread values run the same shape in opposite directions through the selected fixture order.
6. The Control Surface updates continuously while you change the fan values.
7. The affected fixture controls are highlighted in the Control Surface.
8. Save the resulting look with the Scene Toolbox if you want to keep the actual DMX look.

In **Symmetric spread** mode, the **Spread** slider is centered at `0` and can move into positive or negative values. Positive Spread sends the fan in one direction through the selected fixture order; negative Spread sends it in the opposite direction. The **Spread step** field and the `-` / `+` buttons provide fine adjustment of the signed Spread value. Set the step size to the number of DMX increments you want to add or subtract, then click `-` or `+`. The buttons clamp at the valid signed range for the selected Fan Out control. **Start to end** mode hides the Spread nudge row because that mode is edited directly with the **From** and **To** number fields.

The current Fan Out working state is autosaved to the XAMPP server UI-state file. This includes the selected Fan Out control, mode, signed Spread, From/To offsets, and Spread step size, so the toolbox can restore the last working shape after a reload. Saved Fan Out presets are separate: use **Save** when you want to keep a named Fan Out recipe for later recall.

The Fan Out toolbox only shows controls that are available on every fixture in the selected set. For pan/tilt controls, Pan and Tilt appear as separate fan targets. Applying a fan writes the calculated values into the controller just like moving the controls by hand. When a Pico base URL is set, the changed DMX values are also sent to the Pico.

Fan Out calculates from a stored base value for each fixture, control, and axis. **Snapshot** stores those base values. In **Symmetric spread**, the spread is added around the base values:

```
final value = snapshotted base value + spread offset
```

For example, with five fixtures and a spread of `100`, the offsets are `-50`, `-25`, `0`, `+25`, and `+50`. If every fixture has a base value of `128`, the result is `78`, `103`, `128`, `153`, and `178`. With a spread of `-100`, the same offsets run in the opposite direction. If the fixtures have different base values, each fixture still keeps its own base as the starting point.

The offset is assigned by fixture position in the current Fan Out order:

```
fixture position = 0 ... fixture count - 1
```

```
offset position = fixture position / (fixture count - 1)
```

In **Symmetric spread**, the first fixture receives `-spread / 2`, the last fixture receives `+spread / 2`, and fixtures between them are spaced evenly. In **Start to end** mode, the first fixture receives the **From** offset, the last fixture receives the **To** offset, and fixtures between them are spaced evenly.

Fan Out order is important:

- If one or more saved groups are selected, the Controller uses the fixture order stored inside those groups. When several groups are selected, the groups are read in selected group order and duplicate fixtures are skipped after their first appearance.
- If no saved group is selected, the Controller uses the manual fixture selection order from the Control Surface.
- Negative **Spread** reverses the symmetric fan direction. It does not change the saved group, patch order, or fixture selection.

For a physical left-to-right fan, the saved group fixture order must match the physical fixture order. If a group was saved in a different order, the Fan Out math is still correct, but the visible stage result can look crossed or random.

Use **Save** in the Fan Out toolbox to store the fan setup itself: selected group or fixture IDs, selected control, mode, spread, and From/To offsets. Use **Recall** to restore that fan setup later. Recalling a Fan Out preset reapplies the fan to the controller values; it does not create a scene by itself. Use the Scene Toolbox when you want to store the resulting lighting look.

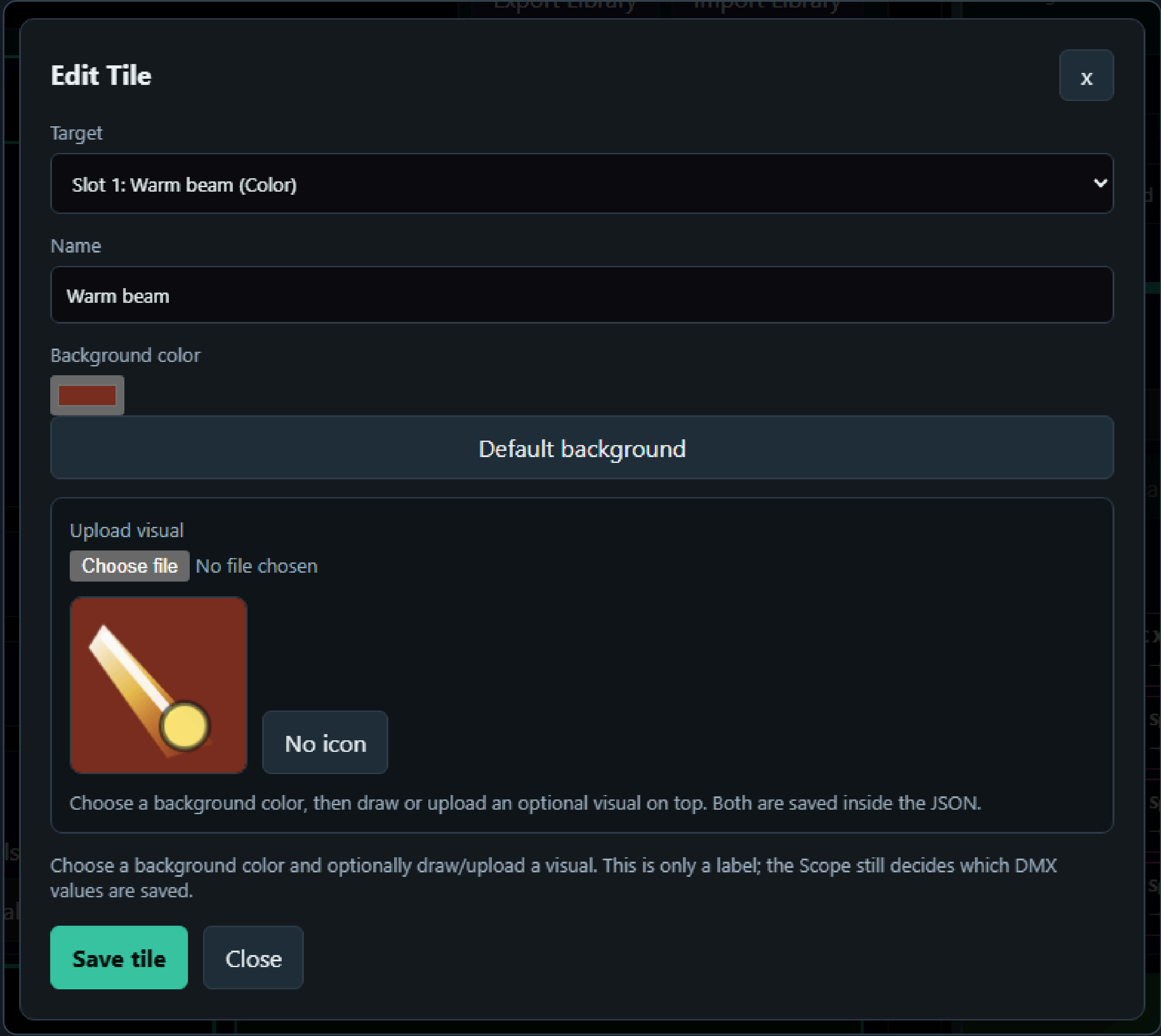
Palette Toolbox

The Fixture Controller also includes a **Palettes** toolbox. Palettes are reusable value fragments, while scenes are complete looks for their saved scope.

Use palettes for building blocks such as:

- Positions
- Colors
- Gobos
- Dimmer levels
- Fan Out results that you want to reuse as an overlay

The small top-left pencil icon on a filled palette slot opens **Edit Tile** for that slot. Use **Name** to rename the tile text, and use the visual controls for a background color plus an optional drawn/uploaded visual on top. The visual is independent from the palette scope. The drawing canvas uses the selected background color, and the brush automatically switches between a light and dark stroke for readable contrast. Use **Default background** to restore the standard slot color, or **No icon** to keep only the colored button. Scope still decides which DMX values are saved; the name and visual are readable labels shown in the palette slot grid and stored inside the palette JSON.



Palette save rules:

- If Fan Out is active, the palette saves only the Fan Out affected controls.
- If Fan Out is not active and fixtures are selected, the palette saves only the selected **Scope** for those fixtures.
- **Position** saves pan/tilt or position controls.
- **Color** saves RGB, RGBW, RGBWA, CMY, CMYK, matrix RGB, and common fixture-library color channels such as red, green, blue, cyan, magenta, yellow, white, amber, UV, lime, hue, saturation, CCT, CTO, or tint.
- **Dimmer** saves controls labeled as dimmer, dimming, or intensity.
- **Shutter / Strobe** saves shutter, strobe, and flash controls.
- **Gobo** saves gobo wheels and gobo rotation/index controls.
- **Prism** saves prism and prism rotation controls.
- **Optics** saves focus, zoom, iris, frost, beam, diffuser, or lens controls.
- **Programs / Effects** saves program, macro, effect, auto show, chase, pattern, scene, or preset controls.
- **All controls** saves every control on the selected fixtures and should be used deliberately.
- If nothing is selected, the palette is not saved; select fixtures or apply Fan Out first.

When saving a **Gobo** palette, the Controller can fill the palette tile visual from the selected fixture's gobo wheel metadata. If all selected fixtures resolve to the same gobo image or color, that shared visual is used automatically. If selected fixtures use different gobos, the current source fixture supplies the tile visual. If no gobo visual metadata exists, the palette uses the normal default visual.

Palette recall rules:

- Recalling a palette applies only the stored values.
- Unrelated controller values are left unchanged.
- The active group filter is cleared.
- The Control Surface is filtered to the fixtures involved in the palette.
- When a Pico base URL is set, only the recalled palette controls are sent to the Pico.

Filled palette slots recall palettes. Merging is done with the separate **Merge** button so recall and edit actions stay distinct.

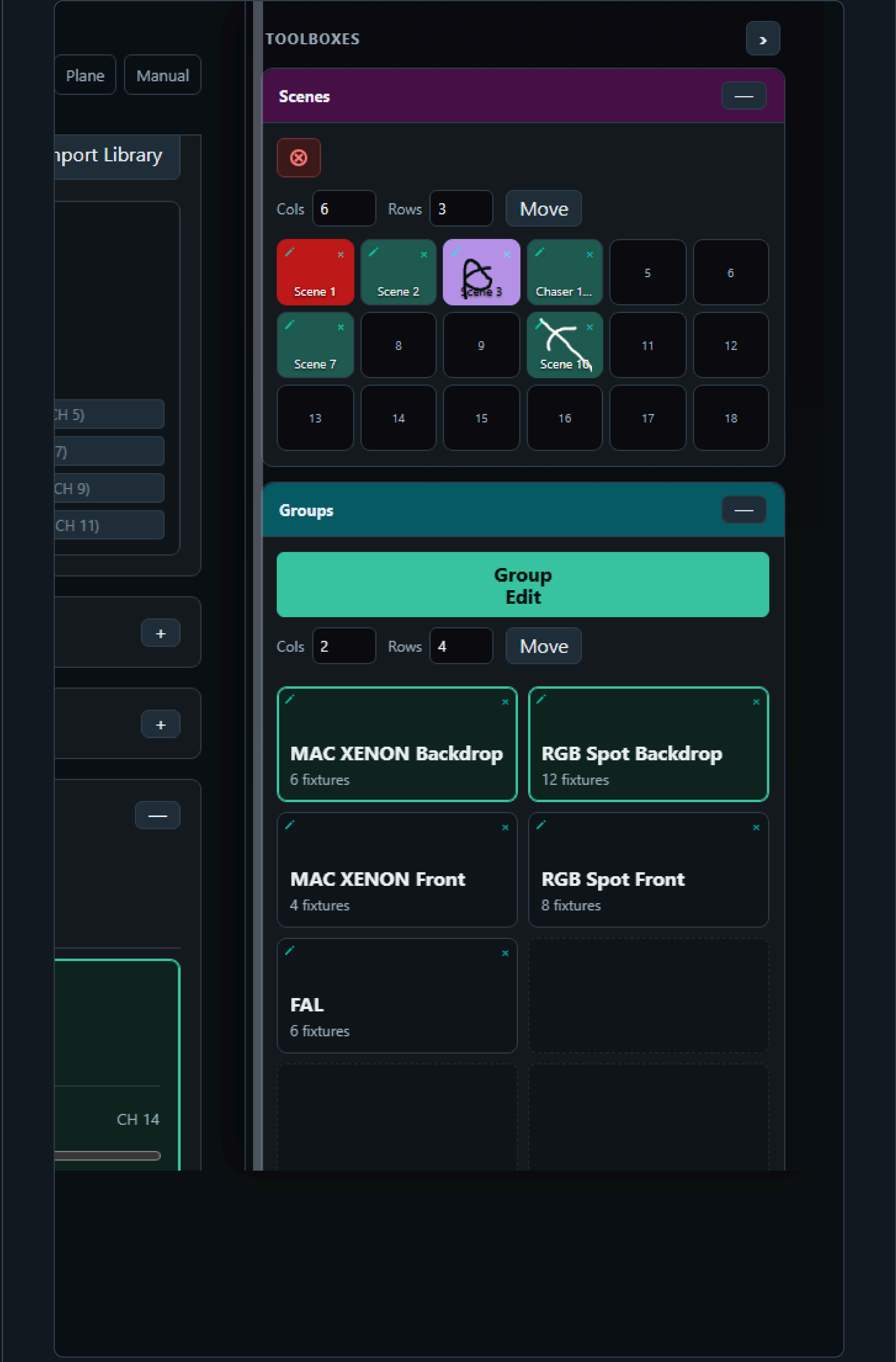
Use **Merge** when you have a current fixture selection or active Fan Out result:

- The controller opens a palette matrix picker, so you choose the saved palette tile visually instead of typing a slot number.
- Empty slots are shown for orientation but cannot be selected as merge targets.
- If the current scope matches the saved palette scope, the values are merged and the palette keeps its scope.
- This allows one **Color** palette to contain RGB controls, color wheels, and RGBWA controls from different fixture types while still remaining a Color palette.
- If the current scope differs from the saved palette scope, the controller asks whether to change the saved palette to **All controls** and merge.
- If you cancel a merge prompt, the palette is not changed.

Use **Move** when you want to reorder palette tiles. The same interaction is also available in the Groups and Scenes toolboxes:

- Click **Move** to enable palette move mode.
- Drag a filled palette tile to an empty slot to move it there; the tile shows the same drag outline and insertion marker used for moving toolboxes.
- Drag a filled palette tile onto another filled palette tile to swap their positions.
- On touch screens, tap a filled tile to select it, then tap the destination slot.
- While Move is active, clicking a palette tile does not recall or save it.
- Click **Move** again when you are done organizing the palette grid.

2. Scenes And Palettes


TOOLBOXES
Plane
Manual
nport Library
CH 5)
7)
CH 9)
CH 11)
CH 14
Scenes
Cols 6
Rows 3
Move
Scene 1
Scene 2
Scene 3
Chaser 1...
5
6
Scene 7
8
9
Scene 10
11
12
13
14
15
16
17
18
Groups
Group Edit
Cols 2
Rows 4
Move
MAC XENON Backdrop
6 fixtures
RGB Spot Backdrop
12 fixtures
MAC XENON Front
4 fixtures
RGB Spot Front
8 fixtures
FAL
6 fixtures

The Scene Toolbox is available on the Fixture Controller page.

Use scenes to store fixture/control looks. A scene stores controller values by fixture/control key, not a raw 512-channel DMX dump.

The small top-left pencil icon on a filled scene slot opens the same **Edit Tile** modal used by palettes. A scene can be renamed and can have a background color plus an optional drawn/uploaded visual in its slot. The canvas background follows the selected background color, and the brush color is calculated for contrast against it. **Default background** restores the standard slot color, and **No icon** removes the drawn/uploaded image. This name and visual are only labels for finding the scene quickly; scene save and recall still use the stored fixture values.

1. Set your desired fixture values.
2. Click an empty scene slot.
3. Enter a scene name.
4. Click a filled slot once to recall it.
5. Use the small `x` on a filled slot to delete it.

Scene Save Rules

Scene saving uses the current working scope:

- If Fan Out is active, the scene saves only the fixtures affected by Fan Out.
- For each Fan Out fixture, the scene saves all controls of that fixture, not only the fanned control. This keeps related values such as Tilt stable when saving a Pan fan.
- If Fan Out is not active and fixtures are selected, the scene saves only the selected fixtures.
- Selected Saved Groups count as selected fixtures, so a group-filtered scene saves only the fixtures in the selected groups.
- If a scene was recalled and the control surface is filtered to that scene's involved fixtures, saving again uses those involved fixtures.
- If nothing is selected, the controller asks before saving all patched fixtures.
- If you cancel that prompt, no scene is saved.

This means a scene is normally scoped to the fixtures you are actually working with. It does not silently save unrelated fixtures unless you explicitly confirm the all-fixtures fallback.

Scene Recall Rules

Scene recall loads the stored values back into the Fixture Controller, redraws the fixture cards, and updates the live-value snapshot used by the Chaser page.

When a Pico base URL is set, scene recall also sends the recalled DMX values to the Pico in one batch. To work browser-only, clear the Pico base URL before recalling or editing looks.

Recalling a scene clears the active group selection and shared group filter. It then selects the fixtures involved in the recalled scene and filters the Control Surface to only those fixtures. This makes the recalled scene visible without leaving an old group filter active, and without showing unrelated fixtures.

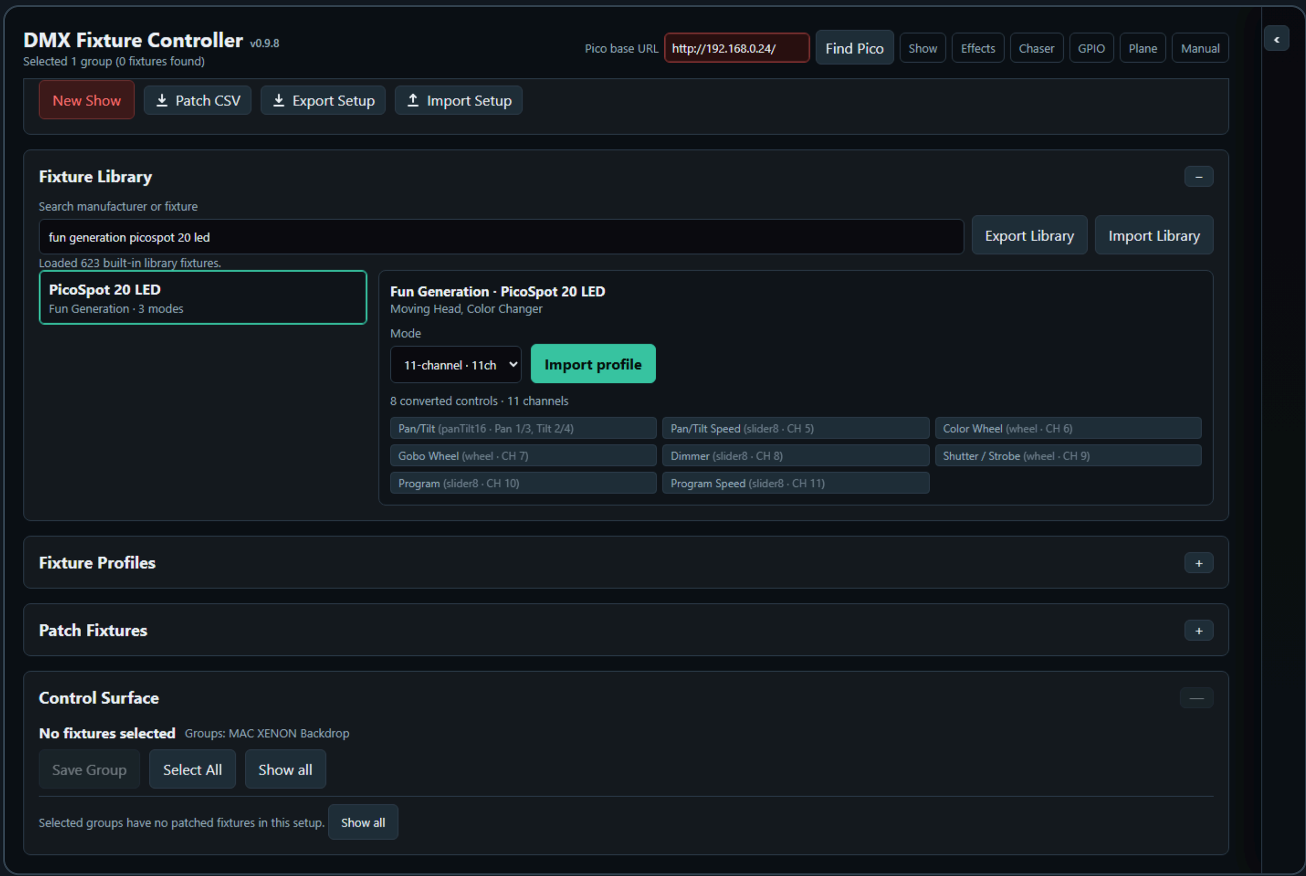
If a saved scene was created before fixtures were repatched, the controller tries to remap old fixture IDs to the current patch by matching the same fixture profile in DMX start-address order. This lets older scenes continue to recall after a fixture run was recreated, as long as the fixture profiles and control IDs still match.

The red **Clear all DMX channels** button clears controller values. When a Pico base URL is set, it also calls `/dmx/clear` on the Pico. This clears both live DMX output and the motion base buffer.

The Scene Toolbox sits in the shared **Toolboxes** sidebar. Its slot grid follows the configured rows and columns, and the tile size expands to the available sidebar width while keeping the old minimum slot size.

---

3. Groups

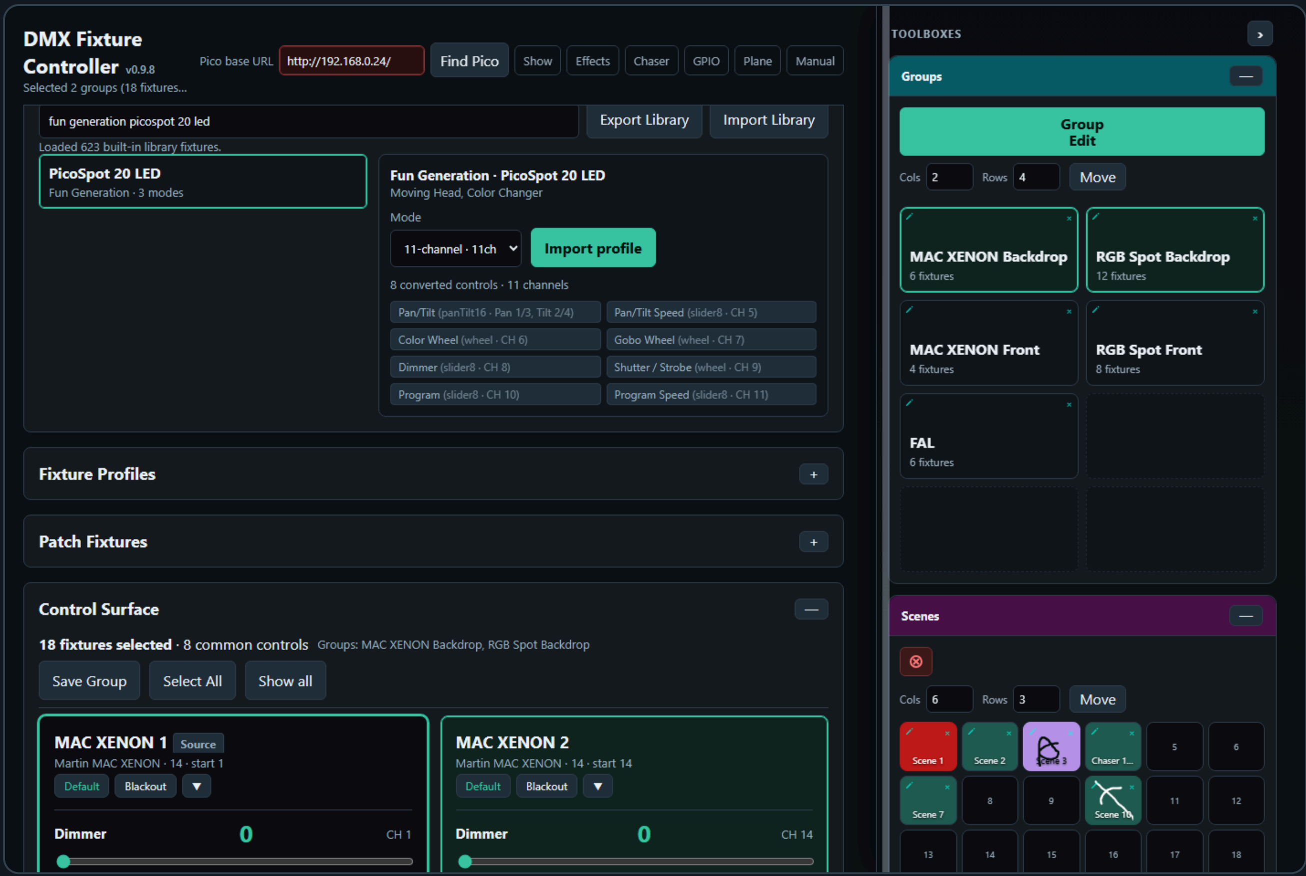


Groups let you control multiple fixtures at once.

Create a Group

1. Select two or more fixture cards.
2. Click **Save Group**.
3. Enter a group name.
4. The group appears in **Saved Groups**.

Saved Group Selection



The Saved Groups card shows groups in a compact matrix. Each group has four actions:

- **Select** adds the group to the active group filter.
- **Deselect** removes only that group from the active filter.
- The small pencil on a group tile opens **Edit Tile** for that saved group's name, background color, and optional drawn/uploaded visual.
- The small `x` on a group tile removes that saved group after confirmation.

More than one saved group can be selected at the same time. The control surface shows the union of all fixtures from the selected groups. This makes it possible to work with several fixture groups together without editing the group definitions.

The group bar above the control surface shows how many fixtures are selected and which saved groups are active. Use **Select All** to select every patched fixture explicitly. Use **Deselect all** to clear a manual fixture selection. When a saved-group filter is active, the same button is shown as **Show all** and clears the saved-group filter to return to the full fixture list.

Group selection is shared across toolbox pages that use the Groups toolbox. It is a working filter: selecting groups limits the visible fixtures for controller editing, Fan Out targeting, and Chaser participating-control setup. Recalling a scene, editing a saved chase step, or loading a saved chase clears the group filter because the recalled data itself becomes the source of truth.

Edit a Group

1. Load a saved group or select multiple fixtures.
2. Click **Group Edit**.
3. Adjust common controls in the modal.

On the Fixture Controller, **Group Edit** lives in the **Groups** toolbox. It requires an explicit fixture selection. If nothing is selected, Group Edit is disabled. Select individual fixture cards, load one or more saved groups, or press **Select All** before opening Group Edit from the Groups toolbox.

The Group Edit modal shows controls that exist in the current edit scope. It can be used for one fixture or many fixtures where the page supports single-fixture editing. Mixed fixture types are allowed; each control is applied only to fixtures that actually have a matching control, so incompatible fixtures are ignored for that control.

The selected **Source** fixture is the template for the modal values. When you open Group Edit, the modal reads the Source fixture's current matching control values and shows those values in the sliders, wheel buttons, color controls, XY pads, and relative nudge controls. Opening the modal does not overwrite the other selected fixtures and does not send DMX by itself.

The source selection is automatic. Loading a saved group makes the first fixture stored in that group the Source. With manual fixture selection, the clicked/selected fixture becomes the Source; if that fixture is removed from the selection, the page picks the next selected fixture that can provide the control.

The values are applied when you actually edit a control in the modal. Moving a slider, typing a direct numeric DMX value, dragging an XY pad, choosing a wheel slot, changing a color, pressing Center, using relative nudges, or using Default/Blackout writes the change to every selected fixture that has the matching control. Absolute edits write the shown Source value to the matching fixtures. Relative nudges keep each selected fixture relative to its own current value, so a group pan/tilt nudge moves the selected fixtures together without forcing them all to the same absolute position. When a Pico base URL is set, each Group Edit change is sent to the Pico immediately. To use Group Edit browser-only, clear the Pico base URL first.

Group Edit remembers the relative step-size fields you set, for example **Pan fine relative** or **Tilt fine relative**. Those settings are autosaved to the XAMPP server UI-state file and restored the next time Group Edit opens, so your preferred nudge sizes survive reloads.

If the controller is currently scoped by a recalled scene, recalled palette, or Fan Out result, **Group Edit** uses that scope. In that case it shows only controls that are part of the active scope and exist in the selected fixture scope. Editing a scoped control writes only to matching fixtures.

Use **Default** or **Blackout** to recall the stored default or blackout values for every fixture in the selected group.

Group Edit Contract

Controller, Chaser, and Effects all use the same basic Group Edit idea: the modal is available when the current page scope contains at least one editable control. Controller and Effects can use the same modal for a single fixture or a group of fixtures; Chaser uses the current participating-control scope for step editing. The fixtures do not need to use the same fixture profile.

Keep these rules as the contract:

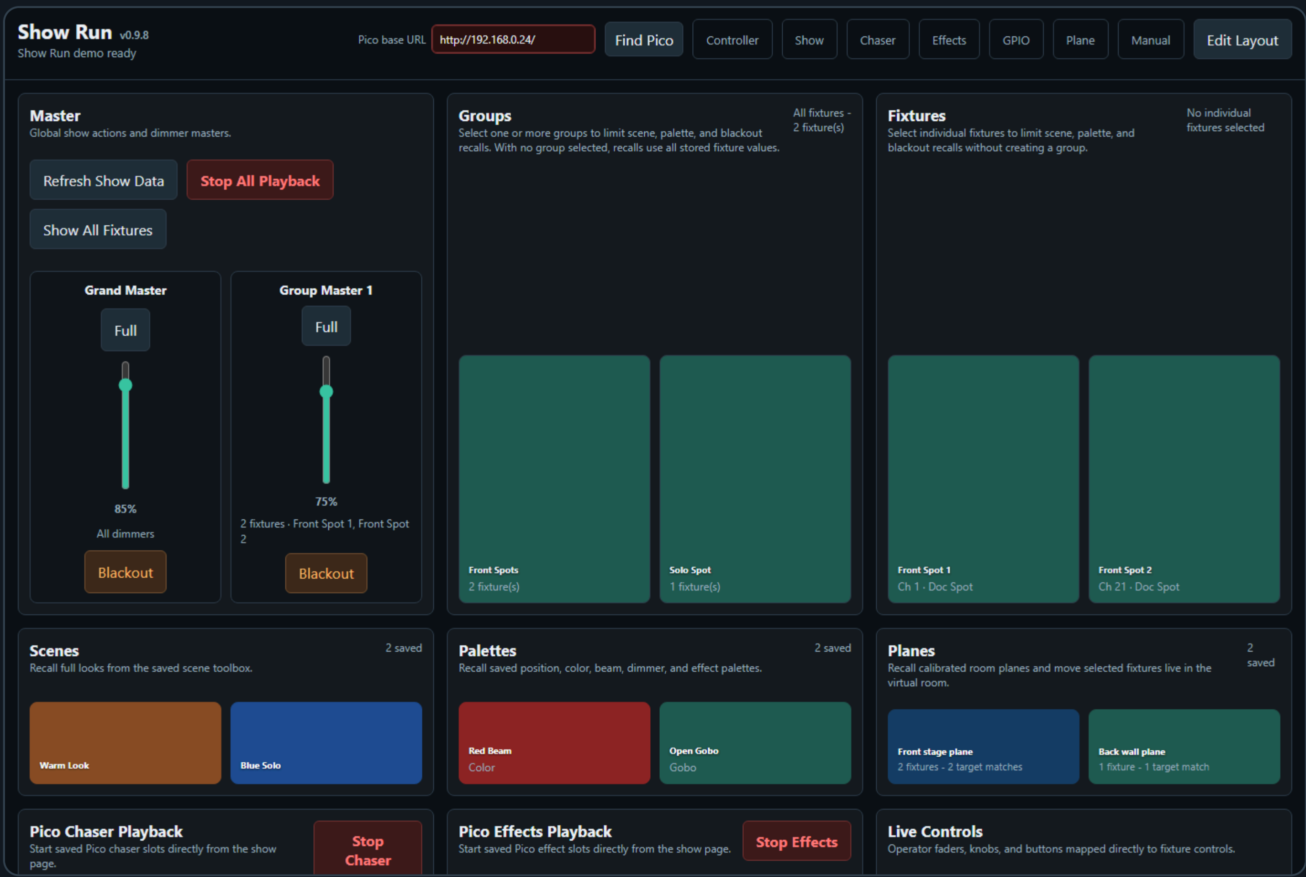
- Mixed fixture types are allowed.
- A control is editable only when the page can match it by control identity, such as type and label.
- Wheel / indexed controls are stricter: same-named wheels are kept separate when their option lists differ, so a MAC Gobo wheel and a Scanner Gobo wheel are not accidentally edited as one control.
- The modal shows the matching fixture/profile scope for each control.
- Opening the modal reads the Source fixture values into the modal but does not apply or send anything yet.
- Editing a control writes only to fixtures that actually have the matching control.
- The first edit in the modal is the moment the value is written to the group; on pages with live output, that edit is also sent to the Pico when a Pico base URL is set.
- Incompatible fixtures remain selected but are ignored for that specific control.

- Group selection is a filter. If groups are selected, Group Edit uses only compatible fixtures inside those groups.
- Direct scope changes, such as Select All, Participating Controls All, or Effects All, clear the saved-group filter and make the page scope the source of truth.

---

4. Show Run

The **Show Run** page is the operator page. Use it when the show has already been prepared on the Fixture Controller, Chaser, and Effects pages and you want fewer editing controls on screen.



Open it from the **Show** navigation link or directly:

```
http://localhost/dmx/dmx_show.html
```

Show Run loads the current XAMPP show data:

- Fixture profiles and patched fixtures from `fixture_setup.php`
- Saved groups from `group_setup.php`
- Saved scenes from `scene_setup.php`
- Saved palettes from `palette_setup.php`
- Saved room planes from `room_plane_setup.php`
- Mirrored Pico chaser slots from `chaser_setup.php?slots`
- Mirrored Pico effect slots from `motion_setup.php?slots`
- Current live values from `fixture_setup.php?livevalues`

Show Run automatically refreshes this show data when the page becomes active again, for example after switching back from the Controller, Chaser, or Effects page. This keeps the operator page current without needing to reload the browser tab. Automatic refresh is skipped while **Edit Layout** is active so card moves, tile moves, and Live Controls edits are not overwritten. Use **Refresh Show Data** only when you want to force the same reload manually.

Master Card

The **Master** card contains the global operator actions and dimmer masters. **Refresh Show Data** reloads the current XAMPP show files manually. **Stop All Playback** stops Pico chaser and effect playback. **Show All Fixtures** clears the current group and fixture target filter.

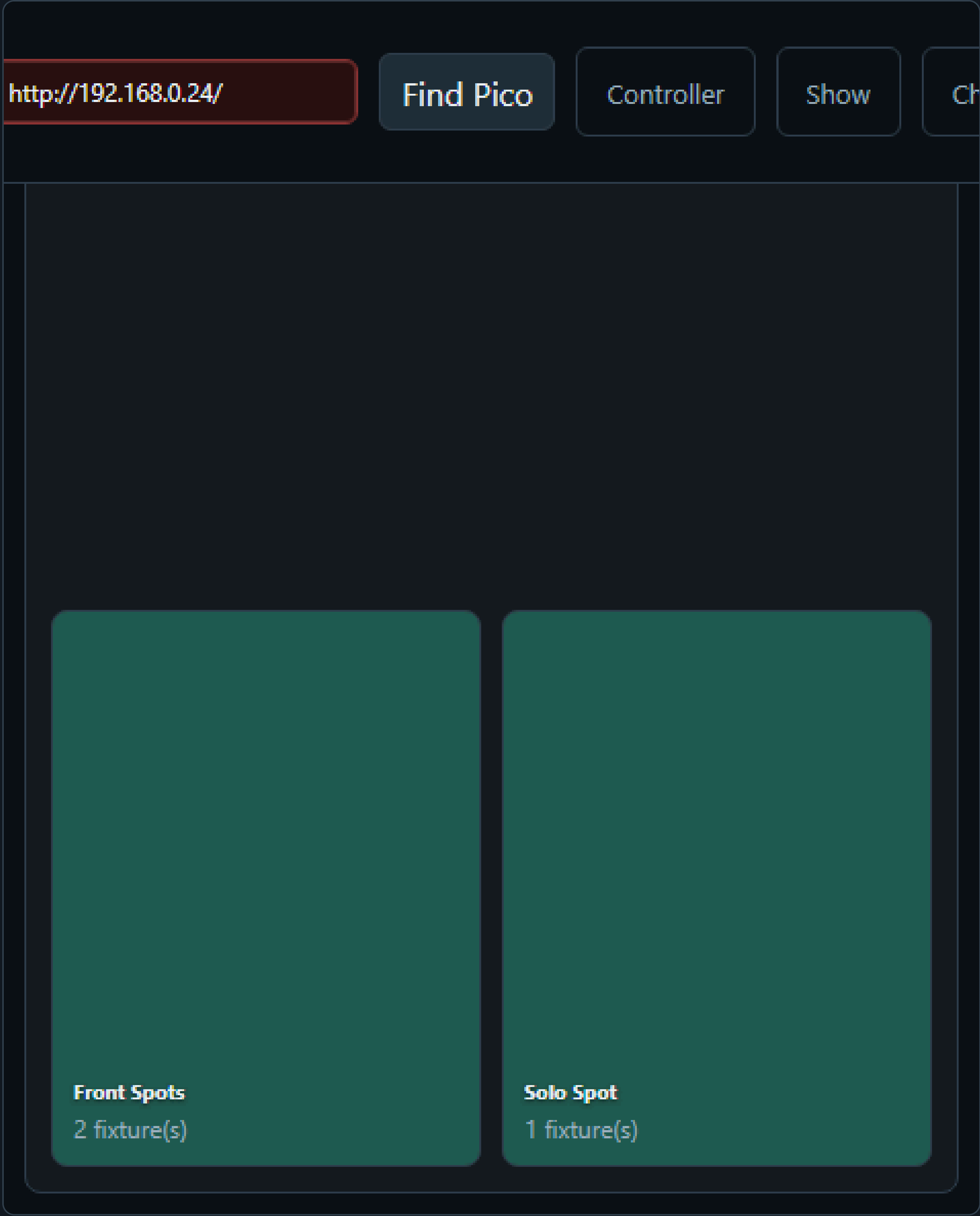
The **Grand Master** is a vertical fader. It does not overwrite stored live values; it multiplies dimmer output by a 0..1 factor, including 16-bit dimmer controls. **Blackout** below the Grand Master sets the Grand Master fader to **0%** and sends the scaled dimmer output immediately. **Group Master** faders work the same way, but only for fixtures assigned to that group master. The **Blackout** button below each Group Master sets only that Group Master to **0%**. When a master is below **100%**, Show Run also locks the affected dimmer channels on the Pico so running Pico chaser/effect playback cannot overwrite the master-limited dimmer output.

To assign a Group Master, first select one or more groups or fixtures, click **Edit Layout**, then click **Assign** on the desired group master fader. **Add Group Master**, **Assign**, and **Clear** are only enabled while **Edit Layout** is active. The Grand Master factor and Group Master assignments are saved to server UI state and restored on page load.

Groups Card

The **Groups** card contains the saved groups from the Controller page. Select one or more groups when you want scene, palette, or Group Master assignment to affect only those fixtures. With no group selected, Show Run uses all fixtures.

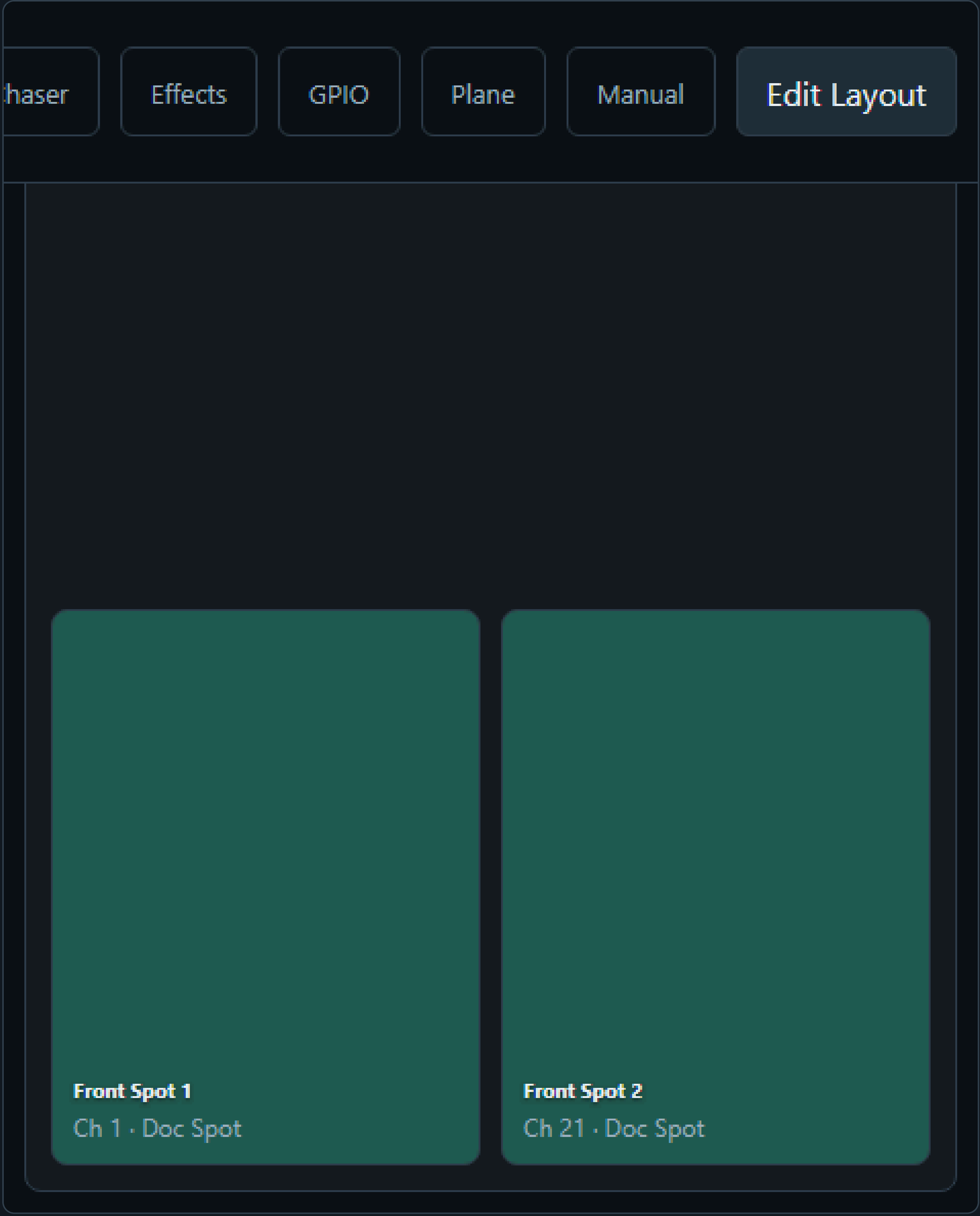
Group selection is an operator filter only. It does not edit the saved group definitions and it does not save setup data.



Fixtures Card

The **Fixtures** card lets you target individual fixtures without creating a saved group. Click a fixture tile to select or deselect it. Fixture selection combines with group selection, so the current show target is the union of selected groups and selected individual fixtures.

Use this when a scene, palette, or Group Master should apply to one fixture or a temporary hand-picked set.

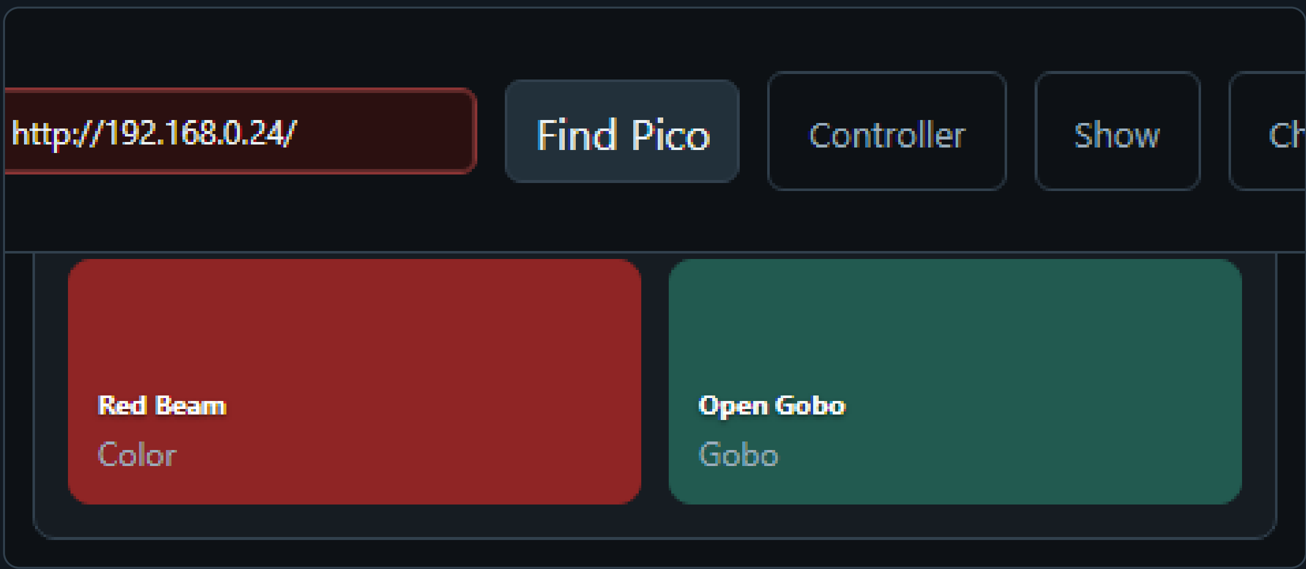


Scenes Card

The **Scenes** card recalls saved scene tiles from the Controller page. Click a scene tile to recall that scene to the current target. Show Run writes the recalled values to the live-value snapshot and sends the matching DMX channels to the Pico in one `/dmx/b` batch when a Pico base URL is set.

Palettes Card

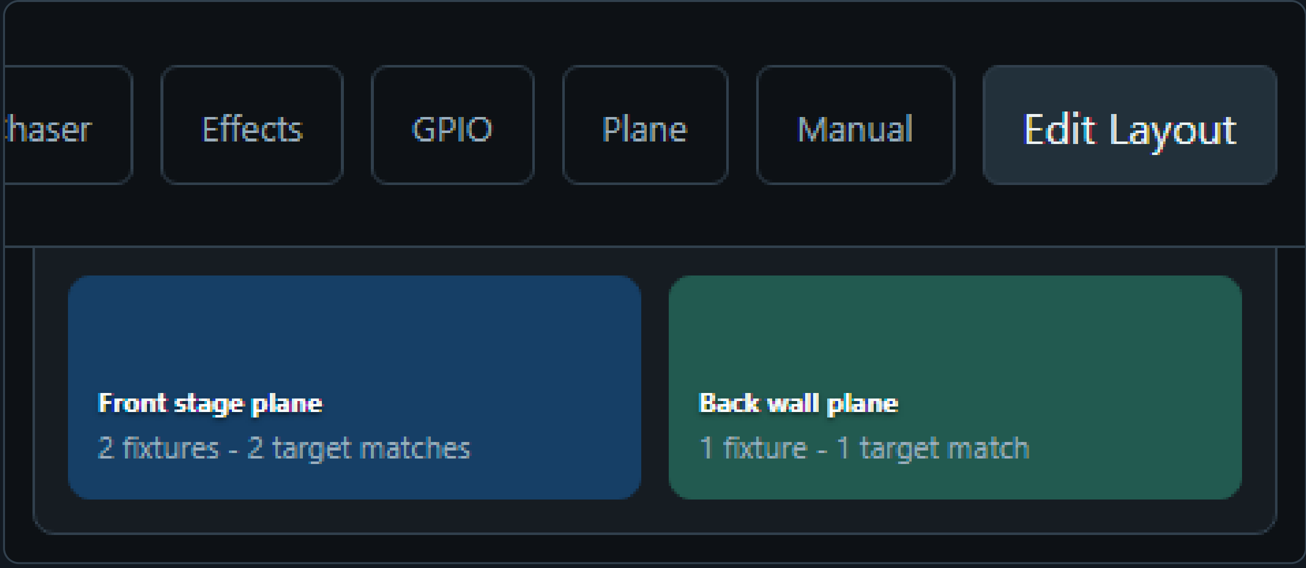
The **Palettes** card recalls saved position, color, beam, dimmer, and effect palettes. If a group or fixture target is active, only stored values for the targeted fixtures are recalled. This lets one saved palette be reused for different parts of the rig.

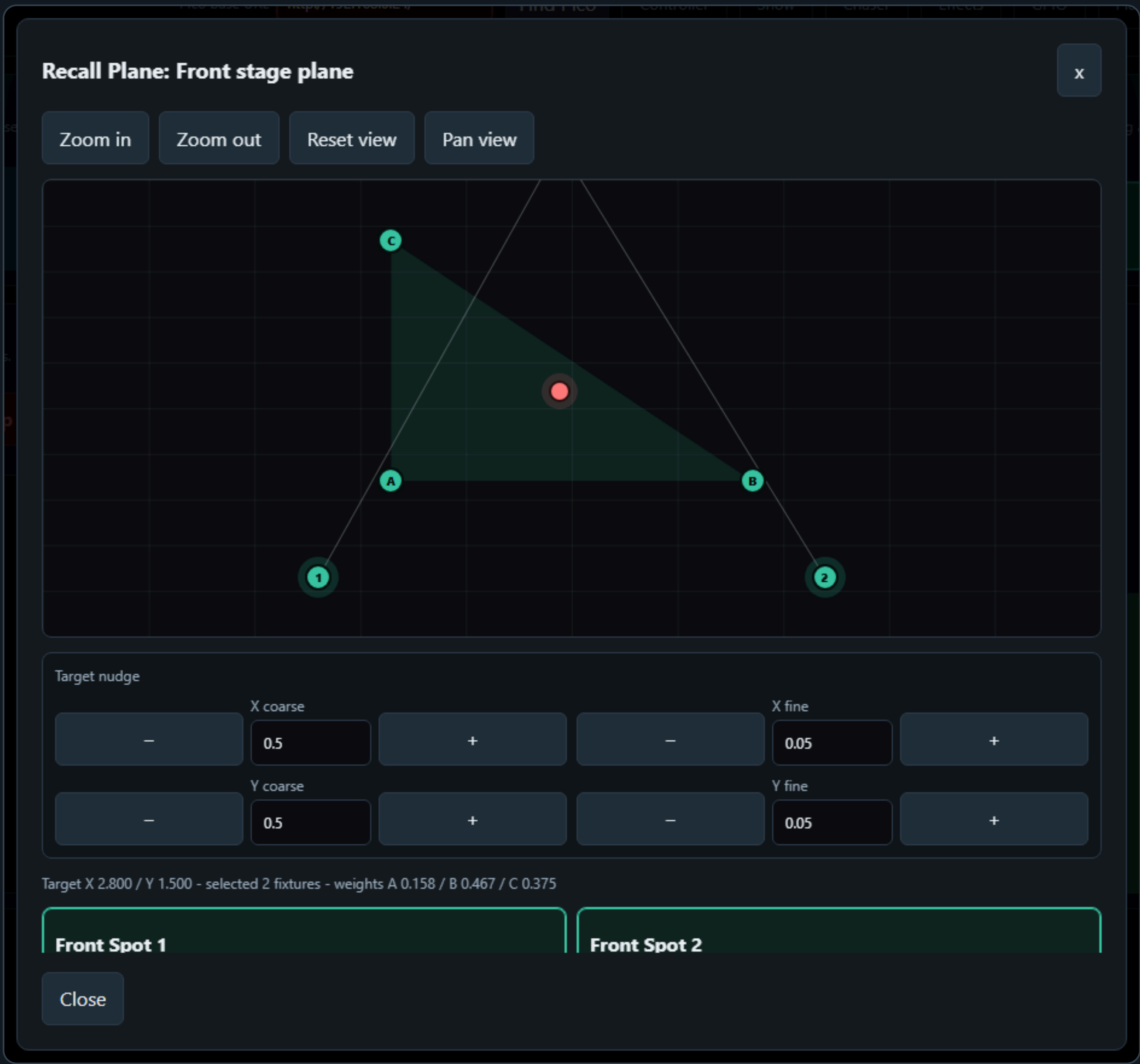


Planes Card

The **Planes** card recalls saved room planes from the Plane page. Click a plane tile to open a virtual room modal without leaving Show Run.

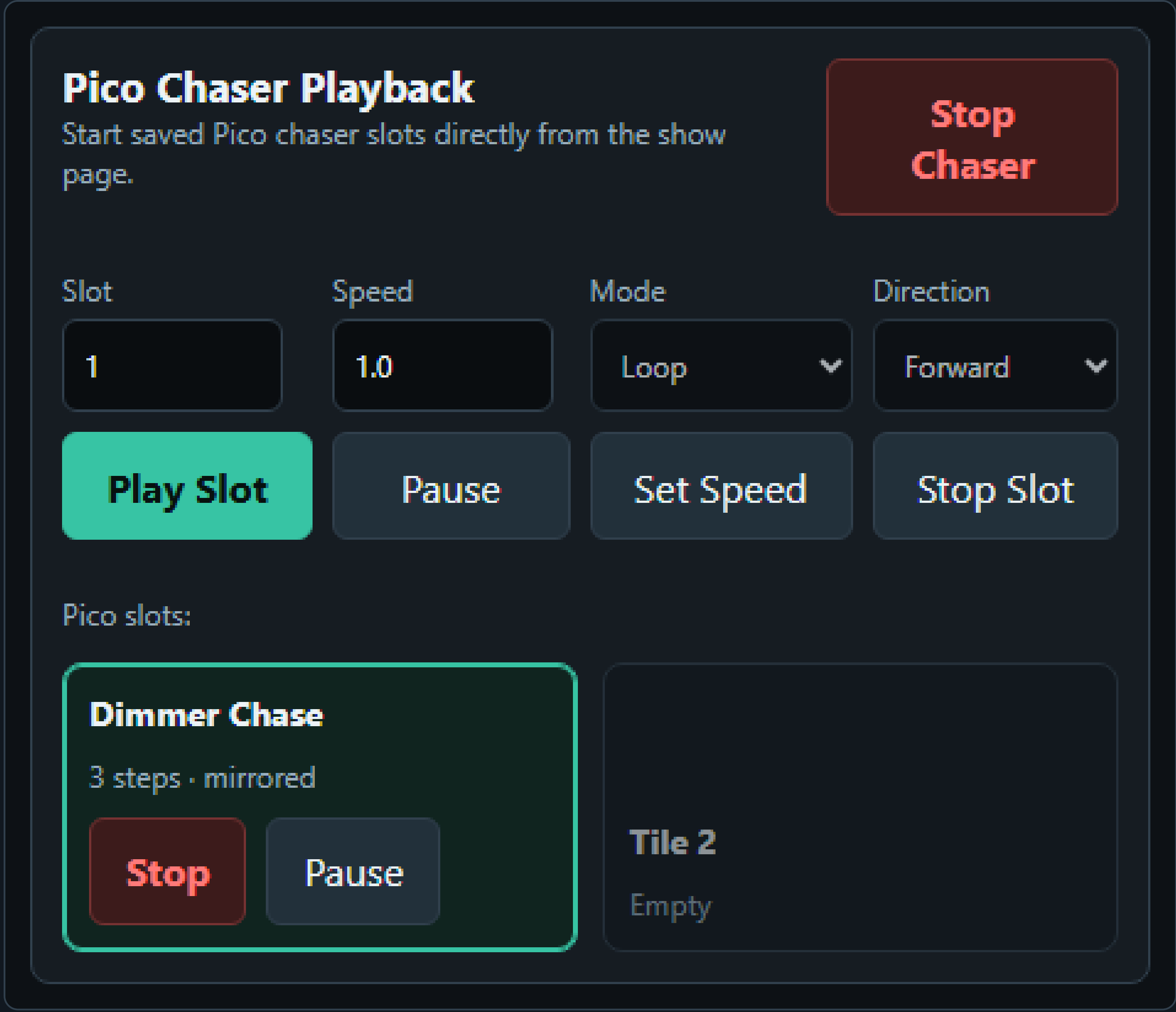
The modal uses the current Show target. Select Groups or Fixtures first when only part of the rig should move; with no target selected, Show Run targets all fixtures that match the recalled plane. Drag or click the red target point, use the X/Y coarse and fine nudge buttons, or use zoom and pan view controls to frame the room. The modal sends calibrated pan/tilt values live to the Pico and updates the shared live-value snapshot so the Controller, Chaser, and Effects pages can see the new position. Recalling a plane from Show Run does not edit the saved plane definition.



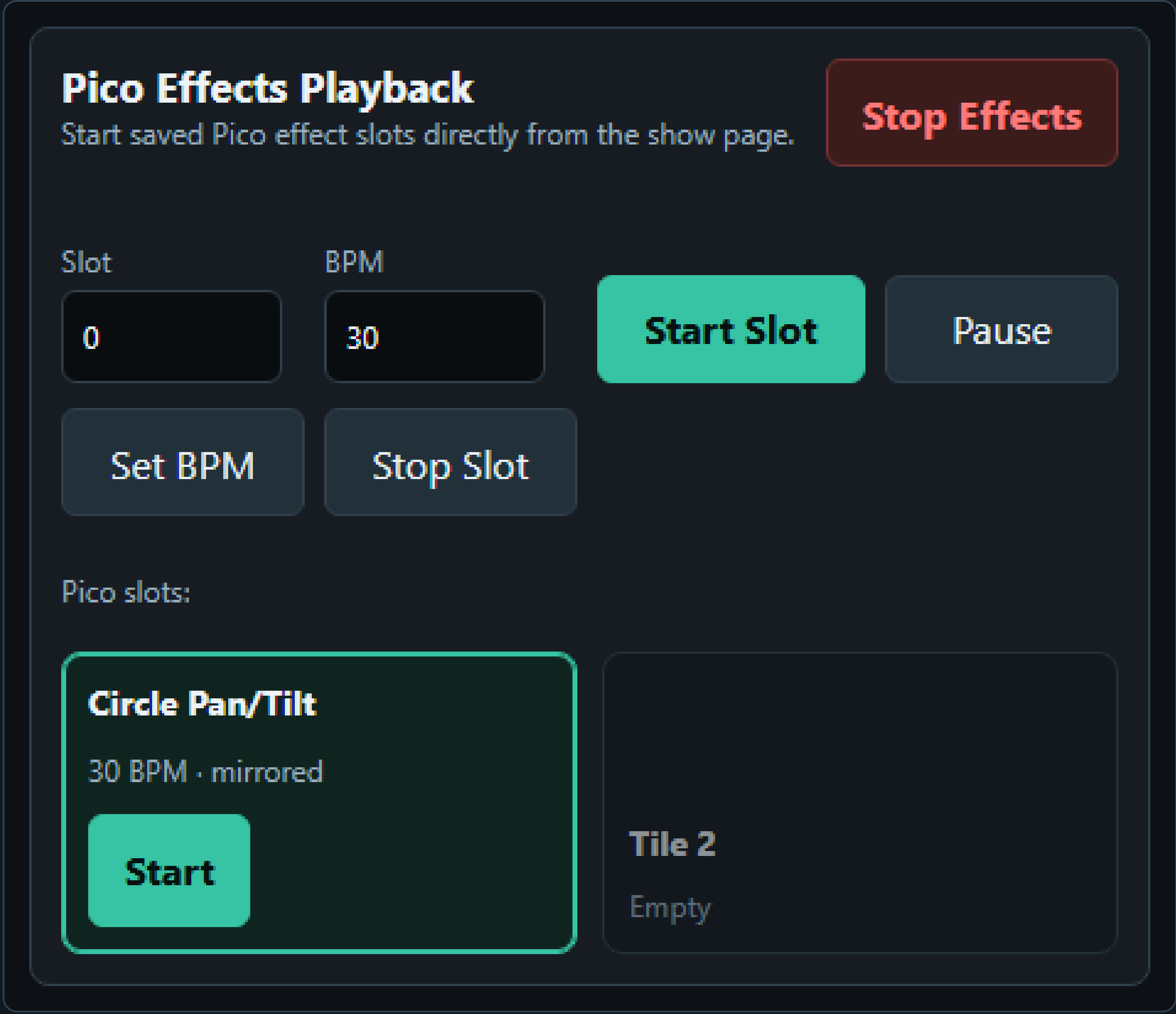


Pico Playback Cards

The **Pico Chaser Playback** card shows chaser slots uploaded from the Chaser page and mirrored to XAMPP. It also reads live Pico slot state, so a slot that is loaded on the Pico can still appear even when the XAMPP mirror is empty. Use the card controls to choose a slot, set speed, play, pause/resume, set speed, or stop.

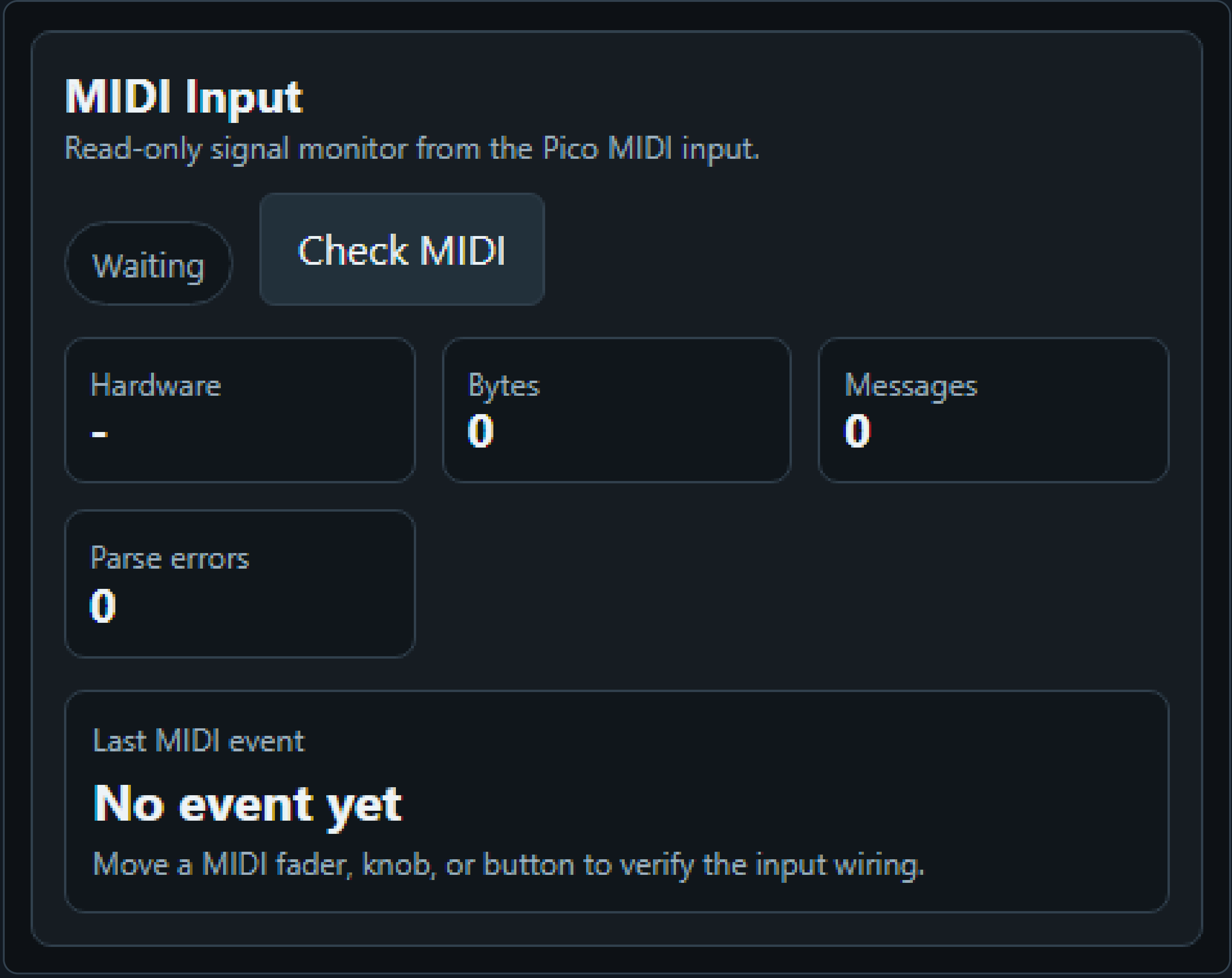


The **Pico Effects Playback** card shows effect slots uploaded from the Effects page. Choose a slot, set **BPM**, then start, set BPM, or stop the slot. Starting a mirrored slot reloads its payload before running it; starting a live-only Pico slot starts the already-loaded Pico slot without overwriting it.



MIDI Input Card

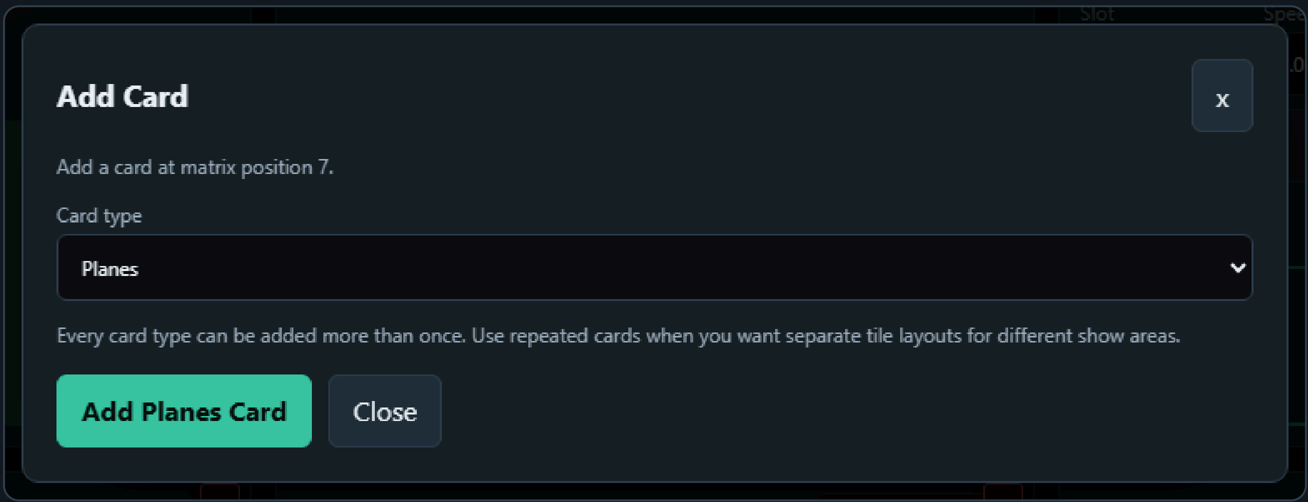
The **MIDI Input** card is a read-only diagnostics card for the Pico MIDI input. It shows whether MIDI is ready, the GPIO/UART/baud configuration, byte and message counters, parse errors, and the last decoded event. Use it to verify MIDI wiring and signal reception before mapping hardware controls to show actions.



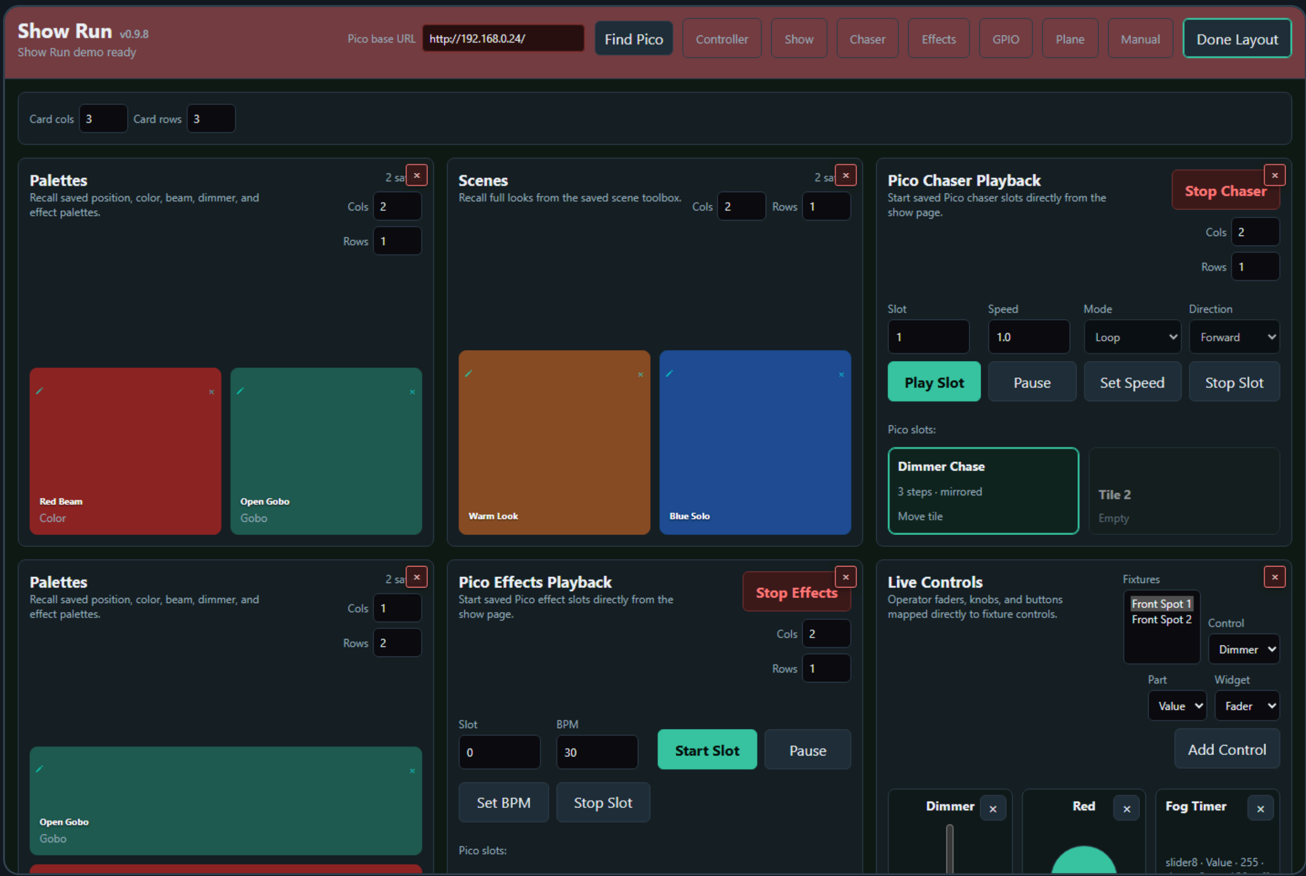
Layout Editing

The sticky title bar has **Edit Layout** on the right. Layout tools are hidden during normal operation. Click **Edit Layout** when you want to show the layout controls, then click **Done Layout** to return to the cleaner operator view.

In layout edit mode, the top of Show Run has a page-level card matrix. Use **Card cols** and **Card rows** to choose how many operator cards fit in the page grid. Empty card positions show a `+` tile; click it to open **Add Card**, choose the card type, then add it at that matrix position. If the matrix is full, Show Run still shows one extra `+` tile at the next position so another card can be added. After a card is added, Show Run reports the matrix position where it was placed. Every card type can be added more than once, so you can keep separate palette, scene, playback, or Live Controls cards with different tile layouts. Each card has a small top-right `x` while layout editing is active. That `x` removes the card from the Show Run page only; it does not delete the saved scenes, palettes, groups, chases, effects, or fixture data behind it.



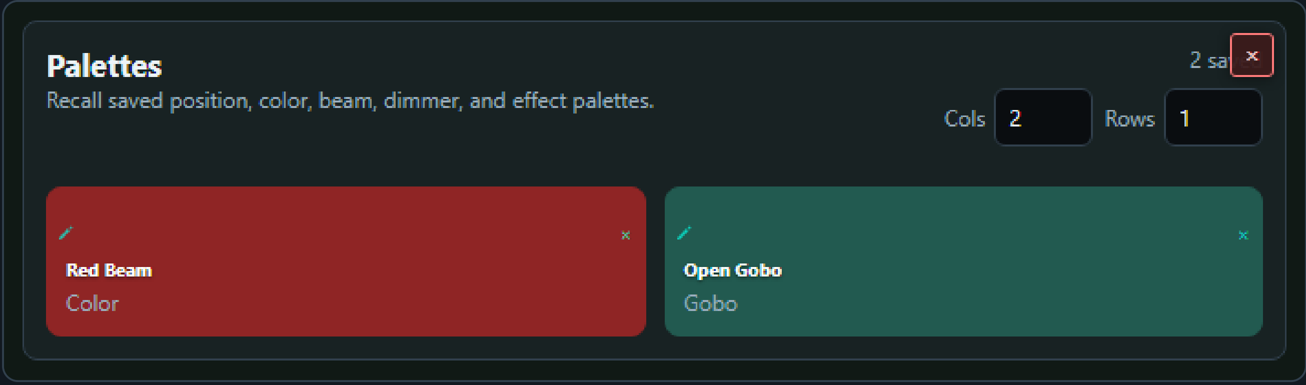
Drag a card by its title/header area to arrange whole cards, for example moving **Palettes** to the top-left and **Pico Effects Playback** to the top-right. Dragging a card to an empty matrix spot places it there and leaves the other card positions unchanged. Dragging a card onto another card swaps only those two cards. The card body remains available for tile editing and other layout controls while **Edit Layout** is active.



Each tile section on Show Run has **Cols** and **Rows** controls like the toolboxes. These controls shape the operator page layout for groups, fixtures, scenes, palettes, planes, Pico chaser slots, and Pico effect slots. While **Edit Layout** is active, tile move mode is active automatically: drag a filled tile to another position, or on touch screens tap the filled tile and then tap the destination tile. Show Run saves card layout, tile layout, and Live Controls configuration to the XAMPP server UI state, so the same operator page is restored on another computer and included in **Export Setup** / **Import Setup**. The layout does not rewrite the saved toolbox setup itself.

If saved groups, scenes, palettes, planes, or loaded Pico playback slots are outside the visible matrix, Show Run opens **Hidden Show Items** and switches into **Edit Layout**. Use **Expand** to increase that section's rows/columns until the hidden item is visible, or use **Place in Free Tile** when an empty visible tile is available. This prevents newly created palettes/scenes/planes or moved playback slots from being silently hidden on a row that is not currently displayed.

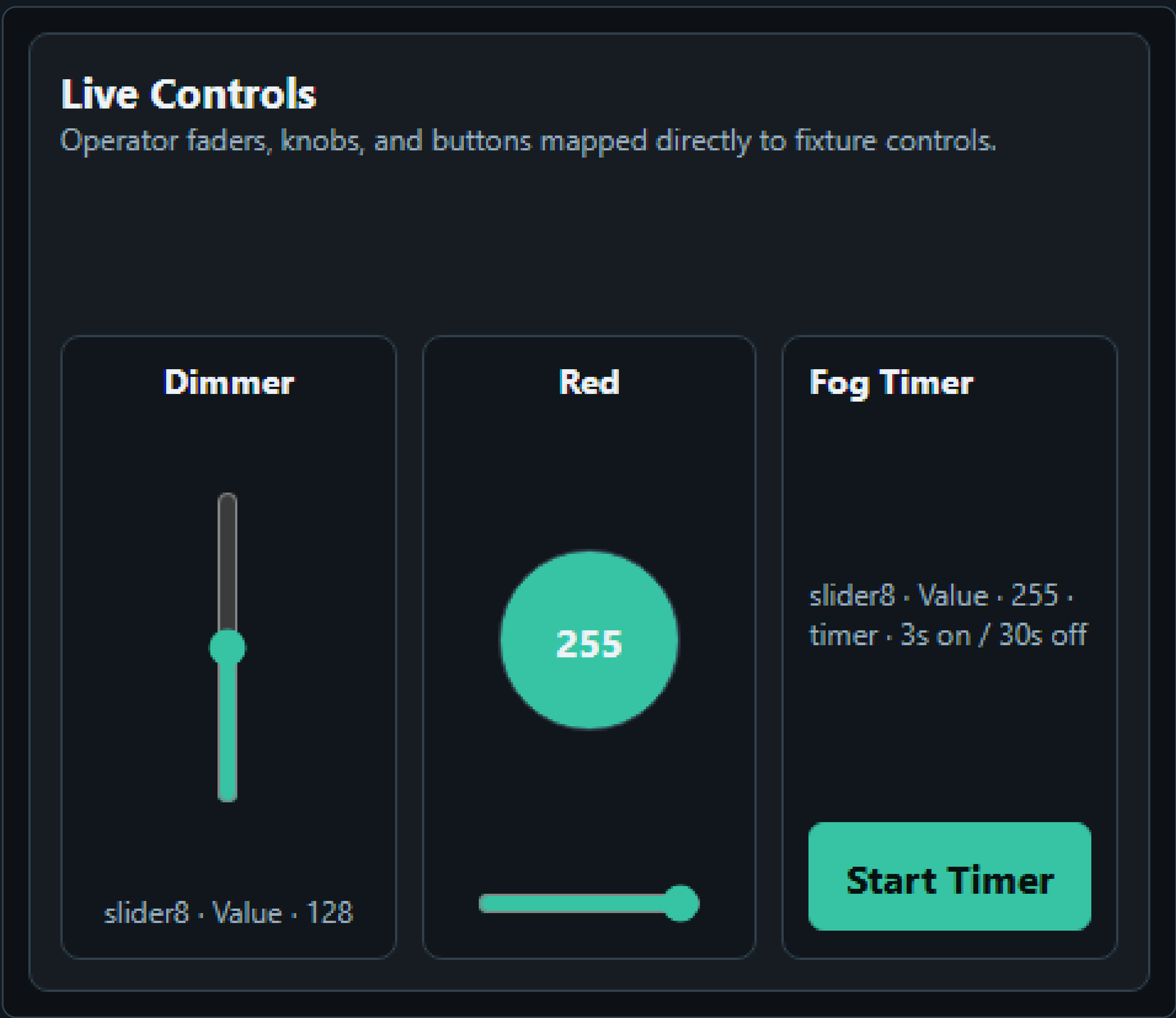
While **Edit Layout** is active, saved group, scene, and palette tiles also show the small pencil and `x` actions. The pencil opens **Edit Tile**, where the tile name, background color, uploaded icon, or drawn icon can be changed with the same visual editor used on the setup pages. These visual edits change the shared saved tile, so the updated name/color/icon appears on the setup page and in every Show Run card that uses that item. The small tile `x` only removes that item from the current Show Run card position. It does not delete the saved group, scene, or palette from the XAMPP setup files, and it does not remove that same item from another repeated Show Run card.

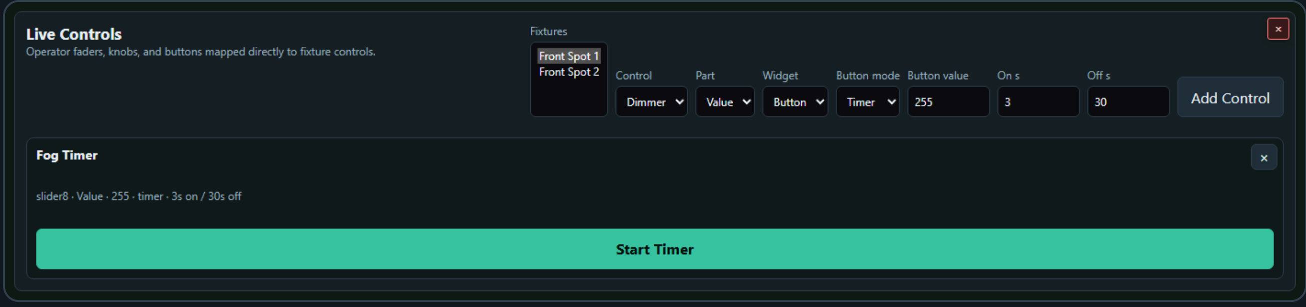


Pico chaser/effect playback tiles use the same tile move behavior during layout editing. Their playback buttons are replaced by **Move tile** while **Edit Layout** is active, so a show cannot accidentally start or stop playback while you are arranging the operator page.

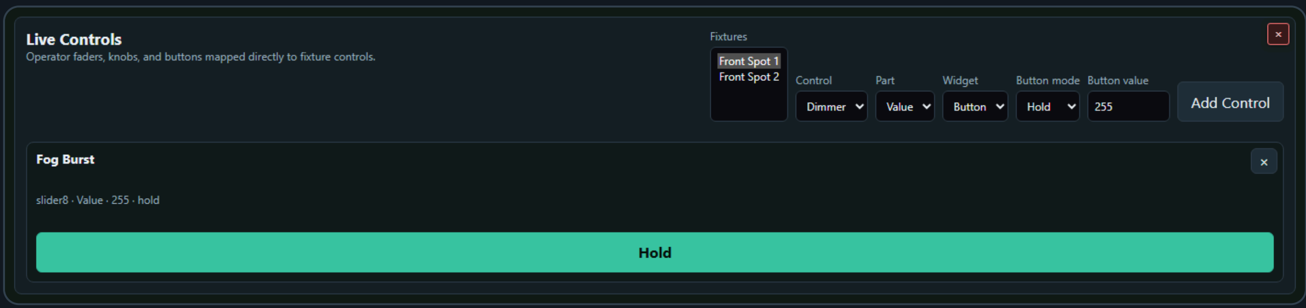
Live Controls

The **Live Controls** card can hold operator faders, knobs, and buttons that write directly to fixture controls without opening the full Controller page. Click **Edit Layout**, choose a patched fixture, control, control part, and widget type, then click **Add Control**. The setup controls are hidden when **Done Layout** is active so the operator page keeps more space for the actual controls. Multiple Live Controls cards can be used to separate faders, buttons, and mixed show controls. Faders and knobs send their value while they are moved. Compound controls are split into clear parts such as Pan, Tilt, Red, Green, Blue, White, or Amber. These widgets update the live-value snapshot and send the matching DMX bytes to the Pico when a Pico base URL is set.

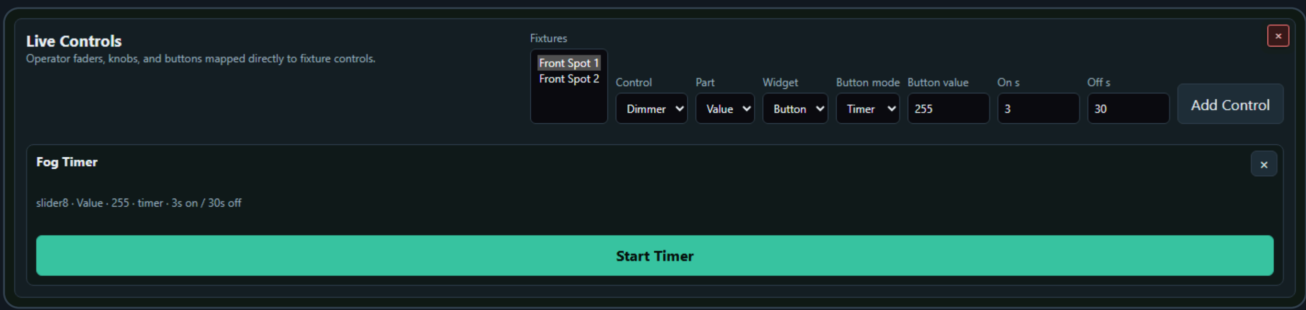




To create a momentary button, set **Widget** to **Button**, set **Button mode** to **Hold**, choose the value to send, then click **Add Control**. **Hold** sends the configured value only while the button is pressed. When the button is released, Show Run restores the value that was active before the press.



To create a fog or haze timer, set **Widget** to **Button**, set **Button mode** to **Timer**, choose the output value, then set **On s** and **Off s**. The timer sends the configured value for the On time, restores the previous/off value for the Off time, and repeats until stopped. While it is running, the widget shows the current On/Off phase, remaining seconds, and a progress bar similar to the Chaser timing display. Stopping or deleting a timer button restores the previous value.



Use **Apply** mode for one-shot commands that should send the configured value once and stay there.

Live Control card configuration is stored with the server-side Show Run layout preferences. Use the small `x` on a live widget while **Edit Layout** is active to remove it from the operator page.

Output Behavior

Scene, palette, Live Control, Grand Master, and Group Master output all use the same current target rules. Selected Groups and Fixtures define the target. With no selection, all fixtures are targeted.

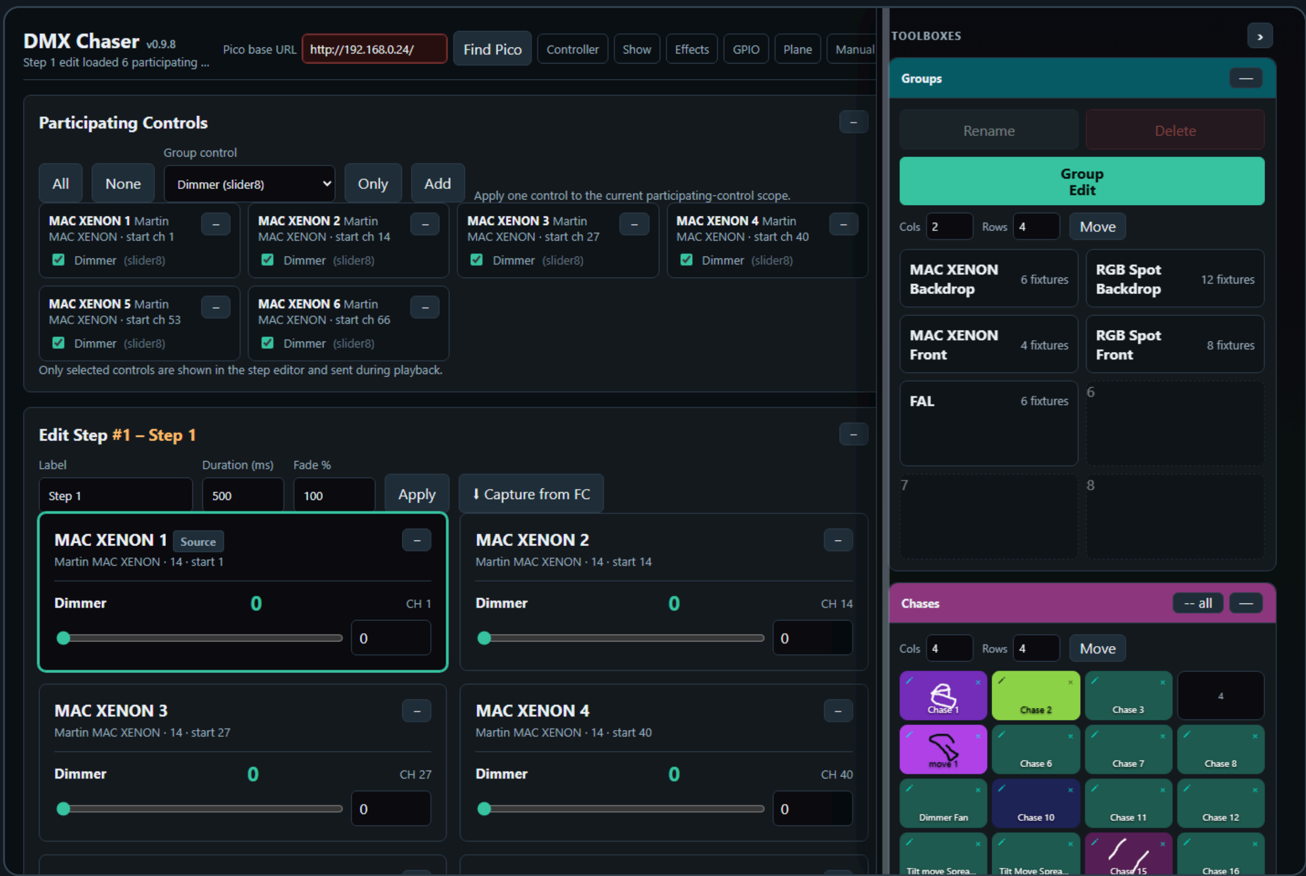
Show Run blackout is handled by the master faders. Click **Full** above a master fader to set that master to **100%**. Click **Blackout** below the Grand Master to set the Grand Master to **0%** for all dimmers. Click **Blackout** below a Group Master to set only that Group Master to **0%**. These buttons use the same master output path as moving the fader manually, so they do not overwrite stored live values. When any Grand Master or Group Master factor is below **100%**, Show Run sends affected dimmer channels to the Pico output-scaling endpoint as `channel:scale` factors. The Pico keeps chaser and effect playback writing their normal raw values, then scales the transmitted DMX output, so a running dimmer sine effect remains visible but follows the Grand/Group Master level. When you enter Show Run, saved Grand Master and Group Master factors are restored to the Pico scaling layer. When you leave Show Run, the page clears the Pico master scale and restores dimmer output to the underlying live values without those multipliers, so setup pages do not inherit the temporary show-level dimming.

The global **Stop All Playback** button calls both Pico stop endpoints.

Show Run is intentionally read-mostly during normal operation. Outside **Edit Layout** it does not create scenes, edit palettes, change fixture profiles, or overwrite the show setup files. In **Edit Layout**, card placement, tile placement, Live Controls, and tile label/visual edits are saved intentionally.

---

5. Chaser



The Chaser page creates step-based sequences. The main page stays focused on **Participating Controls** and **Edit Step**, while the repeated working tools sit in the **Toolboxes** sidebar.

The Chaser screenshot is captured with the important boxes visible on purpose: **Groups**, **Chases**, **Chase Steps**, and **Chase Playback**. Toolbox headers can be dragged up or down to reorder them in the sidebar. The order is shared with the other pages: if a page does not use one of the toolbox types, the next available toolbox moves up.

Basic Workflow

1. Open **Chaser**.
2. Select participating fixture controls.
3. Create steps with **Add step**, **Capture + Add**, or **Capture from FC**.
4. Set step duration and fade in **Edit Step**.
5. Store the chase in the **Chases** toolbox if you want quick recall.
6. Test timing in **Chase Playback**.
7. Click an empty Pico slot to upload the current chase to that slot.
8. Play the slot from the Pico.

Main Work Area

The Chaser page is arranged from top to bottom as **Participating Controls**, **Edit Step**, and **Pico Playback**. Use the cards in that order when building a chase: choose the fixture controls that may participate, edit the selected step values, then upload or play Pico slots after the chase behaves correctly.

The **Toolboxes** sidebar sits beside the main work area on desktop-sized screens. It contains the repeated tools for Groups, Chases, Palettes, Chase Steps, Fan Out, and Chase Playback.

Toolbox Sidebar

Controller, Chaser, and Effects use a shared right-side toolbox sidebar on desktop-sized screens.

- Drag the vertical resize line on the left edge of the sidebar to change its width.
- The sidebar width is shared across all toolbox pages.
- Double-click the resize line to reset the default width.
- The page content and the toolbox sidebar scroll independently. The page scrollbar sits beside the toolbox separation line, while the toolbox scrollbar stays inside the toolbox sidebar.
- Use the arrow button in the Toolboxes header to collapse or reopen the whole sidebar. The collapsed state is shared across toolbox pages.
- Drag a toolbox by its colored header to reorder the sidebar. The toolbox body is not a drag handle, so slot clicks, sliders, and buttons are safe on touch screens. On iPad, toolbox reordering uses the app's pointer drag on the colored header instead of Safari's native drag/drop; the header also suppresses text selection/copy callouts as much as Safari allows.
- On narrow screens, the sidebar changes into a bottom toolbox drawer.

The Chaser page uses several toolboxes:

- **Groups** filters the fixture list by one or more saved fixture groups. It uses the cyan header, like the group tools on the controller page.

GPIO Plane Manual

trol scope.

ON 4 Martin
N · start ch 40
er (slider8)

CH 14
0

CH 40
0

TOOLBOXES

Groups

Rename
Delete
Group Edit

Cols 2 Rows 4 Move

MAC XENON Backdrop 6 fixtures	RGB Spot Backdrop 12 fixtures
MAC XENON Front 4 fixtures	RGB Spot Front 8 fixtures
FAL 6 fixtures	6
7	8

Chases

-- all

Cols 4 Rows 4 Move

Chase 1	Chase 2	Chase 3	4
move 1	Chase 6	Chase 7	Chase 8
Dimmer Fan	Chase 10	Chase 11	Chase 12
Tilt move Sprea...	Tilt Move Sprea...	Chase 15	Chase 16

- **Chases** stores complete editable chases in a slot matrix. Clicking an empty slot saves the current chase. Clicking a filled slot loads that chase.
- The small top-left pencil icon on a filled **Chases** slot opens **Edit Tile** for the chase name, background color, and optional drawn/uploaded visual. **Default background** restores the standard slot color, and **No icon** removes the overlay image. This is only a label; loading a chase still uses the stored chase steps and playback settings.

GPIO Plane Manual

trol scope.

ON 4 Martin
N · start ch 40
er (slider8)

CH 14
0

CH 40
0

TOOLBOXES

Chases
-- all

Cols 4 Rows 4 Move

Chase 1	Chase 2	Chase 3	4
move 1	Chase 6	Chase 7	Chase 8
Dimmer Fan	Chase 10	Chase 11	Chase 12
Tilt move Sprea...	Tilt Move Sprea...	Chase 15	Chase 16

Groups

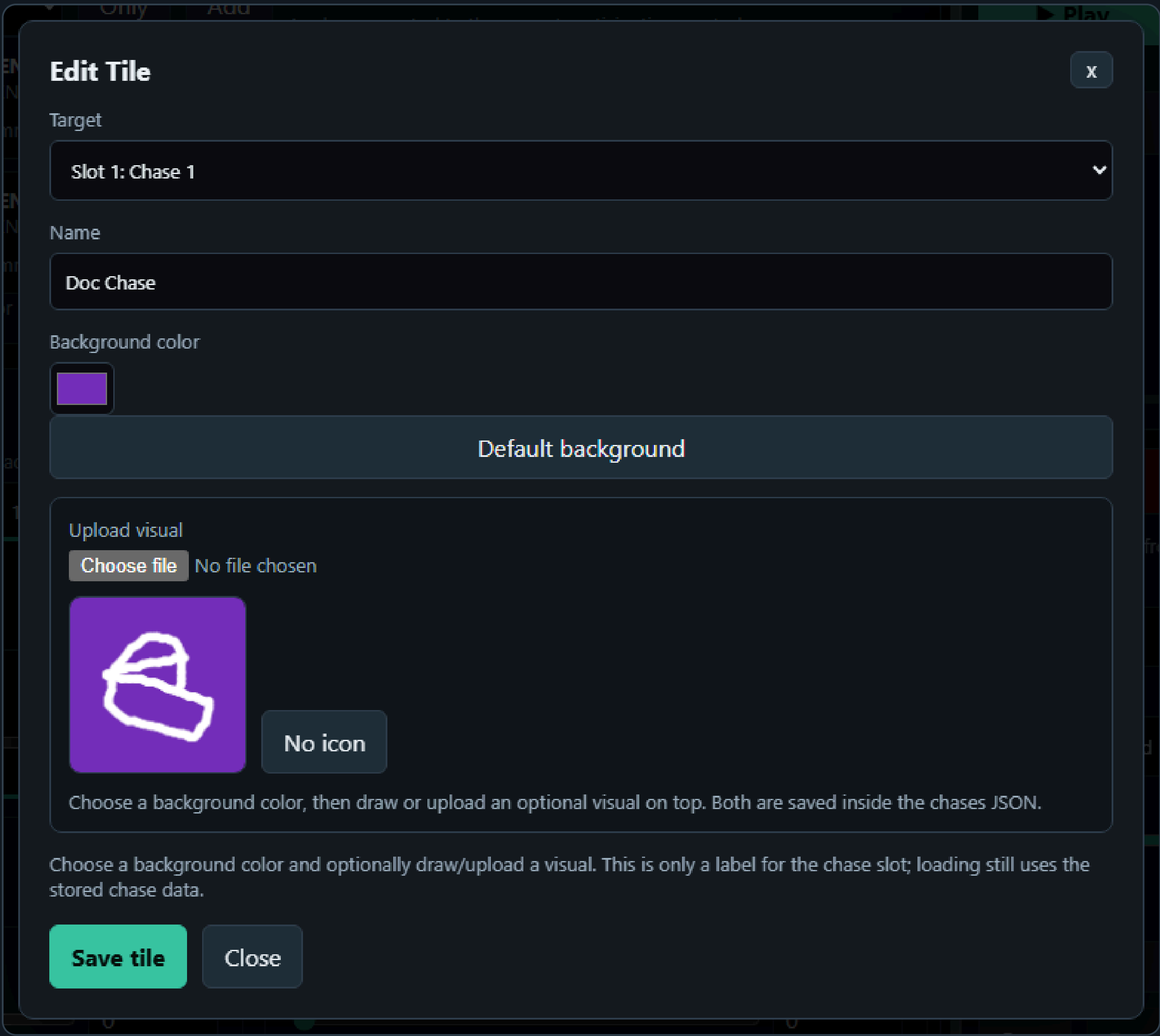
Rename Delete

Group Edit

Cols 2 Rows 4 Move

MAC XENON Backdrop 6 fixtures	RGB Spot Backdrop 12 fixtures
MAC XENON Front 4 fixtures	RGB Spot Front 8 fixtures
FAL 6 fixtures	6
7	8

- **Palettes** stores and recalls reusable step fragments. Clicking an empty palette slot saves the currently selected step's fixture/control values to the shared palette JSON. Clicking a filled palette slot recalls compatible values into the selected step and rebuilds **Participating Controls** from the palette's stored fixture/control keys. If no step is selected, Chaser creates a new selected step from the palette. **Merge** adds the selected step's values into an existing palette; if the palette scope differs, Chaser asks before changing it to **All controls**. The small top-left pencil icon opens **Edit Tile** for a saved palette slot's name and visual appearance. When a Pico base URL is set, recalled palette values are also sent to the Pico.



GPIO Plane Manual

trol scope.

ON 4 Martin
N · start ch 40
er (slider8)

CH 14
0

CH 40
0

TOOLBOXES

Palettes

Merge

Empty slot saves the selected step values. Filled slot recalls compatible values into the selected step. Merge adds selected step values to an existing palette.

Cols 4 Rows 4 Move

Step 1 palette	Step 1 palette	Step 3 palette	Step 3 palette st...
5	6	7	8
9	10	11	12
13	14	15	16

Chase Steps (2/32) -- all

+ Add step ⬇ Capture + Add

Update chase Clear Steps

Dupe Up Down Delete

1 **Step 1**
500ms · fade 100%

2 **Step 2**
500ms · fade 100%

- **Planes** recalls saved room planes into the selected chase step. Select a group first if only part of the rig should be affected. Chaser uses the plane target and calibrated fixtures to write pan/tilt values into the step, rebuilds the participating controls for those pan/tilt channels, and sends the changed step values to the Pico when a Pico base URL is set. The Planes tile matrix uses the same **Cols**, **Rows**, and **Move** behavior as the other saved tile toolboxes.

GPIO | Plane | Manual

trol scope.

ON 4 Martin
N · start ch 40
er (slider8)

CH 14
190

CH 40
190

TOOLBOXES

Planes

Recall a saved room plane into the selected chase step.

Cols 3 Rows 3 Move

1	2	3
4	5	6
7	8	9

Groups

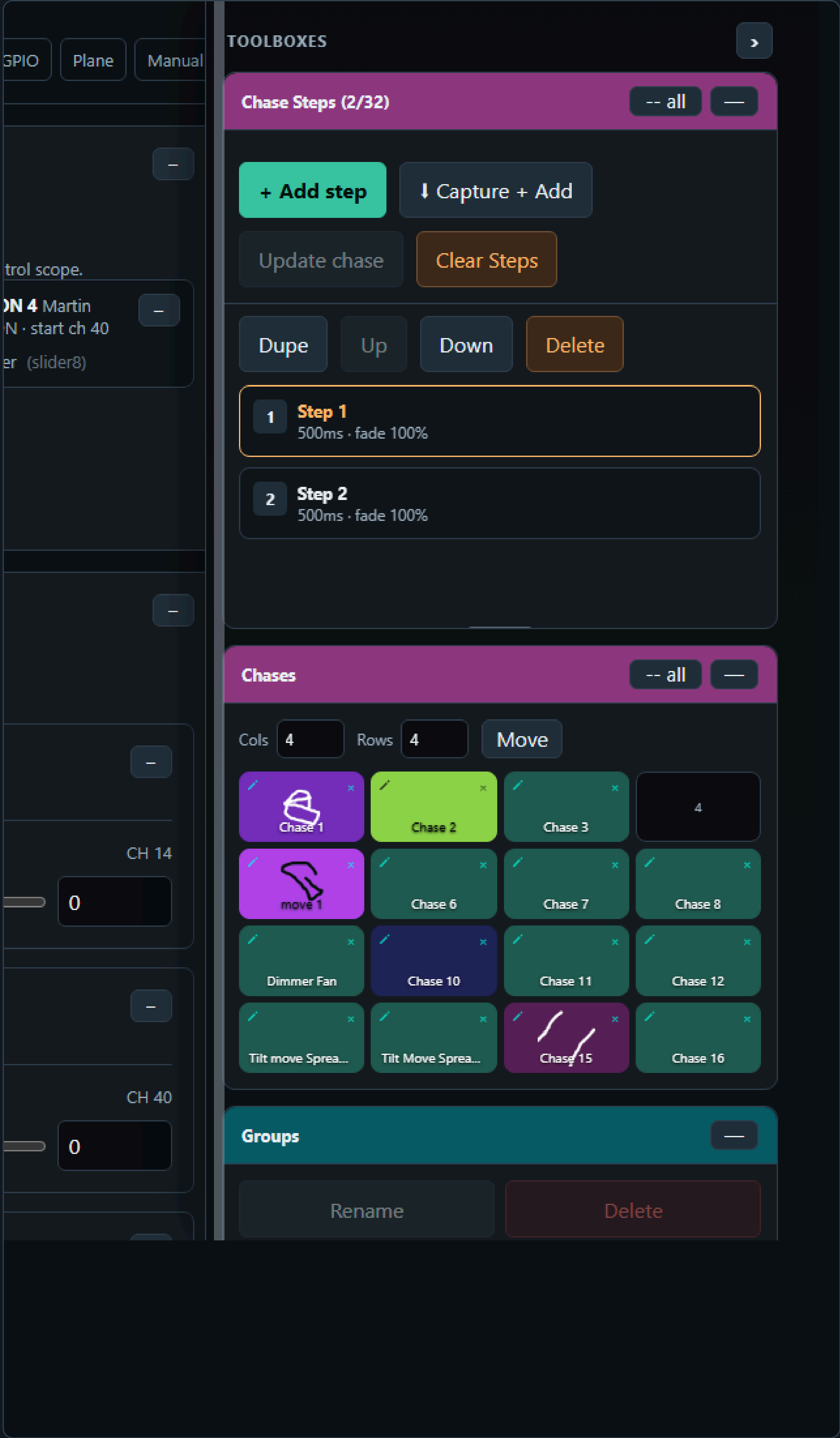
Rename Delete

Group Edit

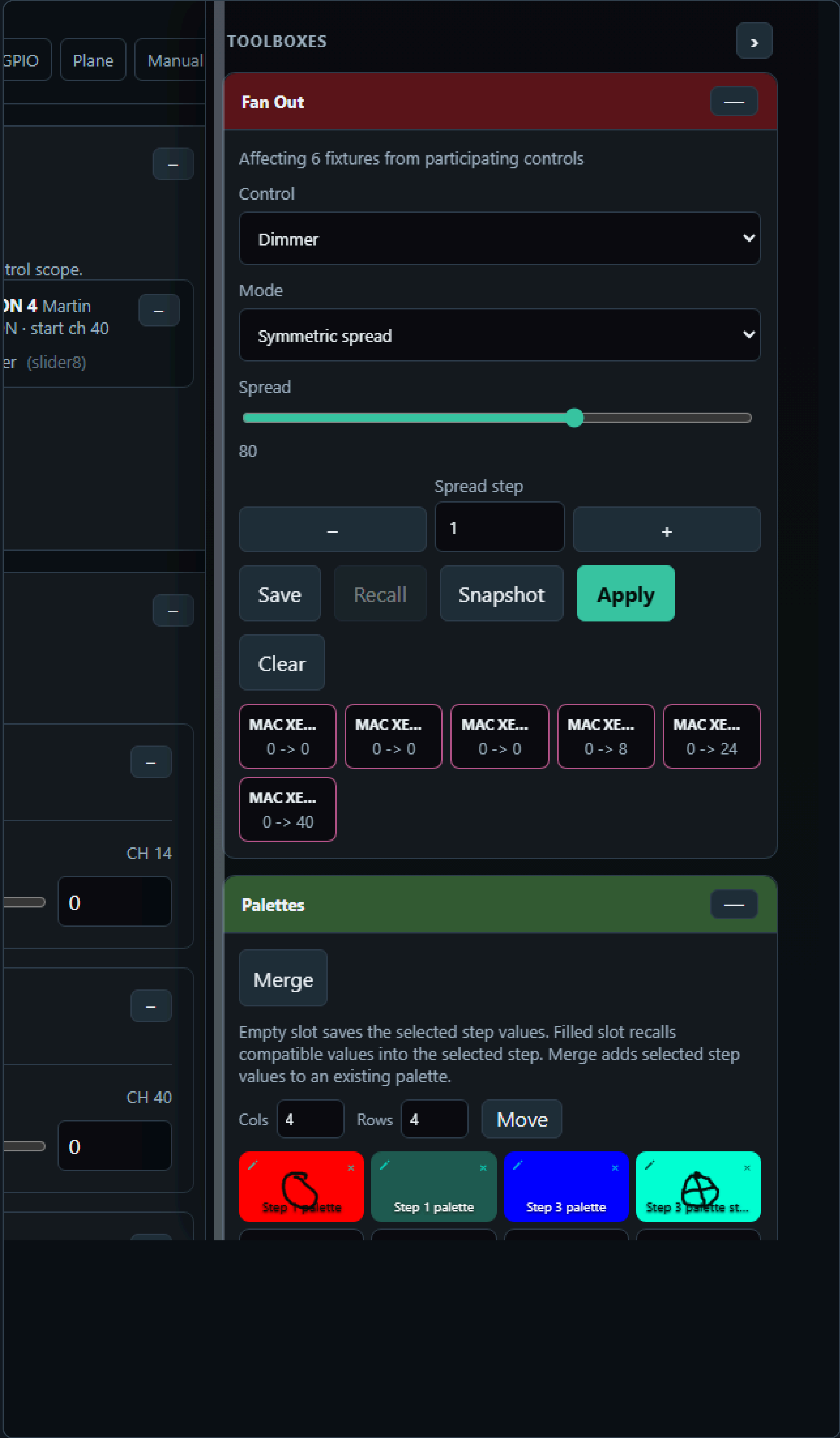
Cols 2 Rows 4 Move

MAC XENON Backdrop 6 fixtures	**RGB Spot Backdrop** 12 fixtures
MAC XENON Front 4 fixtures	**RGB Spot Front** 8 fixtures
FAL 6 fixtures	6
7	8

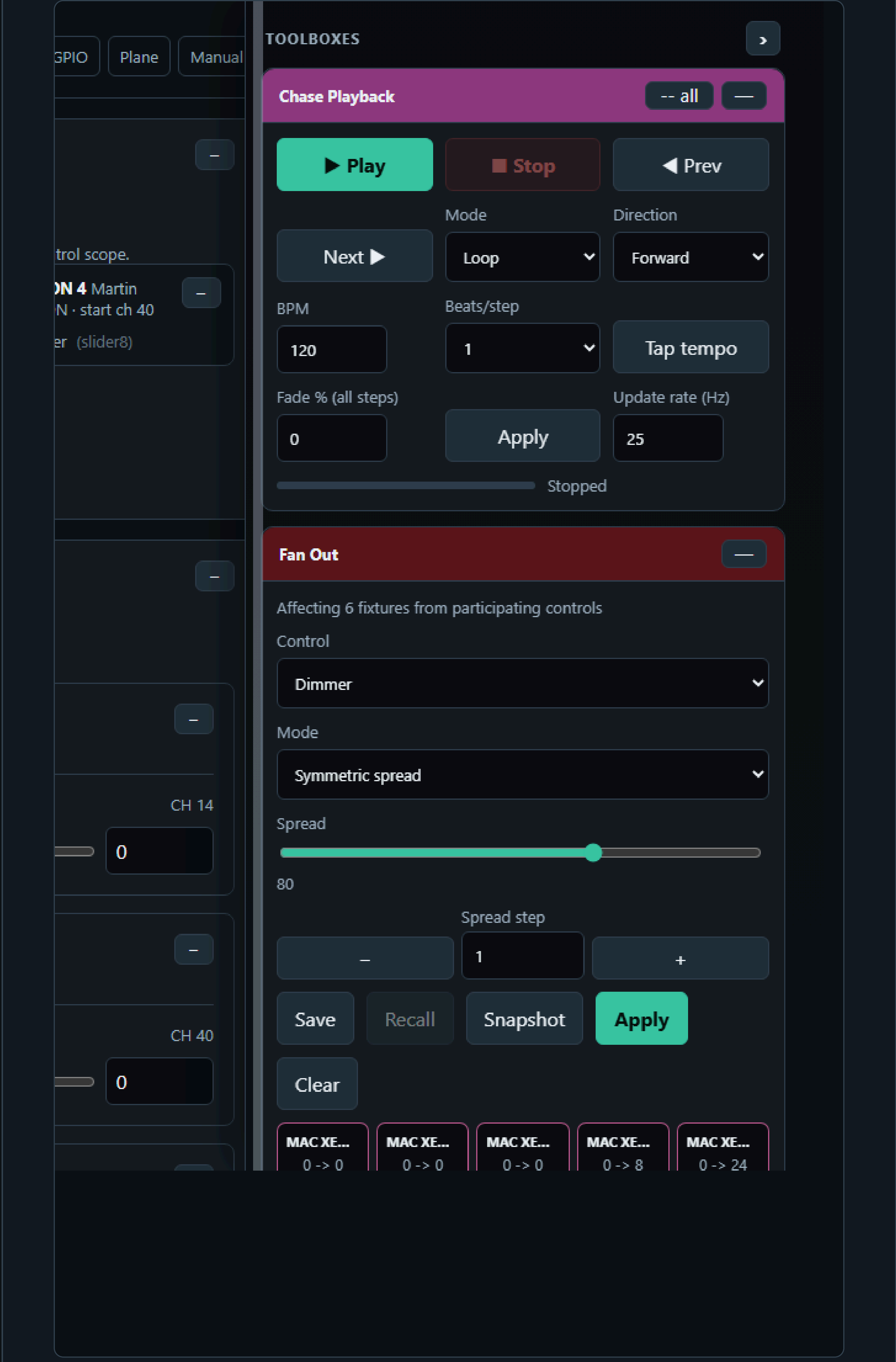
- **Chase Steps** contains the step list and step actions. Use it to add, capture, edit, duplicate, delete, and reorder steps. In the Toolboxes sidebar, drag the lower edge of the Chase Steps box to set its height. The top buttons remain visible while the step list scrolls inside the box.
- **Chases**, **Chase Steps**, and **Chase Playback** share one toolbox color and collapse together. Use **-- all** on any of those boxes to collapse the whole color group, and **+ all** to reopen it.


GPIO
Plane
Manual
TOOLBOXES
trol scope.
ON 4 Martin
N · start ch 40
er (slider8)
CH 14
0
CH 40
0
Chase Steps (2/32)
-- all
+ Add step
⬇ Capture + Add
Update chase
Clear Steps
Dupe
Up
Down
Delete
1
Step 1
500ms · fade 100%
2
Step 2
500ms · fade 100%
Chases
-- all
Cols
4
Rows
4
Move
Chase 1
Chase 2
Chase 3
4
move 1
Chase 6
Chase 7
Chase 8
Dimmer Fan
Chase 10
Chase 11
Chase 12
Tilt move Sprea...
Tilt Move Sprea...
Chase 15
Chase 16
Groups
Rename
Delete

- **Fan Out** is a live step-shaping tool. It works on the currently selected step and writes directly into **Edit Step** as soon as you change the Fan Out mode, spread, or range values. There is no separate Snapshot or Apply action on the Chaser page.
- **Fan Out control selection** is filtered by the selected step. The control dropdown only shows compatible single-value controls that are actually part of the selected step's participating controls and exist on at least two fixtures. If one or more groups are selected, the same rule is applied inside the selected groups only.
- **Fan Out order** comes from the current participating fixture/control list. If a group filter is active, only fixtures inside the selected groups participate, but the calculation still follows the participating order that is shown for the current step/scope. In Symmetric spread mode, negative Spread runs the shape from the opposite side of the fixture row.
- **Fan Out base values** come from the values currently displayed in **Edit Step**. Selecting another step, loading another chase, capturing values, or using **Group Edit** refreshes the Fan Out base from the step values now shown on screen. This keeps spread calculations from drifting away from the edited step.
- **Clear** in the Chaser Fan Out toolbox resets the Fan Out shaping controls to neutral. It does not recall an older preset and it does not undo values that have already been written into the selected step.
- Editing **Edit Step**, using **Group Edit**, and changing **Fan Out** updates the selected step immediately. When a Pico base URL is set, those changed selected-step values are also sent to the Pico. Chase Playback sends the current chase continuously while it runs and therefore requires a Pico base URL.


GPIO
Plane
Manual
trol scope.
N 4 Martin
N · start ch 40
er (slider8)
CH 14
0
CH 40
0
TOOLBOXES
Fan Out
Affecting 6 fixtures from participating controls
Control
Dimmer
Mode
Symmetric spread
Spread
80
Spread step
–
1
+
Save
Recall
Snapshot
Apply
Clear
MAC XE...
0 -> 0
MAC XE...
0 -> 0
MAC XE...
0 -> 0
MAC XE...
0 -> 8
MAC XE...
0 -> 24
MAC XE...
0 -> 40
Palettes
Merge
Empty slot saves the selected step values. Filled slot recalls compatible values into the selected step. Merge adds selected step values to an existing palette.
Cols
4
Rows
4
Move
Step 1 palette
Step 1 palette
Step 3 palette
Step 3 palette st...

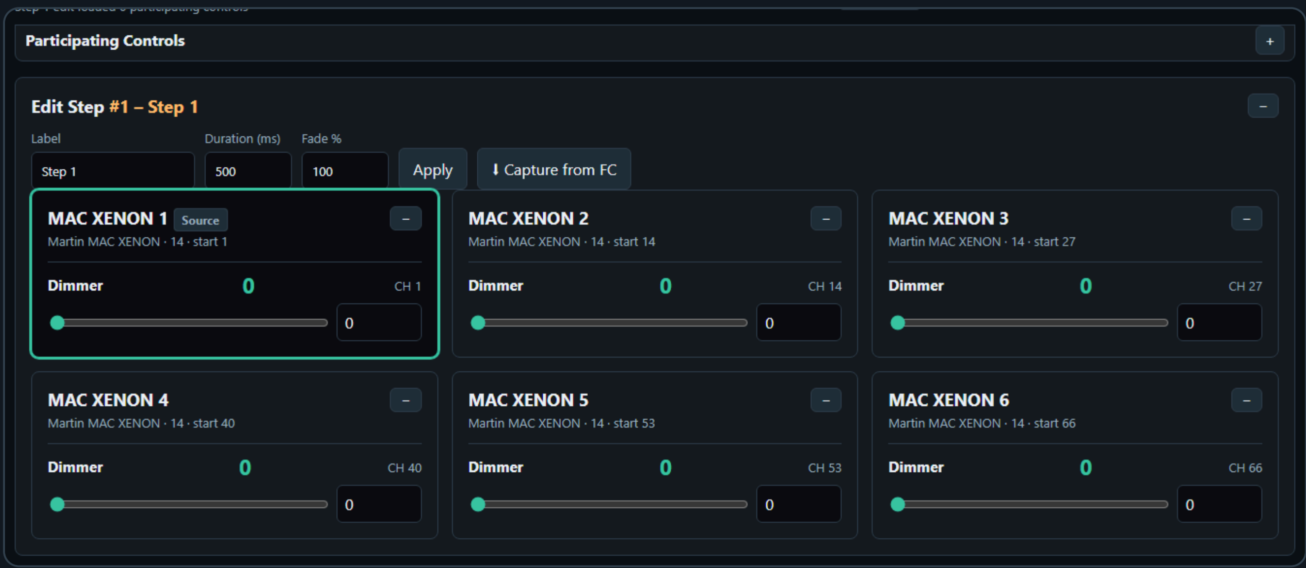
- **Chase Playback** runs the current chase from the browser for checking timing and fades before uploading to the Pico. **Mode** chooses **Single**, **Loop**, **Loop N**, or **Ping Pong**. **Direction** chooses whether playback starts at the first step and moves forward or starts at the last step and moves backward. **Loops** is only used by **Loop N**; normal **Loop** runs forever. **Ping Pong** reverses at the end of the chase instead of jumping back to the first or last step. The **Fade % (all steps)** field applies one fade value to every step immediately. Use **Edit Step > Fade %** when one step needs its own fade value.
- While Chase Playback runs, the currently playing step automatically becomes the selected step. This means **Edit Step** follows the chase visually and always shows the values of the last played step. Recalling a saved chase still selects Step 1 first, so playback starts from a predictable view.
- Selecting a step in **Chase Steps** or manually changing controls in **Edit Step** stops browser Chase Playback. This prevents playback from immediately overwriting the values you are trying to edit.


GPIO
Plane
Manual
trol scope.
ON 4 Martin
N · start ch 40
er (slider8)
CH 14
0
CH 40
0
TOOLBOXES
Chase Playback
-- all
Play
Stop
Prev
Mode
Direction
Next
Loop
Forward
BPM
Beats/step
120
1
Tap tempo
Fade % (all steps)
Update rate (Hz)
0
Apply
25
Stopped
Fan Out
Affecting 6 fixtures from participating controls
Control
Dimmer
Mode
Symmetric spread
Spread
80
Spread step
1
+
Save
Recall
Snapshot
Apply
Clear
MAC XE... 0 -> 0
MAC XE... 0 -> 0
MAC XE... 0 -> 0
MAC XE... 0 -> 8
MAC XE... 0 -> 24

On page load, the Chaser working area starts with no steps selected. Use the **Chases** toolbox to recall a saved chase. Loading a chase from the **Chases** box updates the step list, selects Step 1, and rebuilds participating controls and the currently edited step together. If the chase contains steps, the participating controls are rebuilt from the values stored in the chase, so old fixture/group filters do not hide the controls used by that chase.

Saved chases also recall the playback controls that were saved with them, including browser mode, Loop N count, Ping Pong, direction, BPM, and Pico speed. The Chases toolbox tiles still show only the chase name and visual so they stay consistent with the other toolboxes. Uploading to a Pico slot uses the current **Chase Playback** playmode, loop count, and direction, so the Pico slot behaves like the chase you previewed in the browser. Per-slot playback details are shown in the **Pico Playback** slot tiles after a chase has been uploaded to a Pico slot. Each loaded Pico slot shows compact lines for loop state, direction, and Ping Pong state.

The collapse state, toolbox order, shared sidebar width, and the user-defined Chase Steps box height are stored by the server UI-state file, so the working layout survives reloads. Collapsing **Participating Controls** or **Edit Step** only hides that card body: the sticky page header keeps the same height, and the next card moves up to use the freed space.



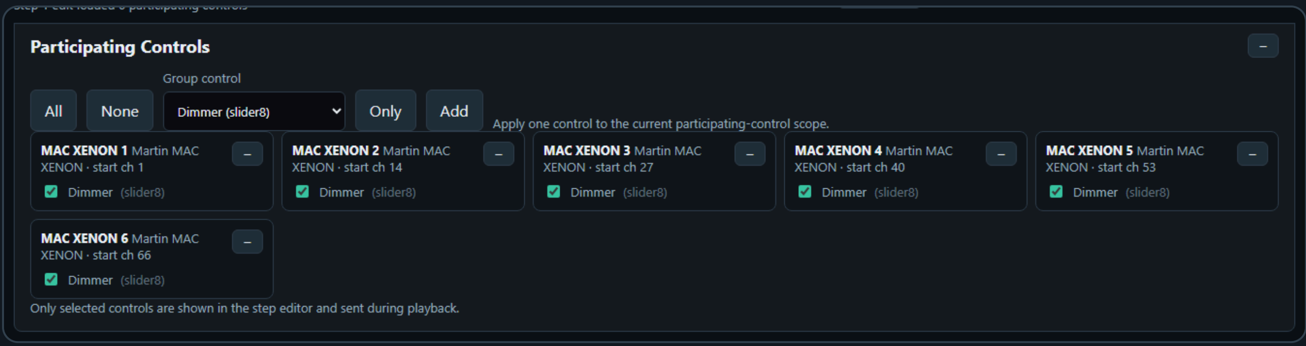
Chase Steps Toolbox Buttons

The **Chase Steps** toolbox uses shared action buttons instead of per-step buttons. Click a step card to select the step first; the selected step is highlighted and loaded into **Edit Step**.

- **Add step** creates a new selected step from the stored default values of the currently participating controls. If a fixture profile has no custom default for a control, Chaser uses the control type fallback.
- **Capture + Add** creates a new selected step only when Fixture Controller live values exist for at least one currently participating control. If no matching live values are available, no empty step is created.
- **Clear Steps** deletes all current working steps after confirmation. It does not delete saved Chases toolbox entries or Pico slot payloads.
- **Dupe** duplicates the selected step directly after itself and selects the copy.
- **Up** and **Down** move the selected step in the step order and keep the same step loaded in **Edit Step**.
- You can also drag a step card with mouse or touch and drop it before or after another step card to reorder the chase. A short click still selects the step. The selected step and the currently playing step stay attached to the same step data after the move.
- **Delete** removes the selected step. If another step remains, it becomes the selected step.

Participating Controls

Participating controls define which fixture controls belong to the chase. This keeps the chaser from editing unrelated channels.



For example, a dimmer chase might include only dimmer controls. A color chase might include only RGB or RGBWA controls.

If no group is selected, all patched fixtures are available. If one or more groups are selected in the Groups toolbox, only fixtures from those groups are shown while you are choosing the scope. Any direct change in **Participating Controls** then clears the group selection. This avoids mixing two different filters: after you press **All**, **None**, **Only**, **Add**, or tick an individual checkbox, the participating-control set becomes the source of truth.

The **Group control** dropdown is built from the fixtures currently visible in the participating-control scope, not from the controls that are already ticked. With no Groups filter, the dropdown stays based on all visible patched fixtures even after **Only** narrows the selected participating controls. With a Groups filter, the dropdown is based on the selected groups. When an edited step is active, the visible step fixtures define the dropdown scope.

All selects every valid participating control for all patched fixtures, clears the Groups filter, expands the participating fixture list, stops browser Chase Playback, clears the selected step, clears **Edit Step**, clears the Source fixture, and resets Fan Out bases. Existing steps in **Chase Steps** are not deleted. After **All**, use **Add step**, **Capture + Add**, or **Group Edit** to create or edit a step from the full participating-control scope.

Group Edit edits the current participating-control scope. It becomes available when the current scope contains at least one matching selected control on two or more participating fixtures, even when those fixtures use different fixture profiles. A step does not need to be selected first. If no step is selected, the first Group Edit value change creates a new step from the current participating controls and writes the edit into that step. If a step is selected, Group Edit edits that selected step.

The fixtures may use different profiles; the modal only shows matching controls that are actually selected as participating controls and exist on at least two fixtures. For example, if only Dimmer is selected, Group Edit opens with Dimmer only and applies a changed Dimmer value to every involved fixture that has a matching Dimmer control.

Group Edit uses the **Source** fixture from **Edit Step** as the reference value. Opening the modal reads the Source fixture's current selected-step values into the modal controls. It does not automatically overwrite the other fixtures, does not create a new step, and does not send output by itself.

The values are written when you change a control in the modal. If no step is selected, the first Group Edit value change creates a new step from the current participating controls and writes the edited value into that step. If a step is selected, the edit writes into that selected step. When a Pico base URL is set, changed selected-step values are also sent to the Pico.

Chaser Selection Rules

The Chaser page has two different selection modes: defining a new participating-control set, and editing or recalling an existing step. They intentionally behave differently.

Keep these rules as the contract for the Chaser page:

- **Participating Controls** define the fixture/control scope for new steps and for Group Edit.
- **Edit Step** shows the currently selected step, not a separate preview copy.
- If a saved chase is recalled, Step 1 is selected immediately and becomes the visible **Edit Step**.
- If **Chase Playback** is running, playback overrules manual step rendering: every played step automatically becomes the selected step and redraws **Participating Controls** plus **Edit Step**.
- During fades, **Edit Step** updates the existing sliders, readouts, XY dots, colors, and wheel highlights continuously from the interpolated playback values. The full card is only rebuilt when the played step changes.
- When playback stops or pauses, the last played step remains selected.
- Manual previous/next playback also selects the stepped-to chase step and redraws **Edit Step**.
- Group filters are only temporary scope builders. Direct Participating Controls changes clear the group filter.
- **All** clears the selected step/edit context but keeps the existing step list unchanged.
- **Group Edit** does not require a selected step. If no step is selected, the first Group Edit value change creates a new step from the current participating controls. If a step is selected, Group Edit edits that step.
- Opening **Group Edit** loads the current Source fixture values into the modal only. The output or selected step changes only after you edit a modal control.

When you define participating controls manually:

- If no group is selected, the Participating Controls panel shows all patched fixtures.
- If one or more groups are selected, the panel is temporarily filtered to the fixtures in those groups.
- **All** selects every currently visible control and clears the group selection.
- **None** clears the participating-control selection, collapses the fixture list, and clears the group selection.
- The control dropdown lists controls from the selected groups while a group filter is active. If no group is selected, it lists controls from the current participating-control scope.
- The control dropdown plus **Only** selects one matching control type from the current scope, clears all other controls, and then clears the group selection.
- The control dropdown plus **Add** adds one matching control type from the current scope without clearing existing participating controls, then clears the group selection.
- Ticking or unticking an individual participating-control checkbox also clears the group selection.

When you click a step in the **Chase Steps** toolbox:

- The step values are checked against the current fixture setup.
- Invalid fixture/control references are removed from the edited step.
- The Participating Controls panel is rebuilt from the chase values.
- If no group is selected, only fixtures and controls that are actually stored in the selected step are shown, and **Group Edit** becomes step-dependent.
- If a group is selected, the selected step is filtered to that group and **Group Edit** uses the grouped fixtures as its edit scope.

- The Edit Step card shows the same scoped fixture/control set, so editing Step 2 cannot accidentally show or edit unrelated controls from another group or old filter.

When you click a saved chase in the **Chases** toolbox:

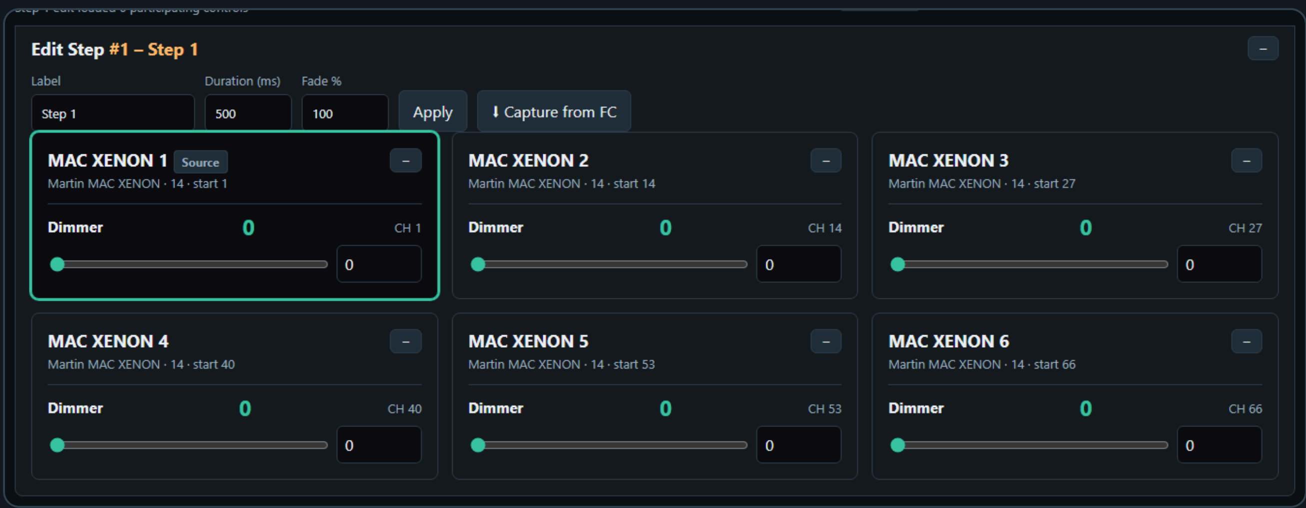
- The selected Groups filter is cleared first.
- The saved steps, playback settings, and Chase Playback settings are loaded.
- The first step becomes the edited step.
- Participating Controls and Edit Step are rebuilt from that loaded step.
- If the saved chase was made before fixtures were repatched, the page tries to remap old fixture IDs to the current setup by matching the same fixture profile in DMX start-address order.
- If no valid step values can be found, the page falls back to the saved participating-control map stored with the chase.

This means group selection is a tool for building, filtering, and group-editing a step. Loading a saved chase clears the group filter because the recalled chase data becomes the source of truth. After recall, **Group Edit** can still become available without a selected group because it uses the fixtures participating in the selected step.

Create Chase Steps

A chase step stores values for the selected **Participating Controls**. It does not store every patched fixture automatically.

To create a step manually:



1. Select the controls that should participate in **Participating Controls**.
2. In the **Chase Steps** toolbox, click **Add step**.
3. Click the new step in the **Chase Steps** list.
4. In **Edit Step**, set **Label**, **Duration (ms)**, and **Fade %**.
5. Adjust the controls shown in **Edit Step**. Those control values are written into the selected step immediately. When a Pico base URL is set, they are also sent to the Pico.
6. Click **Apply** after changing the label, duration, or fade.

Add step starts from the stored default values of the selected participating controls. If a fixture profile has no custom default for a control, Chaser uses the control type fallback: centered pan/tilt, zero for sliders, wheels, and color channels.

After **Add step** or **Capture + Add**, the first fixture in the new step is automatically marked as the **Source** fixture in **Edit Step**. Click another fixture card in **Edit Step** to make it the Source fixture. Group Edit uses the Source fixture's current step values as its starting values before writing changes to the matching participating fixtures.

Use the **Palettes** toolbox when the selected step should become a reusable fragment. Empty palette slots save exactly the selected step's stored values, not the whole chase and not all DMX channels. Filled palette slots recall compatible palette values into the selected step and make the palette's stored controls the active participating-control scope. Other values already stored in the step remain stored, but **Edit Step** focuses on the recalled palette controls. If no step is selected, recalling a palette creates a new selected step from that palette.

To capture a new step from the Fixture Controller:

1. Open **Fixture Controller** in another browser tab or window.
2. Build the look there by moving controls, recalling a scene, or recalling a palette.
3. Return to **Chaser**.
4. Make sure **Participating Controls** contains the controls you want to capture.
5. In the **Chase Steps** toolbox, click **Capture + Add**.

Capture + Add creates a new step and copies the current Fixture Controller live values for the selected participating controls.

To capture into an existing step:

1. Click the step in the **Chase Steps** list.
2. Build the wanted look in **Fixture Controller**.
3. Return to **Chaser**.
4. In **Edit Step**, click **Capture from FC**.

Capture from FC updates the selected step with the current Fixture Controller live values for the selected participating controls.

Capture From Fixture Controller

Use **Capture from FC** or **Capture + Add** to read the current Fixture Controller live values and use them as chase step values.

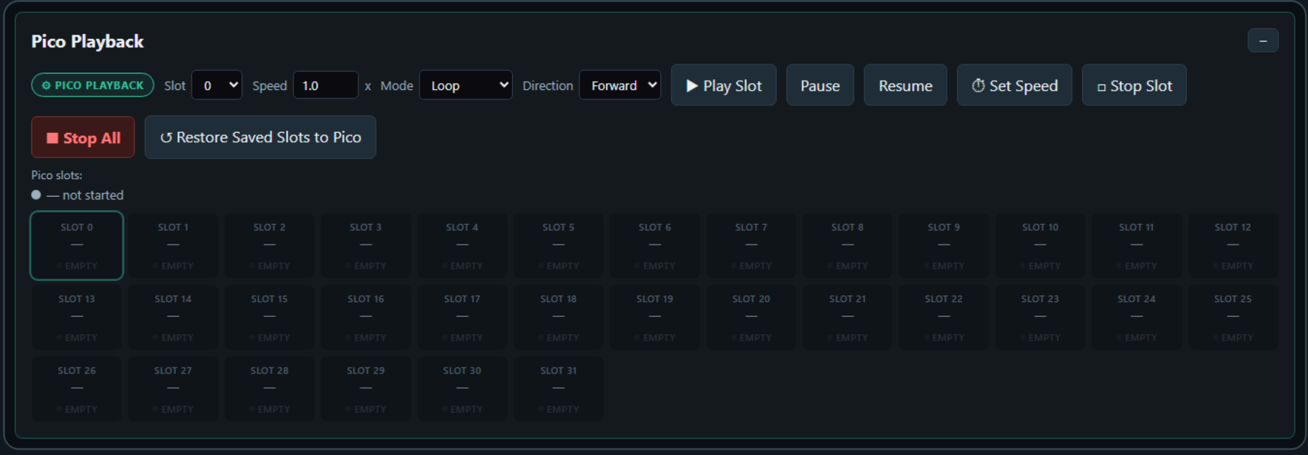
This is useful when you want to build a chase visually:

1. Create a look on the Fixture Controller.
2. Capture it into the Chaser.
3. Change the look.
4. Capture the next step.

If Chaser reports that no Fixture Controller values are available, open the Fixture Controller page and move a control or recall a scene first. Capture only reads controls that are selected as participating controls, so unselected controls are ignored.

Pico Slot Playback

Chasers can be uploaded to 32 Pico slots. Each slot can run on the Pico without the browser staying open. The Pico slot strip is the upload target: click an empty slot to upload the current chase. Click a loaded slot once to select it for play, pause, resume, speed, or stop. Click the selected loaded slot again if you want to replace it with the current chase; the page asks before overwriting.



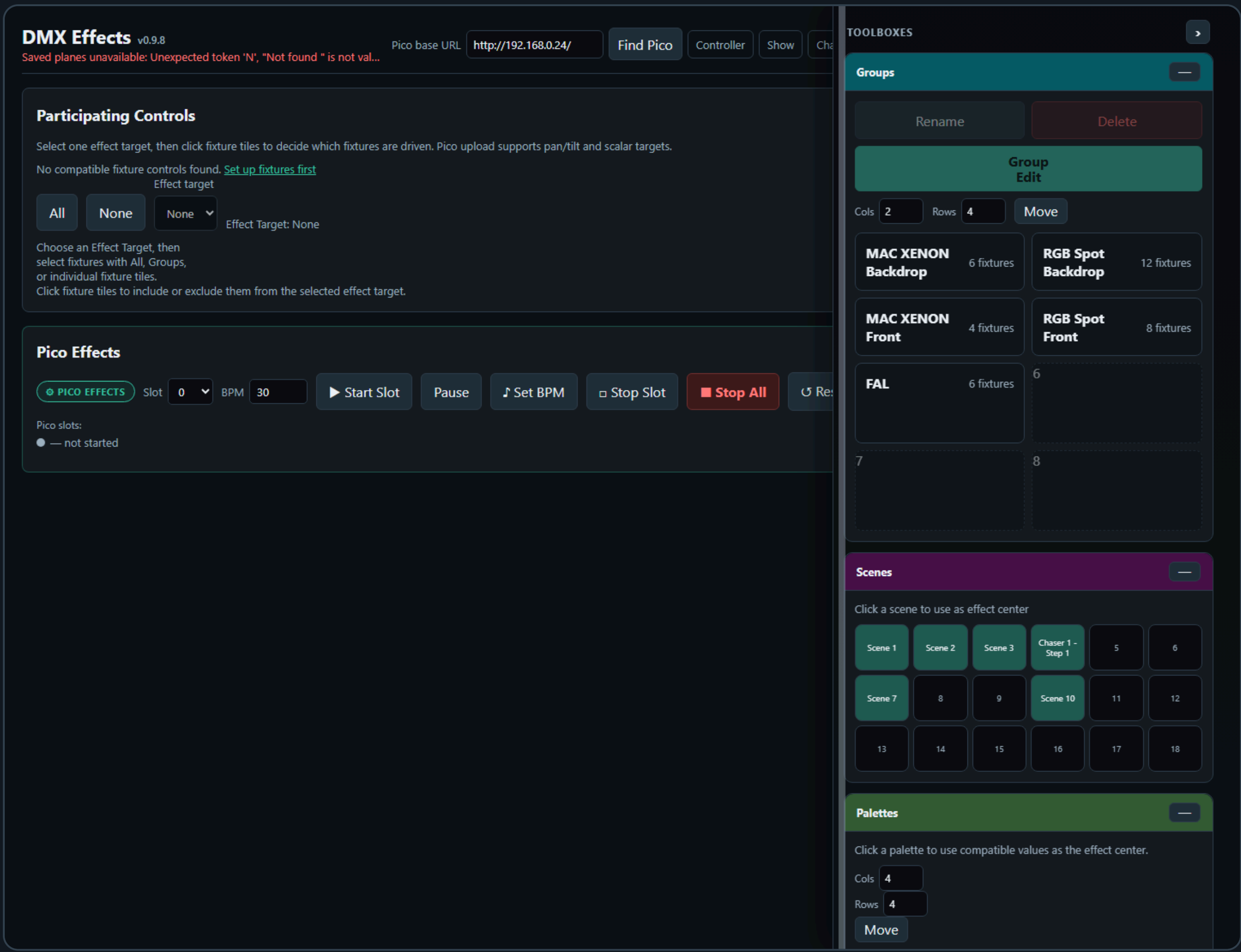
Supported playback options:

- Single run
- Loop
- Loop N times
- Ping Pong
- Forward or reverse direction
- Speed multiplier
- Pause and resume

Direction sets the first playback direction for the slot. In Ping Pong mode the Pico reverses direction whenever it reaches either end of the chase.

Each chaser slot supports up to 32 steps.

6. Effects



Effects creates continuous effects for one selected effect target at a time. It can drive a combined pan/tilt target, or one scalar DMX target such as dimmer, prism, gobo, zoom, or iris.

Supported effects include:

- Pan/tilt targets: Circle, Figure-8, Pan swing, Tilt swing
- Scalar targets: Sine, Pulse

Pan/tilt is treated as one combined two-axis target. Pan/tilt effects are relative to the current scene position, so the effect moves around the position that was last written into the Pico base buffer. Scalar controls are one-axis targets and use their displayed center value plus the **Amplitude** slider as the effect depth. Browser playback is only an overlay for testing; when you press **Stop**, Effects restores the stored base values to the output so the last moving effect position does not become the next center.

Participating Controls

The **Participating Controls** panel uses an **Effect target** dropdown. The default target is **None**. With **None** selected, no fixture tiles are enabled, Group Edit is disabled, and no effect can be played or uploaded.

Hard reload, including Ctrl+F5, resets **Effect target** to **None** and clears playback participation. Normal navigation away from Effects and back in the same browser tab restores the current working target, fixture participation, and parameters from session state. The saved server preset is not auto-applied on page load; use **Load**, import an Effects JSON file, or recall a saved **Effect** tile when you want to explicitly restore saved target and participant data.

One effect can only target one control type at a time: either pan/tilt, or one scalar control type. Choosing an effect target filters the fixture matrix to compatible fixtures, but it does not automatically enable those fixtures. This keeps target choice separate from fixture participation and prevents mixed targets such as dimmer plus gobo plus pan/tilt in one effect.

The **Participating Controls** card can be collapsed when you only need the toolboxes. In collapsed mode it stays compact and does not change the sticky page-header height.

The fixture matrix is a selection and preview surface:

- Click a fixture tile to include or exclude it from the effect.
- Use **All** to clear any group filter and enable every fixture for the current target.
- Use **None** to disable every visible fixture for the current target.
- Selecting groups applies an additional filter and enables compatible fixtures inside the selected groups.
- Pan/tilt targets show a small XY plot with the current position.
- Scalar targets show a small value bar with the current value.

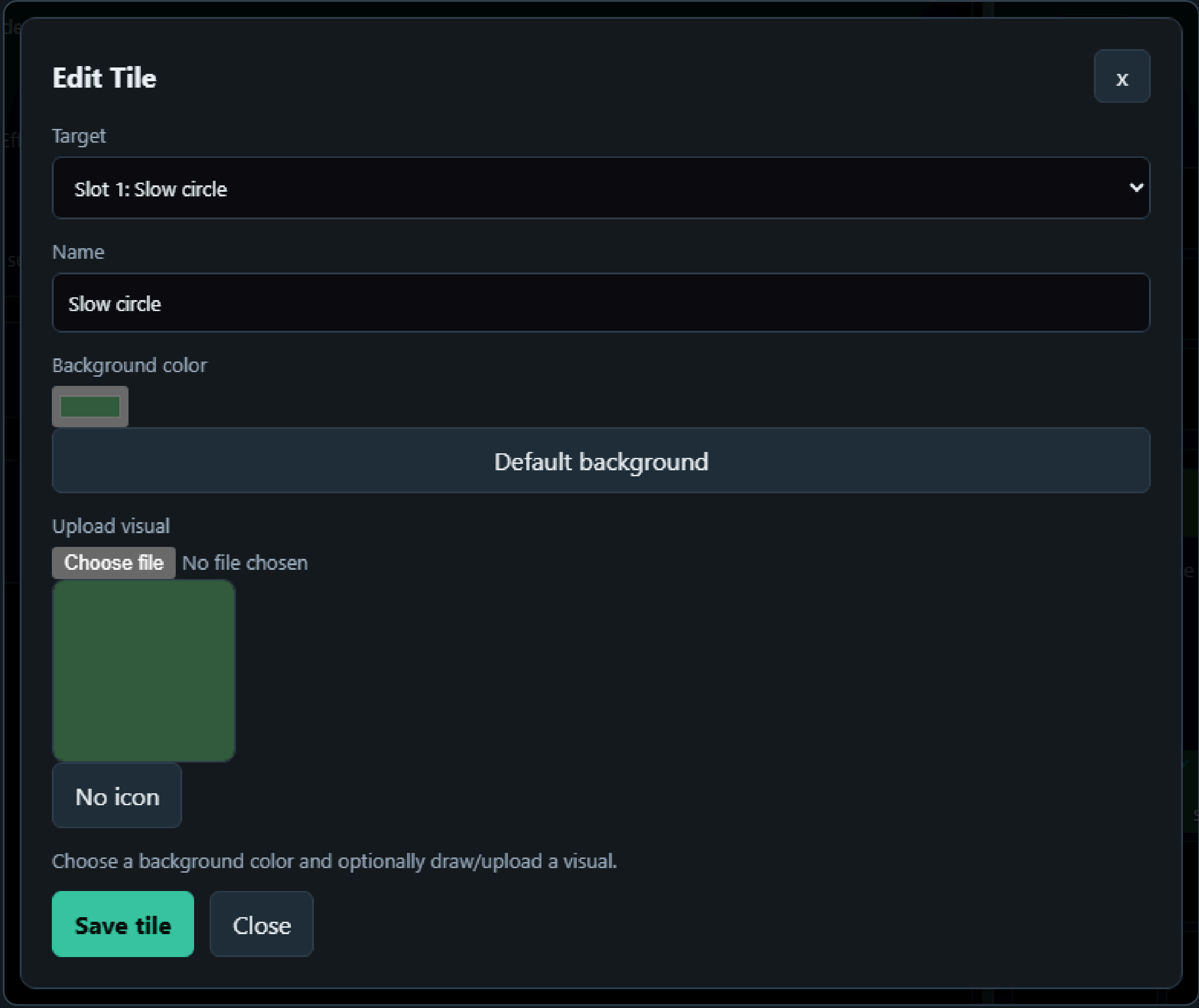
Center values come from the current base buffer or from recalling a scene as the effect center. Phase spread, amplitude, BPM, and effect shape are set in the **Effect Parameters** toolbox. The amplitude controls are target-aware and effect-aware. Circle and Figure-8 show **Pan amp** and **Tilt amp**. Pan Swing shows only **Pan amp**. Tilt Swing shows only **Tilt amp**. Scalar effects show one **Amplitude** slider. Hidden axes are forced to zero for preview and Pico upload, but their last two-axis values are remembered when you return to Circle or Figure-8.

The **Effect** dropdown is target-aware. It only shows effects that make sense for the selected **Effect target**.

The same target rules are used for Pico upload. Pan/tilt and scalar effects can be uploaded to one of the Pico effect slots, and the Pico reads the effect center from its base buffer while playing. Pan/Tilt profile mapping is included in the uploaded target: swapped axes use the swapped physical channels, and reversed axes invert the effect offset around the current base value.

The Effects page also includes the shared **Palettes** toolbox. Clicking a palette recalls any values that are compatible with Effects and uses them as the current effect center. For example, a position palette can set pan/tilt centers, while a dimmer or beam palette can set scalar centers. The small pencil opens **Edit Tile** so palette names and visuals can be adjusted from Effects too. Effects recalls, imports, exports, and saves the shared palette JSON.

The **Effects** toolbox stores reusable effect recipes. Click an empty effect slot to save the selected Effect target, participating fixtures, effect type, BPM, amplitudes, spread, and phase offsets. Effects do not store the current center/base values, so the same saved effect can be reused with different scene or palette centers. Clicking a saved effect recalls the recipe without sending DMX and without uploading to a Pico slot. The small pencil opens **Edit Tile** for the effect name, background, and optional drawing/upload.



Effect Parameters and **Effects** share one toolbox color and collapse together. Use **-- all** on either box to collapse the whole effect group, and **+ all** to reopen it.

Pico Effects Slots

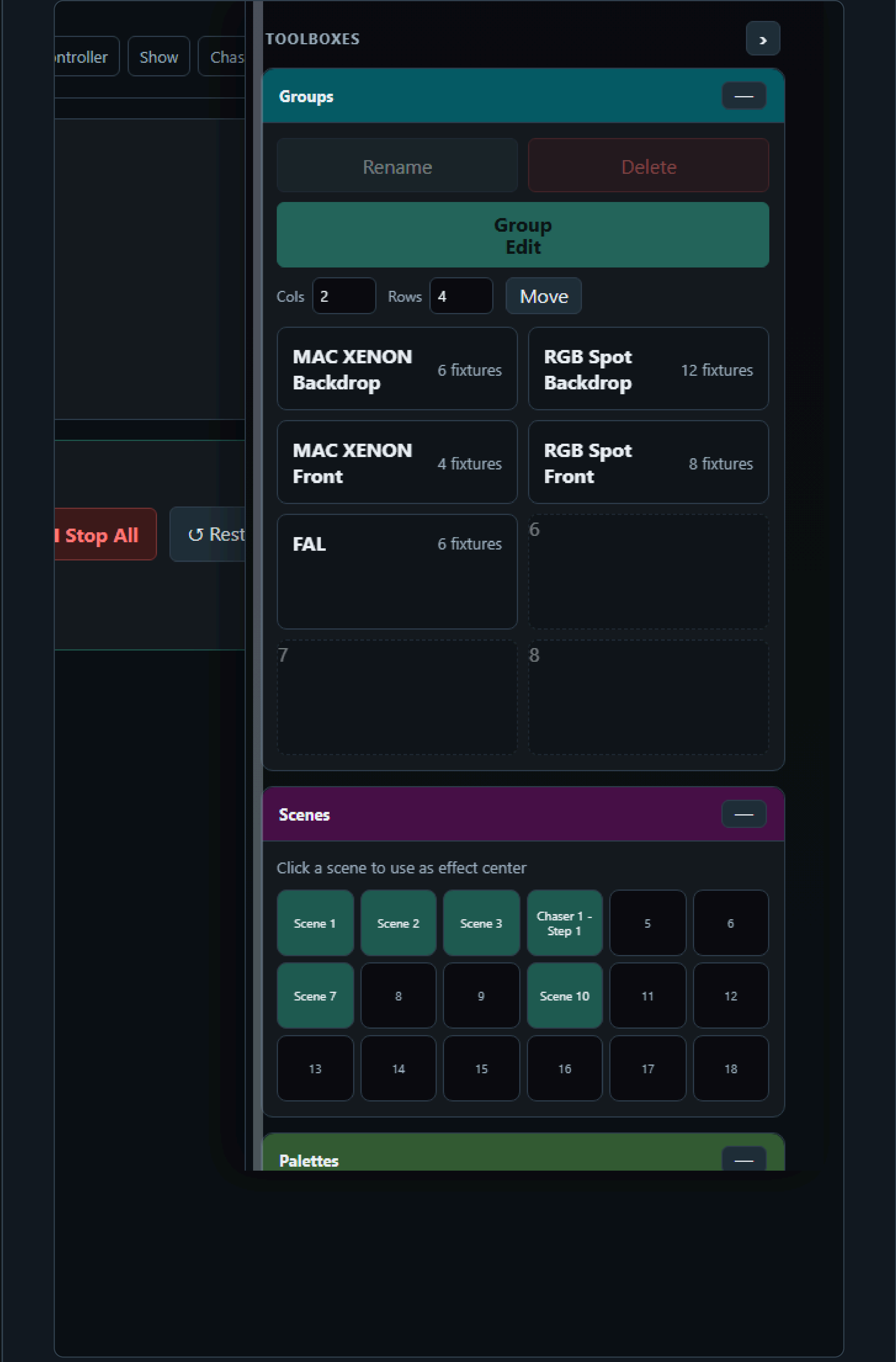
The Pico effect slot upload uses the same selected **Effect target** as the browser page.

- If the target is pan/tilt, the uploaded slot stores pan and tilt channel addresses and plays two-axis effects.
- If the target is pan/tilt, one-axis effects still store the pan/tilt channel addresses, but unused amplitude axes are uploaded as zero: Pan Swing uses `AMP1` and zero `AMP2`; Tilt Swing uses zero `AMP1` and `AMP2`.
- If the target is scalar, the uploaded slot stores that one control's DMX channel address and plays one-axis effects such as sine or pulse. Scalar uploads use `AMP1` for **Amplitude** and force `AMP2` to zero.
- Click an empty Pico slot to upload the current effect to that slot.
- Click a loaded slot once to select it for start, stop, or BPM changes.
- Click the selected loaded slot again to replace it with the current effect; the page asks before overwriting.
- Slots store channel mappings, BPM, amplitude, spread, effect type, and target phase. They do not store fixed center values.
- The center value is read from the Pico base buffer during playback. This means a scene recall or live controller change can define the center before the slot starts.
- Up to 64 effect slots can be loaded on the Pico.

For scalar targets, set the current value first, then upload/start the slot. For example, set a dimmer to its desired base brightness and use Sine if you want the Pico to pulse above and below that base value.

Effects Toolboxes

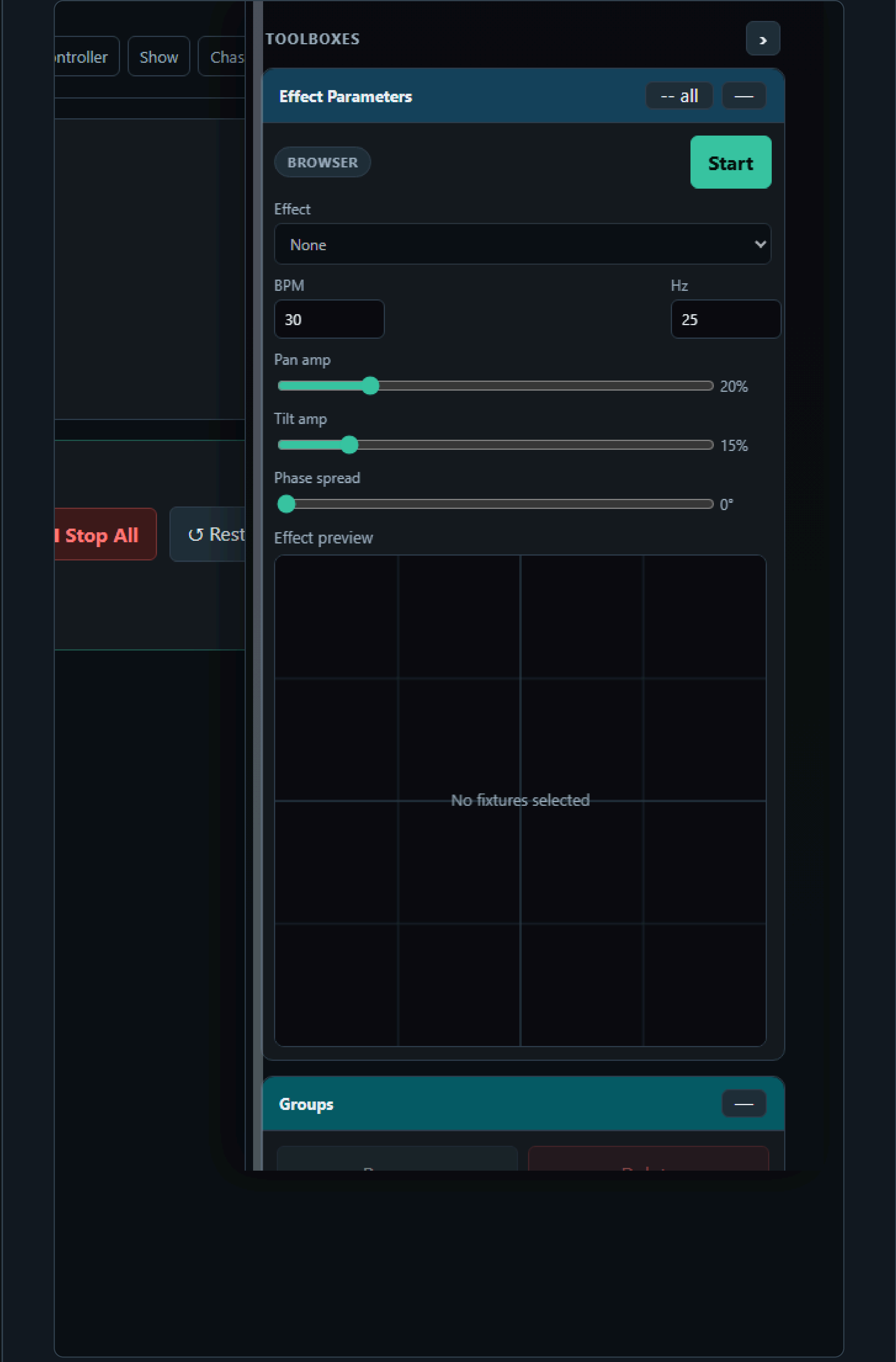
The Effects page uses five toolboxes in the shared sidebar.


ntroller
Show
Chas
Stop All
↺ Rest
TOOLBOXES
Groups
Rename
Delete
Group Edit
Cols 2
Rows 4
Move
MAC XENON Backdrop 6 fixtures
RGB Spot Backdrop 12 fixtures
MAC XENON Front 4 fixtures
RGB Spot Front 8 fixtures
FAL 6 fixtures
6
7
8
Scenes
Click a scene to use as effect center
Scene 1
Scene 2
Scene 3
Chaser 1 - Step 1
5
6
Scene 7
8
9
Scene 10
11
12
13
14
15
16
17
18
Palettes

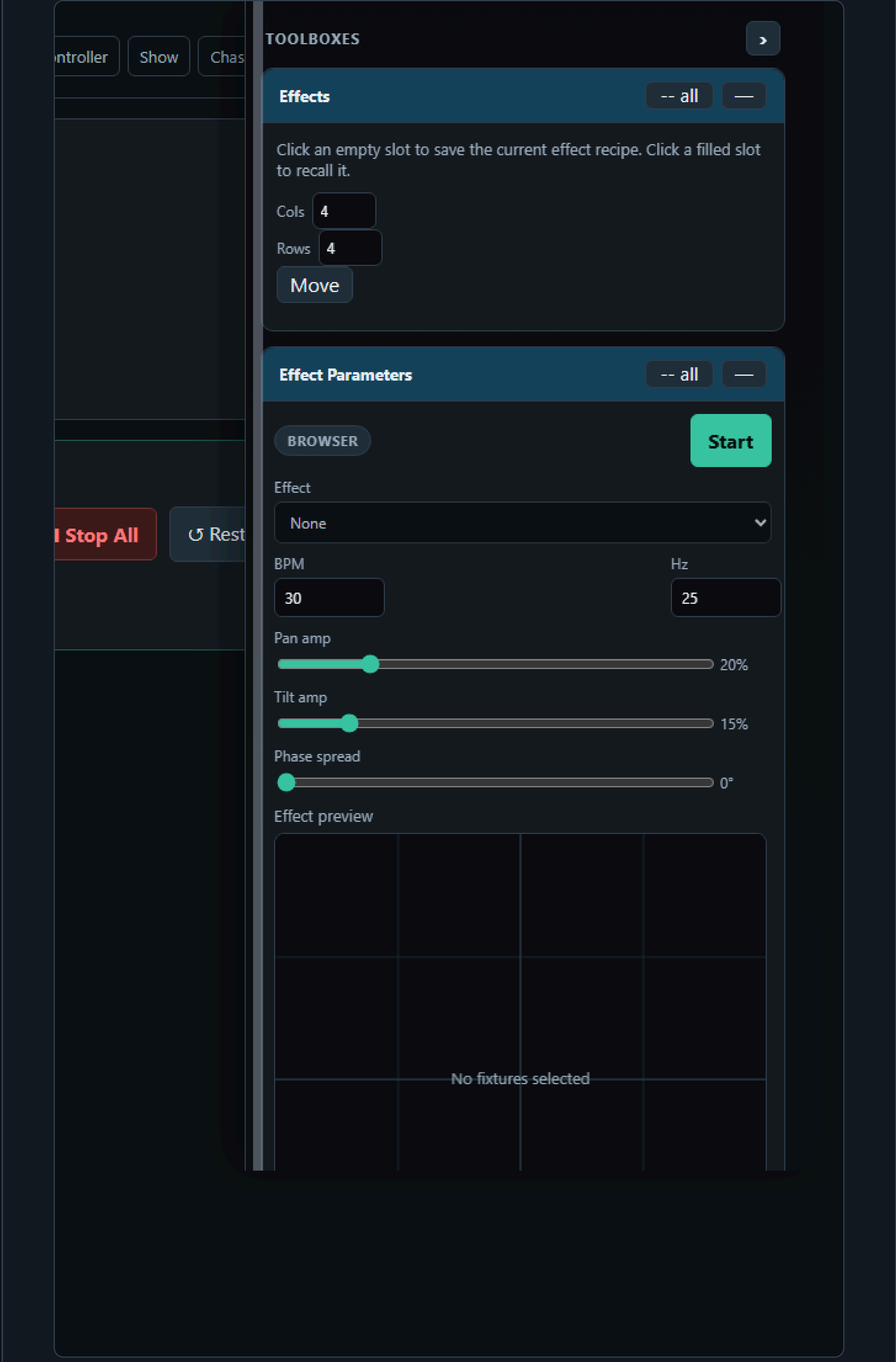
Groups filters the fixture matrix for the selected effect target. When **Effect target** is **None**, Group Edit is disabled. Choosing a real target does not automatically enable fixtures for playback, but it does make Group Edit available as soon as at least one fixture has that target. For example, after a hard reload, choosing **Dimmer** lets Group Edit work across every MAC and RGB Spot fixture that has a Dimmer control while playback participation remains off. Choosing **Pan/Tilt** can also open Group Edit for a single moving light. Pressing **All** clears the group filter and enables every fixture available for the current target. Selecting one or more groups enables compatible fixtures inside those groups. If some fixtures are already enabled, Group Edit uses that enabled subset; if none are enabled yet, Group Edit uses all fixtures compatible with the selected target.

The Effects fixture grid marks the current **Source** fixture with the same highlighted selection language used by the Controller. The Source fixture supplies the values shown in the Group Edit modal. Click another enabled fixture tile to make it the Source; click the current Source tile again when you want to remove that fixture from participation.

Effects Group Edit uses the Controller-style controls: pan/tilt edits use the XY pad for absolute center placement and relative nudge rows for movement from the current values. The modal no longer exposes separate absolute Pan/Tilt sliders. Use **Pan coarse relative**, **Pan fine relative**, **Tilt coarse relative**, and **Tilt fine relative** for 16-bit moving lights when you want to move one fixture or a selected group without forcing every fixture to the same absolute value. Relative nudges keep each fixture offset from its own current center, then send the changed center values to the Pico when a Pico base URL is set.

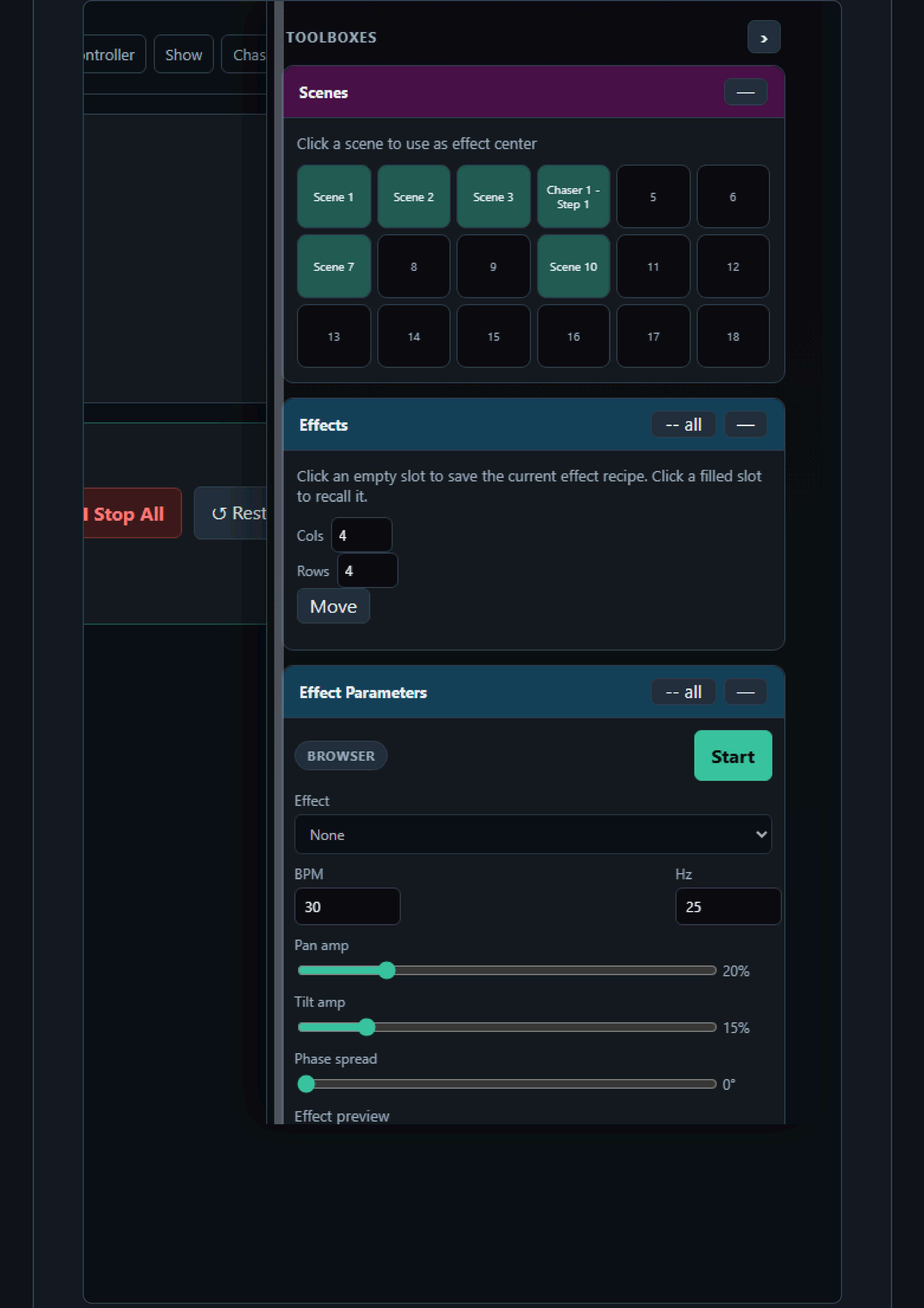

ntroller
Show
Chas
TOOLBOXES
›
Effect Parameters
-- all
—
BROWSER
Start
Effect
None
BPM
30
Hz
25
Pan amp
20%
Tilt amp
15%
Phase spread
0°
Effect preview
No fixtures selected
Stop All
↺ Rest
Groups
—

Effect Parameters is the live effect editor. It contains the target-aware effect dropdown, BPM, amplitude, phase spread, browser preview controls, and the Pico slot upload/play controls. The shown amplitude controls follow the selected effect: two-axis effects show both axes, Pan Swing and Tilt Swing show only the moving axis, and scalar targets show one **Amplitude** control.



ntroller
Show
Chas
Stop All
↺ Rest
TOOLBOXES
Effects
-- all
Click an empty slot to save the current effect recipe. Click a filled slot to recall it.
Cols 4
Rows 4
Move
Effect Parameters
-- all
BROWSER
Start
Effect
None
BPM
30
Hz
25
Pan amp
20%
Tilt amp
15%
Phase spread
0°
Effect preview
No fixtures selected

Effects stores reusable effect recipes. It saves the selected target, fixture participation, effect type, BPM, amplitude, spread, and phase offsets. It does not store fixed center values.


ntroller
Show
Chas
TOOLBOXES
Scenes
Click a scene to use as effect center
Scene 1
Scene 2
Scene 3
Chaser 1 - Step 1
5
6
Scene 7
8
9
Scene 10
11
12
13
14
15
16
17
18
Stop All
↺ Rest
Effects
-- all
Click an empty slot to save the current effect recipe. Click a filled slot to recall it.
Cols 4
Rows 4
Move
Effect Parameters
-- all
BROWSER
Start
Effect
None
BPM
30
Hz
25
Pan amp
20%
Tilt amp
15%
Phase spread
0°
Effect preview

Scenes is read-only on Effects. Recalling a scene updates the effect center/base values used by effects. When a Pico base URL is set, the stored values are also sent to the Pico.

ntroller
Show
Chas

Stop All
↺ Rest

TOOLBOXES

Palettes

Click a palette to use compatible values as the effect center.

Cols 4

Rows 4

Move

Step 1 palette	Step 1 palette	Step 3 palette	Step 3 palette st...
5	6	7	8
9	10	11	12
13	14	15	16

Scenes

Click a scene to use as effect center

Scene 1	Scene 2	Scene 3	Chaser 1 - Step 1	5	6
Scene 7	8	9	Scene 10	11	12
13	14	15	16	17	18

Effects

-- all

Click an empty slot to save the current effect recipe. Click a filled slot to recall it.

Cols 4

Rows 4

Palettes recalls compatible palette values into Effects. A position palette can set pan/tilt centers, while color, dimmer, or beam palettes can set scalar centers when the selected effect target matches.

ntroller Show Chas

I Stop All ↺ Rest

TOOLBOXES ›

Planes —

Recall a saved room plane as the effect center for calibrated pan/tilt fixtures.

Cols 3

Rows 3

Move

1	2	3
4	5	6
7	8	9

Palettes —

Click a palette to use compatible values as the effect center.

Cols 4

Rows 4

Move

Step 1 palette	Step 1 palette	Step 3 palette	Step 3 palette st...
5	6	7	8
9	10	11	12
13	14	15	16

Scenes —

Planes recalls saved room planes as the current pan/tilt effect center. Select **Pan/Tilt** as the effect target, optionally select one or more groups, then click a filled plane tile. Effects interpolates the calibrated pan/tilt values for matching fixtures, enables those fixtures for the current target, sends the new center values to the Pico, and keeps the effect recipe itself separate from the recalled center. The Planes toolbox uses the same **Cols**, **Rows**, and **Move** tile behavior as Groups, Palettes, and Effects.

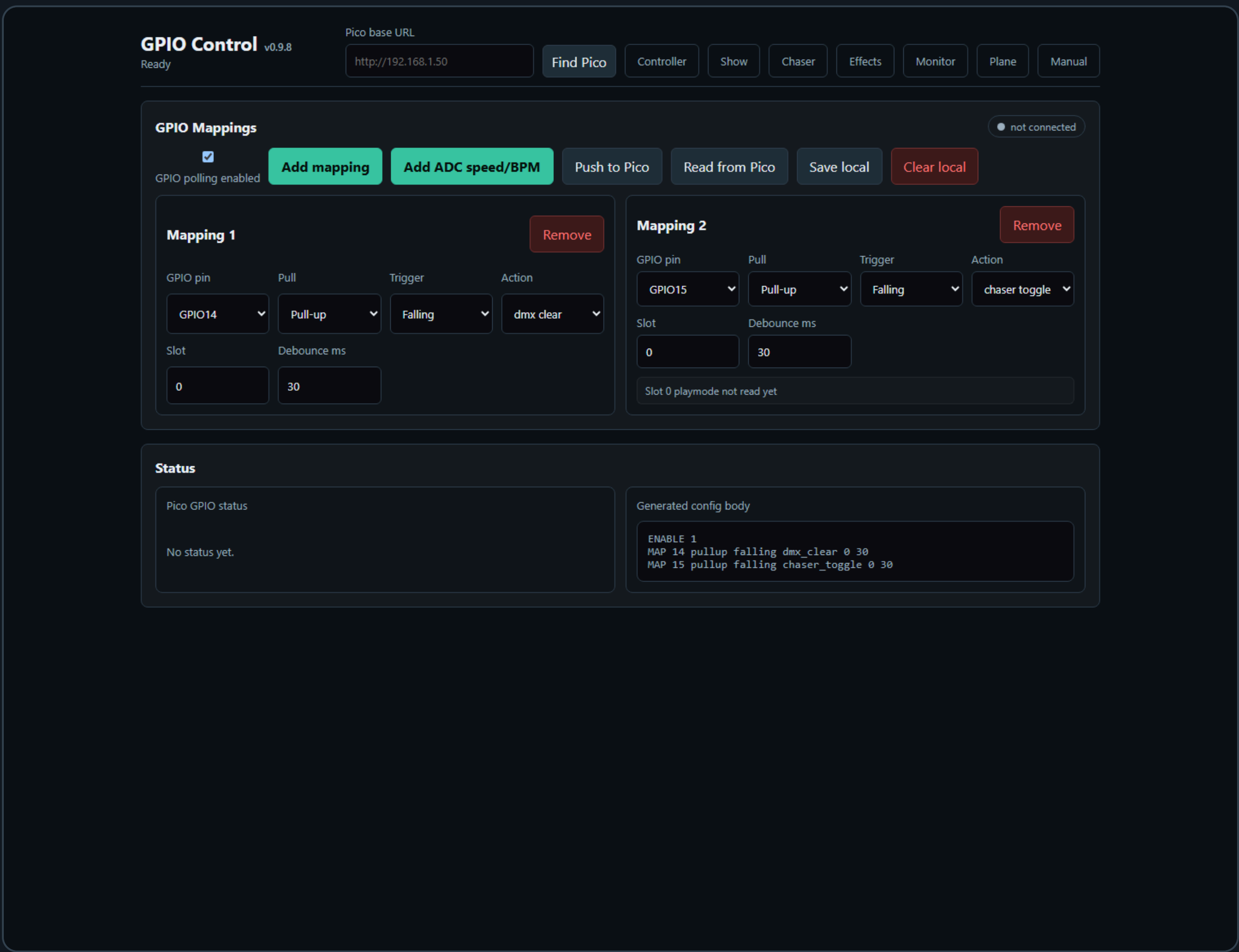
Recommended Workflow

1. Use the Fixture Controller to position the fixtures.
2. Save or recall a scene.
3. Open Effects.
4. Select one **Effect target**.
5. Set BPM, effect shape, amplitude, and spread in the **Effect Parameters** toolbox.
6. Optionally recall a palette from the **Palettes** toolbox to set the center for the selected target.
7. Optionally save or recall the recipe from the **Effects** toolbox.
8. Click an empty Pico slot to upload the effect, then start the slot when you want autonomous playback without browser timing jitter.
9. Use browser playback from the **Effect Parameters** toolbox when you want quick live testing before uploading.

The Effects page also has a read-only scene toolbox. Clicking a scene updates the effect center. When a Pico base URL is set, it also sends the position to the Pico.

---

7. GPIO Control



GPIO Control maps physical Pico inputs to lighting actions.

GPIO Control does not use the shared toolbox sidebar. Its setup is kept in normal page panels because GPIO mapping is configuration work, not a live fixture/scene/palette workflow.

The GPIO editor loads its mapping setup from the XAMPP server first, using `gpio_setup.php` and `data/gpio_setup.json`. Browser storage is only a fallback if the server file is not available. Adding, removing, or changing a mapping autosaves the setup back to the server, so a PC and iPad should show the same mappings after reload.

Chaser GPIO actions do not define their own playmode. They start, stop, pause, resume, toggle, or tap the selected Pico chaser slot. The playmode belongs to that slot, so the button follows whatever was uploaded from the Chaser page: **Single**, **Loop**, **Loop N**, or **Ping Pong**, plus forward/reverse direction. When a Pico base URL is set, the GPIO page reads the chaser slot status and shows the slot mode, direction, loop state, step count, and live/ready state beside chaser mappings and chaser speed ADC mappings.

Digital GPIO pins can trigger:

- DMX clear
- Output-only clear
- Stop all
- Chaser play, stop, toggle, pause, resume, pause toggle
- Chaser tap tempo
- Effects start, stop, toggle
- Effects tap tempo

ADC pins can control:

- Chaser speed multiplier
- Effects BPM

Only GPIO26, GPIO27, and GPIO28 support ADC input on the Pico 2 W.

Add a Digital Button Mapping

1. Open **GPIO Control**.
2. Click **Add mapping**.
3. Select a free GPIO pin.
4. Choose pull mode and trigger edge.
5. Choose the action.
6. Set the slot number if the action needs one.
7. Upload the config to the Pico.

The page disables reserved pins and pins already used by other mappings.

Editing the mapping saves the setup on the XAMPP server, but the Pico only receives it after **Push to Pico**. Use **Read from Pico** when the Pico already has the mapping you want to bring back into the editor.

Add an ADC Mapping

1. Add an ADC mapping.
2. Select GPIO26, GPIO27, or GPIO28.

3. Choose `chaser_speed` or `motion_bpm`.
4. Set the target slot.
5. Set the min and max value.
6. Upload the config.

ADC values are filtered on the firmware side with a short mean filter to reduce ripple.

Tap Tempo

Tap actions use the time between button presses.

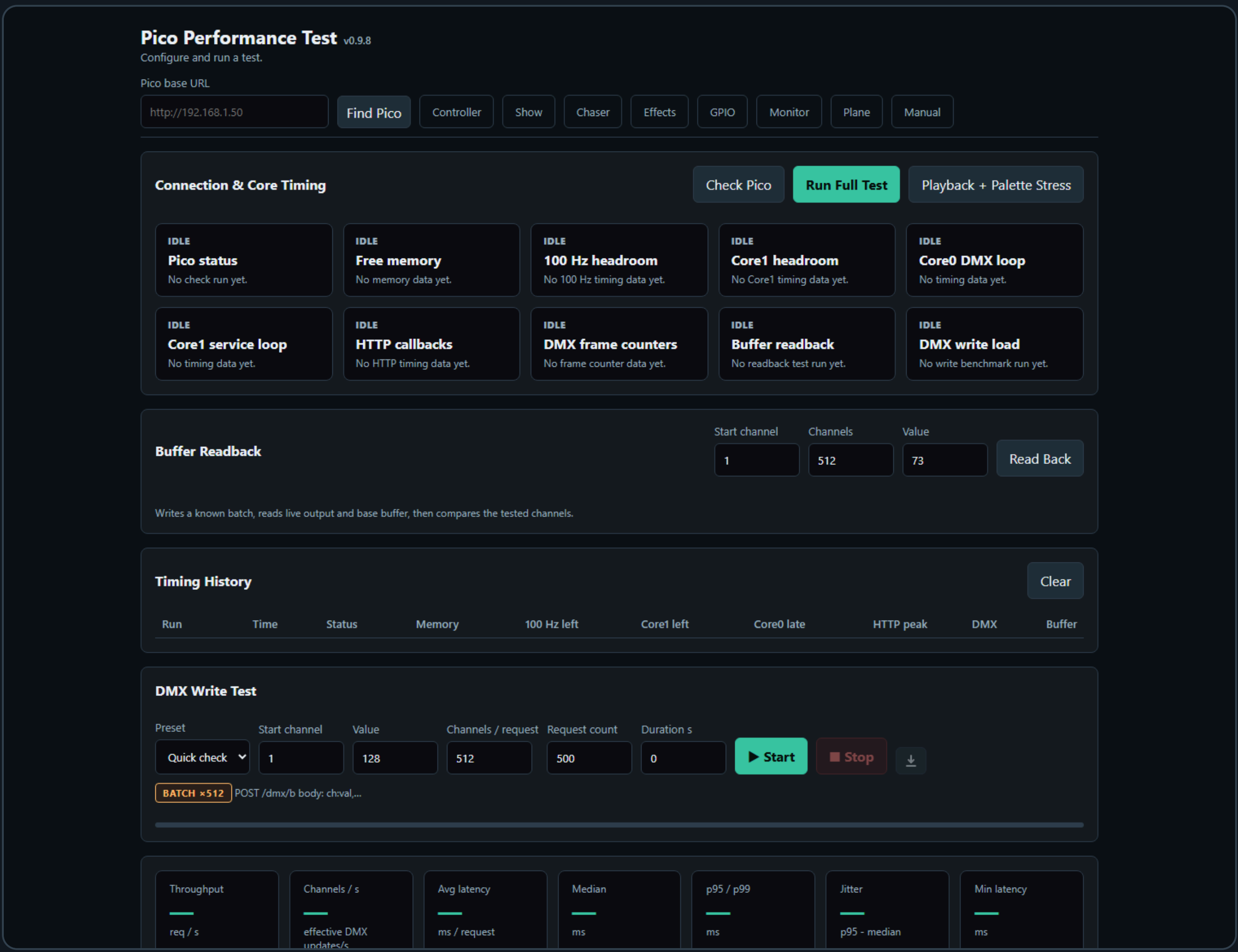
The beat divider can be set to:

- 1 beat
- 1/2 beat
- 1/4 beat
- 1/8 beat
- 1/16 beat

The firmware ignores very long gaps between taps so stopping for a while does not create an extremely slow tempo when tapping resumes.

---

8. Pico Performance Test



The Pico Performance Test page checks the full browser-to-Pico control path.

Pico Performance Test does not use the shared toolbox sidebar. It is a developer/test page for measuring request performance, buffer readback, and firmware timing health.

Use **Check Pico** to read Pico status and firmware performance telemetry. Current firmware exposes `/perf/status.json`, which reports free RAM, Core0 100 Hz playback-loop timing, Core1 service-loop timing, HTTP callback timing, and DMX frame counters. Older firmware falls back to parsing `/logs.txt`.

The **Free memory** check shows the firmware heap/stack gap. The **100 Hz headroom** check shows the minimum time left before the Core0 playback loop would miss its 10 ms update budget. The **Core1 headroom** check shows the minimum time left before the network/service loop would miss its 2 second service budget. Healthy runs should show no late Core0 or Core1 cycles and comfortable remaining slack.

Use **Buffer Readback** to write a known batch with `/dmx/b`, then compare the tested channels from both `/dmx/output.json` and `/dmx/base.json`.

Use **DMX Write Test** to compare:

- Single-channel updates
- Scene-sized batch updates
- Large batch updates
- Longer soak tests

The write-test result panel shows:

- Throughput
- Effective DMX channel updates per second
- Average latency
- Median latency
- p95 and p99 latency
- Jitter
- Errors

Use **Run Full Test** to run the Pico status/performance check, buffer readback, write test, and a final timing check in one sequence.

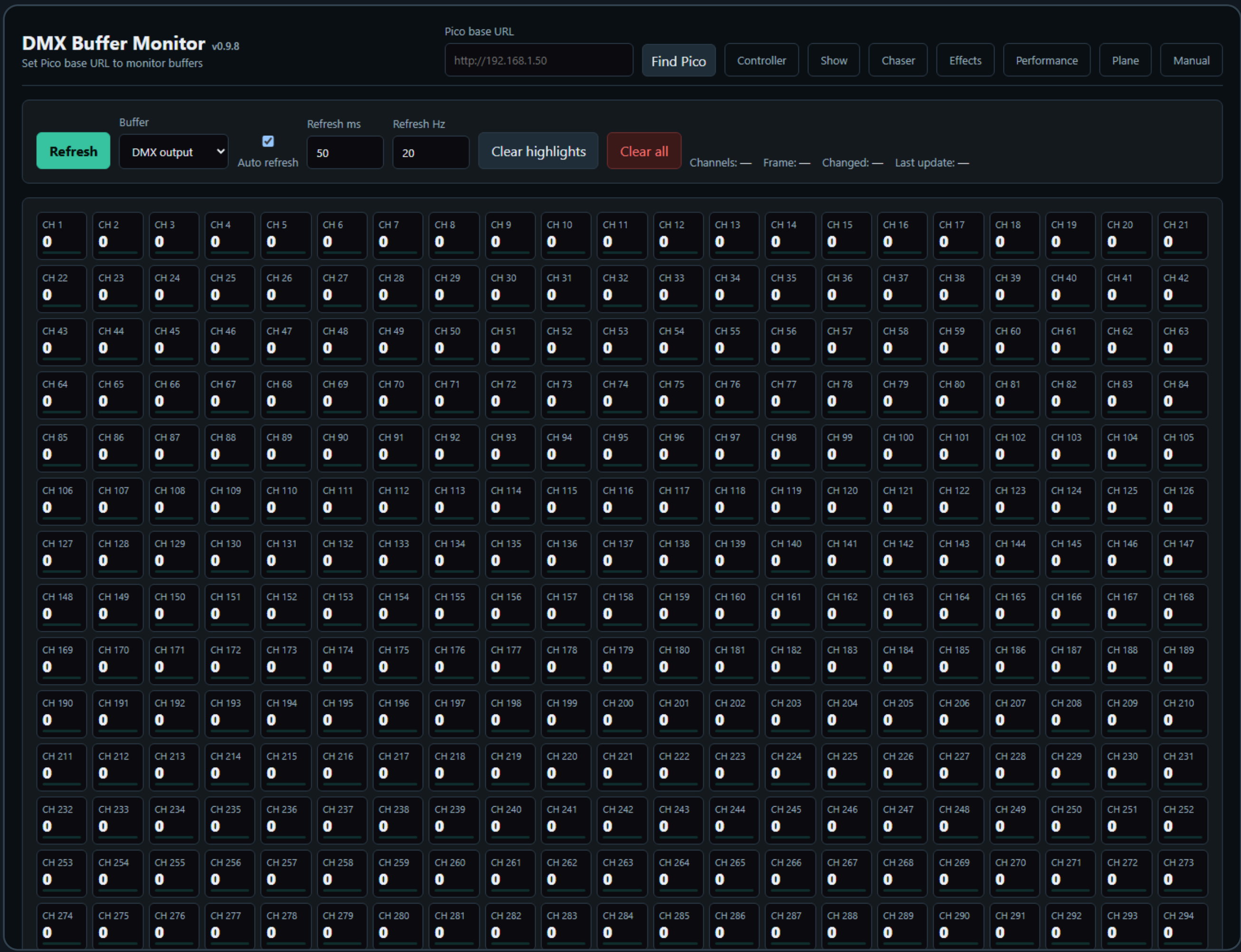
Use **Playback + Palette Stress** to stress playback without overwriting saved Pico slots. The page starts chaser/effect slots that are already loaded, adds temporary demo data only to slots that are currently empty, stores those temporary slot numbers in the server UI state, then sends repeated full 512-channel palette-style `/dmx/b` recalls while playback is running. When the run finishes, the page stops playback, clears only the temporary demo slots, and removes the temporary-slot marker from the server. If a browser session is interrupted, the next stress run first reads the marker and clears the previously recorded temporary slots.

The **Timing History** table records each Pico timing check. A manual **Check Pico** adds one row immediately. **Run Full Test** and **Playback + Palette Stress** record a final timing sample, so the row reflects Core0/Core1 slack after the selected checks have run. The 100 Hz and Core1 columns use the same wording as the status cards, for example "Minimum 9419us left before missing the 10ms update budget".

Use **Export CSV** to save results for later comparison.

Pico Performance Test and GPIO Control link to the **DMX Buffer Monitor**.

9. DMX Buffer Monitor



The monitor displays all 512 channels as tiles and reads one Pico buffer at a time.

Use **DMX output** to see the actual output frame currently held by the DMX engine, including values produced by Pico-side Chaser or Effects playback.

Use **Base / position** to inspect the scene base buffer used as the center for Effects.

Use **Refresh ms** to control the polling interval directly. Use **Refresh Hz** when you prefer rate instead. Both fields stay synchronized, so `500 ms` is shown as `2 Hz`. Longer intervals are calmer for observation; shorter intervals are useful when checking fast changes. **Auto refresh** starts or stops polling without changing the selected buffer.

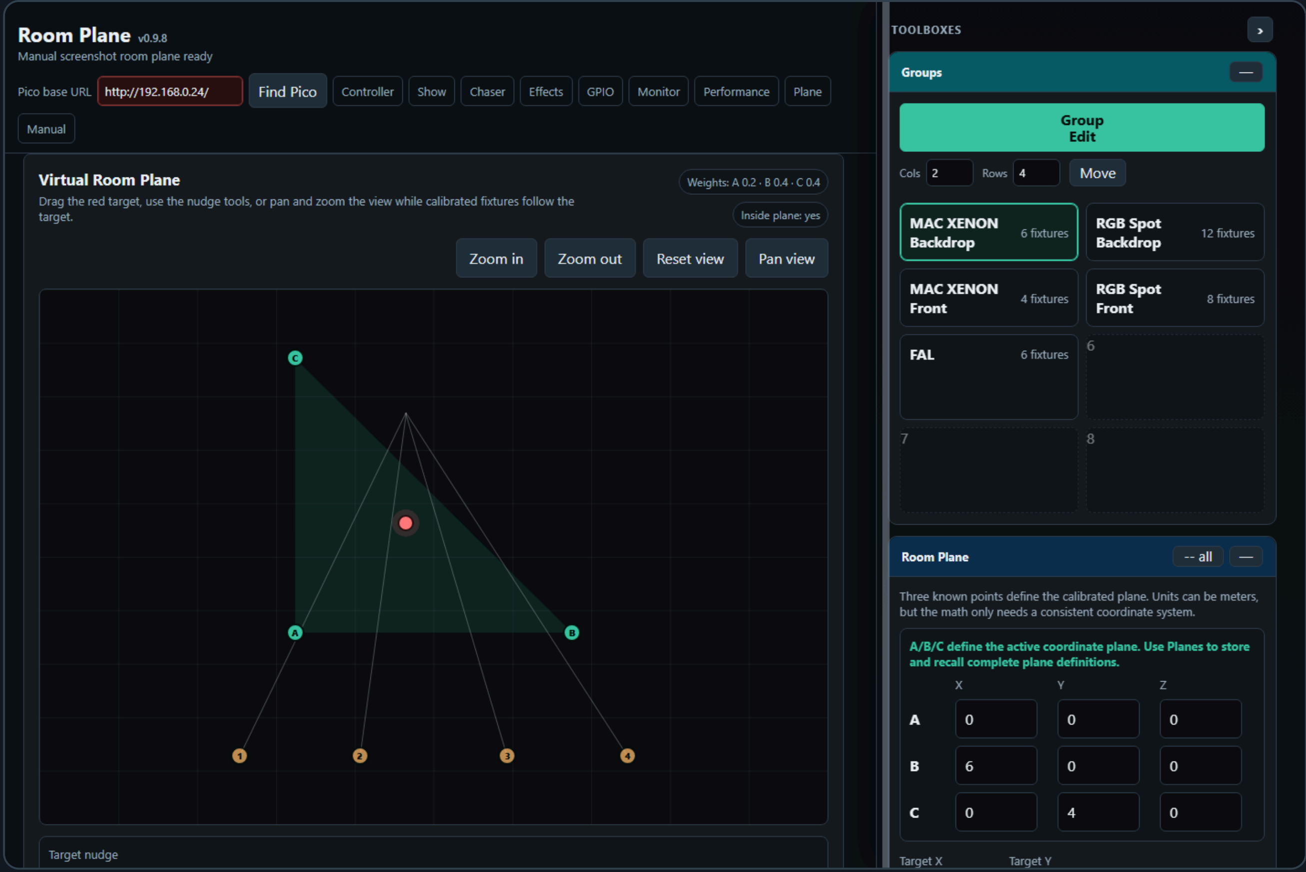
Use **Clear all** when you want to clear both Pico buffers from the monitor page. It calls the Pico clear endpoint immediately, clears the DMX output buffer and the base/position buffer, and then refreshes the displayed tiles.

10. Room Plane

The **Room Plane** page is a calibration page for moving-light position mapping. It stores saved room planes, fixture mount positions, and A/B/C calibration values on the server.

Open it from the **Plane** navigation link or directly:

```
dmx_room_plane.html
```



The idea is to describe a real stage or room as a 2D coordinate plane. Points **A**, **B**, and **C** are measured points in that room. Each moving light is then calibrated by storing the pan/tilt values that hit A, B, and C. When the red target is moved in the virtual plane, the page calculates a weighted pan/tilt value for every selected fixture and sends the resulting DMX values.

10.1 Room Plane Workflow

1. Patch the moving lights on the Fixture Controller page.
2. Save groups if you want to select fixtures by group on the Room Plane page.
3. Open **Plane**.
4. In the **Room Plane** toolbox, enter the measured coordinates for A, B, and C.
5. Use **Load patched moving lights** in the **Fixtures** toolbox, or recall a saved plane that already contains fixtures.
6. Select fixtures directly in the fixture table, or use the **Groups** toolbox to select a whole fixture group.
7. Edit one fixture at a time. Move the real moving light to point A, store A, then repeat for B and C.
8. Move the red target in the virtual room plane. The selected fixtures update automatically.
9. In the **Planes** toolbox, click an empty tile to save the current plane definition and fixture calibration.

The target can be moved by dragging the red dot, by clicking in the plane, or by using the coarse/fine target nudge buttons below the plot. The view itself can be zoomed with the zoom buttons or mouse wheel. Use **Pan view** when you want to drag the coordinate view instead of the target.

10.2 Room Plane Toolbox

mance
Plane

A 0.2 · B 0.4 · C 0.4
Inside plane: yes
Pan view

TOOLBOXES

Room Plane

-- all

Three known points define the calibrated plane. Units can be meters, but the math only needs a consistent coordinate system.

A/B/C define the active coordinate plane. Use Planes to store and recall complete plane definitions.

	X	Y	Z
A	0	0	0
B	6	0	0
C	0	4	0

Target X: 2.4
Target Y: 1.6

Reset calibration
Demo values

Groups

Group Edit

Cols 2 Rows 4 Move

MAC XENON Backdrop 6 fixtures	**RGB Spot Backdrop** 12 fixtures
MAC XENON Front 4 fixtures	**RGB Spot Front** 8 fixtures
FAL 6 fixtures	6

The **Room Plane** toolbox contains the measured A/B/C points and the current target controls.

Field	Meaning
A, B, C X/Y/Z	Known physical points in the room. Use one consistent unit, for example meters.
Target X/Y	Numeric target position in the plane.
Reset calibration	Marks all fixture A/B/C calibration points as missing.
Demo values	Loads the small built-in demo plane and demo fixtures.

The Z value is stored with the plane and fixture mount positions. The current target interpolation uses X/Y coordinates on the plane. Z is kept so the same saved data can later support full 3D fixture-position calculations.

Use **-- all** in any of the Room Plane, Planes, or Fixtures toolbox headers to collapse those three toolboxes together. When they are all collapsed, the button changes to **+ all**.

10.3 Planes

mance
Plane

A 0.2 · B 0.4 · C 0.4
Inside plane: yes
Pan view

TOOLBOXES

Planes

-- all

A saved plane includes A/B/C room points, target/view, fixture mount positions, and fixture A/B/C calibration. Click an empty tile to save the current plane.

Cols 3 Rows 3

Main stage plane 4 fixtures · calibrated	Save current plane +	Save current plane +
Save current plane +	Save current plane +	Save current plane +
Save current plane +	Save current plane +	Save current plane +

Room Plane

-- all

Three known points define the calibrated plane. Units can be meters, but the math only needs a consistent coordinate system.

A/B/C define the active coordinate plane. Use Planes to store and recall complete plane definitions.

	X	Y	Z
A	0	0	0
B	6	0	0
C	0	4	0

Target X 2.4
Target Y 1.6

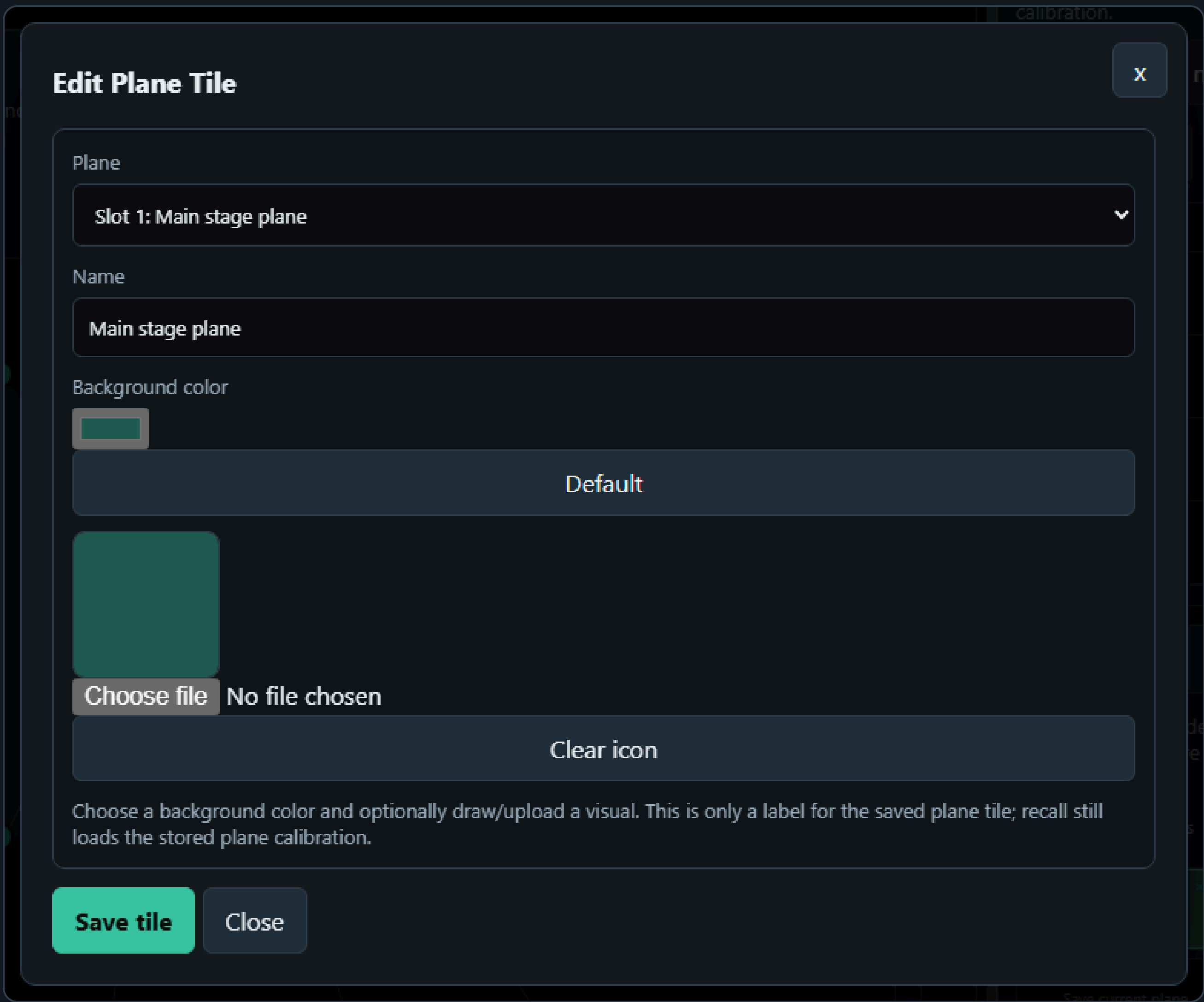
Reset calibration

Demo values

Groups

The **Planes** toolbox stores complete plane definitions as tiles, using the same visual language as scene, palette, group, and chase tiles.

Field	Meaning
Filled plane tile	Recalls that saved plane, including A/B/C points, target, view, fixtures, mount positions, and fixture calibration.
Empty `+ Save current plane` tile	Saves the current plane into that tile and opens **Edit Plane Tile** so it can be named and styled.
Pencil	Opens **Edit Plane Tile** for the selected plane tile name, background color, uploaded icon, or drawn icon.
`x`	Deletes the saved plane after confirmation.
Cols / Rows	Shapes the Planes tile matrix.



10.4 Fixture Calibration

mance | Plane

A 0.2 · B 0.4 · C 0.4

Inside plane: yes

Pan view

TOOLBOXES ›

Fixtures -- all —

Mount X/Y/Z and A/B/C pan/tilt calibration are stored inside the active saved plane. Recalling another plane recalls its own fixture calibration.

Load patched moving lights | Add fixture

Remove last

Edit	Select	Fixture	Calibration	Live Pan	Live
Edit	☑	**MAC XENON 1**	Calibrated	28400	382(
Edit	☑	**MAC XENON 2**	Calibrated	32600	374(
Edit	☐	**MAC XENON 3**	Calibrated	37100	369(
Edit	☐	**MAC XENON 4**	Calibrated	41400	365(

Planes -- all —

A saved plane includes A/B/C room points, target/view, fixture mount positions, and fixture A/B/C calibration. Click an empty tile to save the current plane.

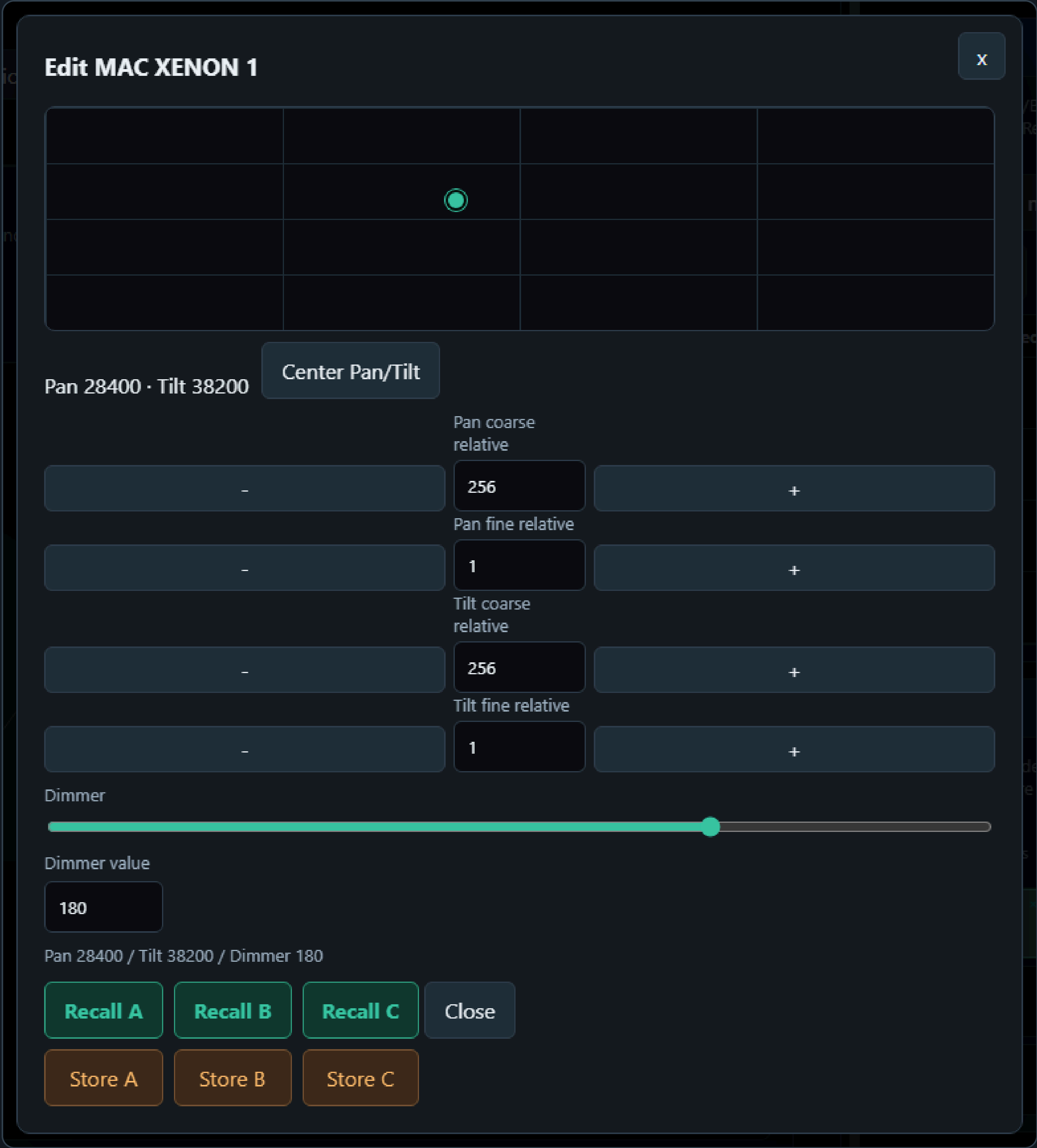
Cols 3 Rows 3

Main stage plane 4 fixtures · calibrated	Save current plane +	Save current plane +
Save current plane +	Save current plane +	Save current plane +
Save current plane +	Save current plane +	Save current plane +

Room Plane -- all —

The **Fixtures** toolbox stores the relationship between each moving light and the room plane.

Column	Meaning
Edit	Opens the pan/tilt/dimmer editor for this fixture. Only one fixture is edited at a time.
Select	Includes the fixture when the target point is applied.
Calibration	Shows whether A, B, and C have been stored.
Live Pan / Live Tilt / Dimmer	Current fixture values used by the editor and output.
Last send / channels	The last DMX channels written for this fixture.
Mount X/Y/Z	The fixture's physical mounting position in room coordinates. Today this is used for the plot, saved plane context, and future 3D calibration. It is not yet used to calculate pan/tilt output.
A/B/C pan/tilt	Stored calibration values for the three known points.



In the fixture editor, **Recall A/B/C** loads an already stored calibration point into the editor. **Store A/B/C** writes the current pan/tilt editor value into that calibration point. Store buttons are separated from recall buttons and use the warning color because they overwrite calibration data.

The editor sends live DMX for patched fixtures. If the fixture profile has a 16-bit pan/tilt control, the full 0...65535 value is used. If the fixture profile only has an 8-bit pan/tilt control, the value range is 0...255.

10.5 Coordinate Math

The three known room points define a triangle in the room plane:

```
A = (Ax, Ay)
B = (Bx, By)
C = (Cx, Cy)
P = (Px, Py)   target point
```

The page calculates barycentric weights for the target point. The determinant is:

```
D = (By - Cy) * (Ax - Cx) + (Cx - Bx) * (Ay - Cy)
```

If `D` is very close to zero, A, B, and C are on one line and the plane is invalid.

The weights are:

```
wA = ((By - Cy) * (Px - Cx) + (Cx - Bx) * (Py - Cy)) / D
wB = ((Cy - Ay) * (Px - Cx) + (Ax - Cx) * (Py - Cy)) / D
wC = 1 - wA - wB
```

The readout shows these as **Weights: A / B / C**. The target is inside the triangle when all three weights are greater than or equal to zero:

```
wA >= 0 and wB >= 0 and wC >= 0
```

The page still allows positions outside the triangle. In that case the same formula extrapolates beyond the calibrated area, and the **Inside plane** readout changes to `no`.

10.6 Pan/Tilt Interpolation

For each fixture, the stored calibration values are:

```
panA,  tiltA    values that hit point A
panB,  tiltB    values that hit point B
panC,  tiltC    values that hit point C
```

The target output is the weighted blend:

```
pan(P)  = wA * panA  + wB * panB  + wC * panC
tilt(P) = wA * tiltA + wB * tiltB + wC * tiltC
```

For 16-bit pan/tilt controls, this interpolation is done in the full 16-bit range:

```
0 ... 65535
```

The 16-bit value is then split into DMX coarse and fine bytes:

```
coarse = value >> 8
fine   = value & 255
```

For example, a pan value of `32768` becomes:

```
coarse = 128
fine   = 0
```

The final DMX channel numbers come from the fixture profile and the patched start address:

```
absoluteChannel = fixtureStart + relativeChannel - 1
```

The page sends selected fixtures as a batch update. While the target is being dragged or nudged, output is automatically applied after a short debounce so movement feels live without flooding the Pico with unnecessary requests.

10.7 Screen Transformation

The virtual room drawing uses the same room coordinates as the math, but it transforms them into screen pixels for display.

First the visible bounds are calculated from A/B/C, the target, and fixture mount positions. Then the page chooses one common scale for both axes:

```
scale = min(screenWidth / roomWidth, screenHeight / roomHeight)
```

Using one common scale is important. It means 1 meter in X and 1 meter in Y draw with the same visual length, so the room shape is not stretched on wide screens.

Room coordinates are converted to screen coordinates like this:

```
screenX = offsetX + (roomX - minX) * scale
screenY = offsetY + (maxY - roomY) * scale
```

Y is inverted because screen coordinates grow downward, while room Y grows upward in the virtual plane.

The view zoom changes the visible bounds before this transformation. The pan view changes the center point of those bounds. The saved plane stores the current view, so a recalled plane opens with the same framing.

10.8 What Is Saved

Room Plane data is saved to the server in `room_plane_setup.json`. A saved plane contains:

- A/B/C room point coordinates
- Target position
- View zoom and pan center
- Fixture list for that plane
- Fixture mount X/Y/Z
- Fixture A/B/C pan/tilt calibration
- Active saved plane ID

This means you can create multiple room planes for different physical setups or stage areas. Recalling a plane recalls its own fixture calibration.

10.9 Current Limits

The current implementation is calibration-based. It does not yet calculate pan/tilt from fixture mount position, fixture orientation, beam length, and full 3D geometry. Mount X/Y/Z is saved and drawn, but the output is currently calculated from the measured A/B/C pan/tilt values.

Use three non-collinear points. If A, B, and C are on one line, the determinant becomes zero and the page cannot calculate a valid plane.

11. Backup and Import

Use **Fixture Controller > Show > Export Setup** before large changes or before moving the show to another computer.

The exported setup file includes:

- Fixture profiles and patched fixtures
- Current live fixture values
- Groups
- Scenes
- Palettes
- Saved chases and saved Pico chaser slot payloads
- Effects settings, saved effects, and saved Pico effect slot payloads
- Saved room planes and fixture calibration
- GPIO mappings
- Custom imported fixture library catalog, if one is saved on the XAMPP server
- Toolbox layout, collapse state, grid sizes, and other saved UI state

The file also stores the project name, project version, export time, and setup format version. During import, the controller checks the setup format before writing anything to the server. Older supported formats are upgraded automatically to the current format. If a future setup file uses a newer format than the installed software understands, import stops and asks you to update the controller software first.

Use **Import Setup** on the Fixture Controller to restore the complete setup file. Import replaces the saved setup on the XAMPP server and reloads the controller page after the restore.

Patch CSV is separate. It exports the patched DMX channel table for documentation or troubleshooting, not for restoring the show.

---

11. Clear Functions

There are two different clear actions:

Action	What it clears	When to use
DMX clear	Live DMX output and the motion base buffer	Full reset of output and base position
Output-only clear	Live DMX output only	Black out output while keeping the stored motion center

Use output-only clear when you want Effects to resume around the same stored center after the blackout.

Troubleshooting

The UI does not control the Pico

- Check that the Pico is powered and connected to WiFi.
- Open the Pico URL directly in a browser, for example `http://192.168.0.24/`.
- Make sure the base URL in the UI ends with `/`.
- Make sure the Fixture Controller has the correct Pico base URL.

Chaser capture does not contain the expected values

- Move or recall the values on the Fixture Controller first.
- Make sure the participating controls include the controls you want to capture.
- Save or reload the Fixture Controller setup if fixtures were changed recently.

Effects move around the wrong center

- Recall the desired scene first.
- On the Effects page, click the same scene in the scene toolbox.
- Then start the effect slot.

GPIO mapping does not react

- Check `/gpio/status` or the status panel on the GPIO page.
- Confirm that the selected pin is not reserved or already used.
- Check pull mode and trigger edge.
- For ADC, use only GPIO26, GPIO27, or GPIO28.

A chaser has more than 32 steps

The firmware supports 32 steps per chaser slot. Keep the chase within this limit before uploading to the Pico.